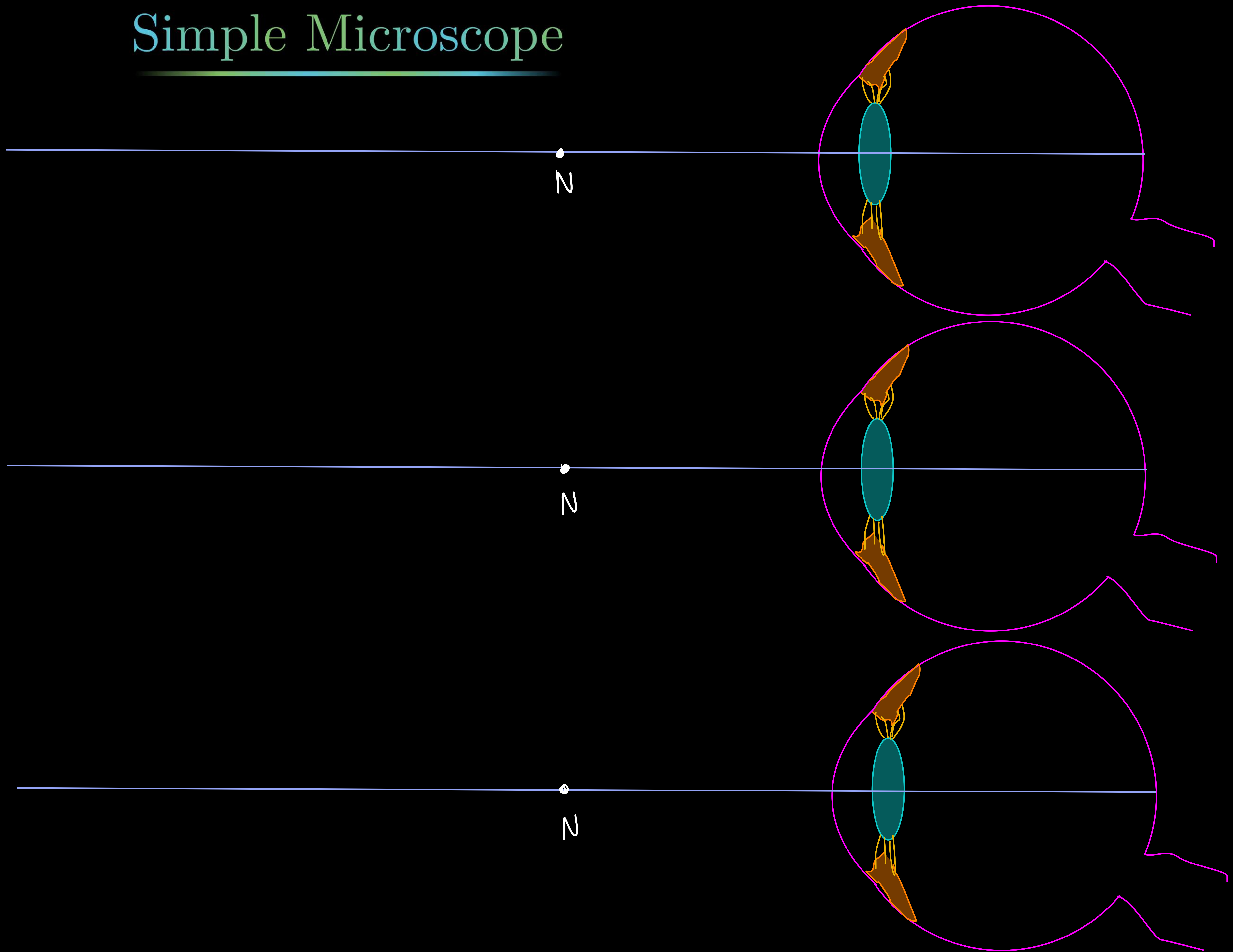
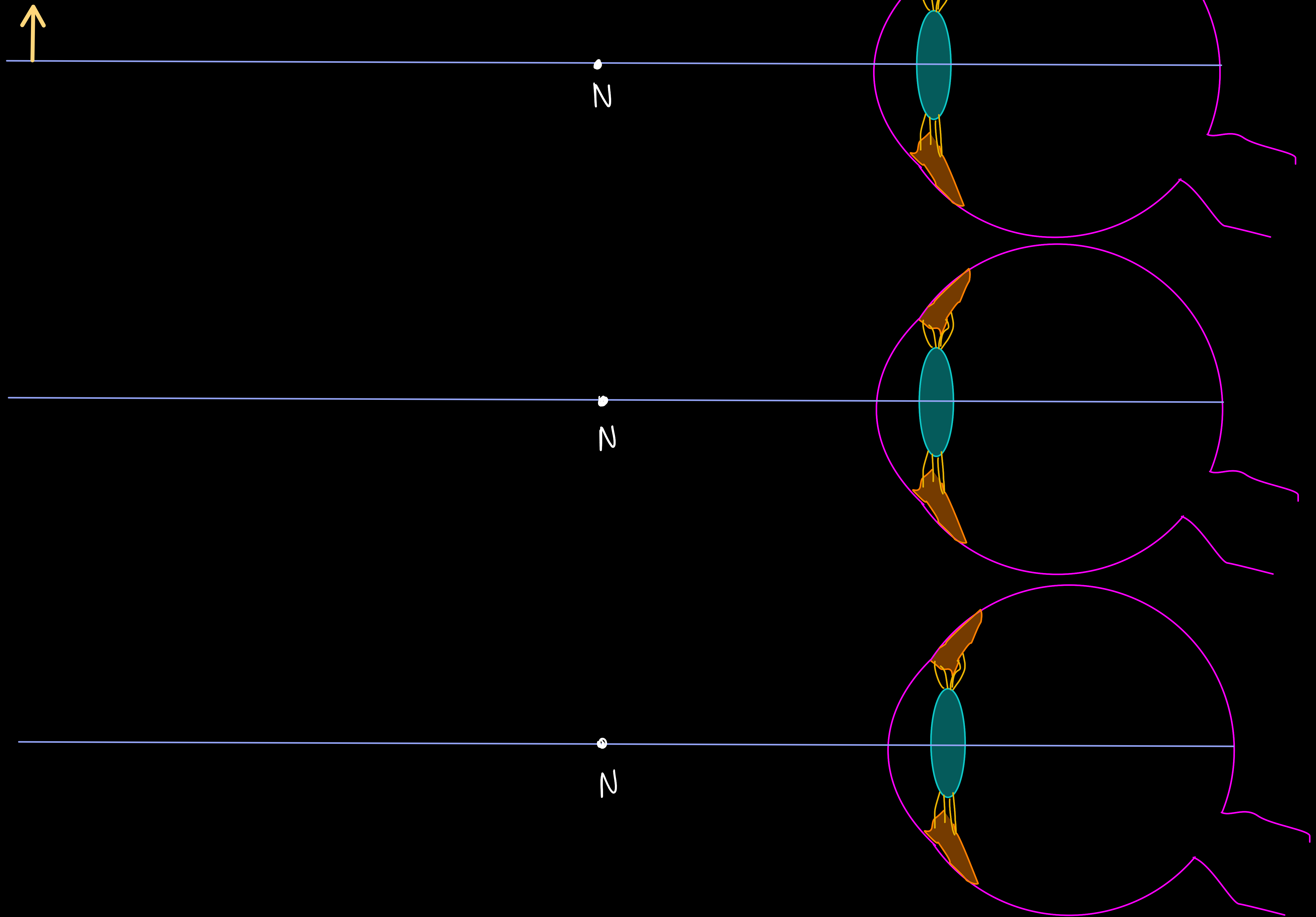


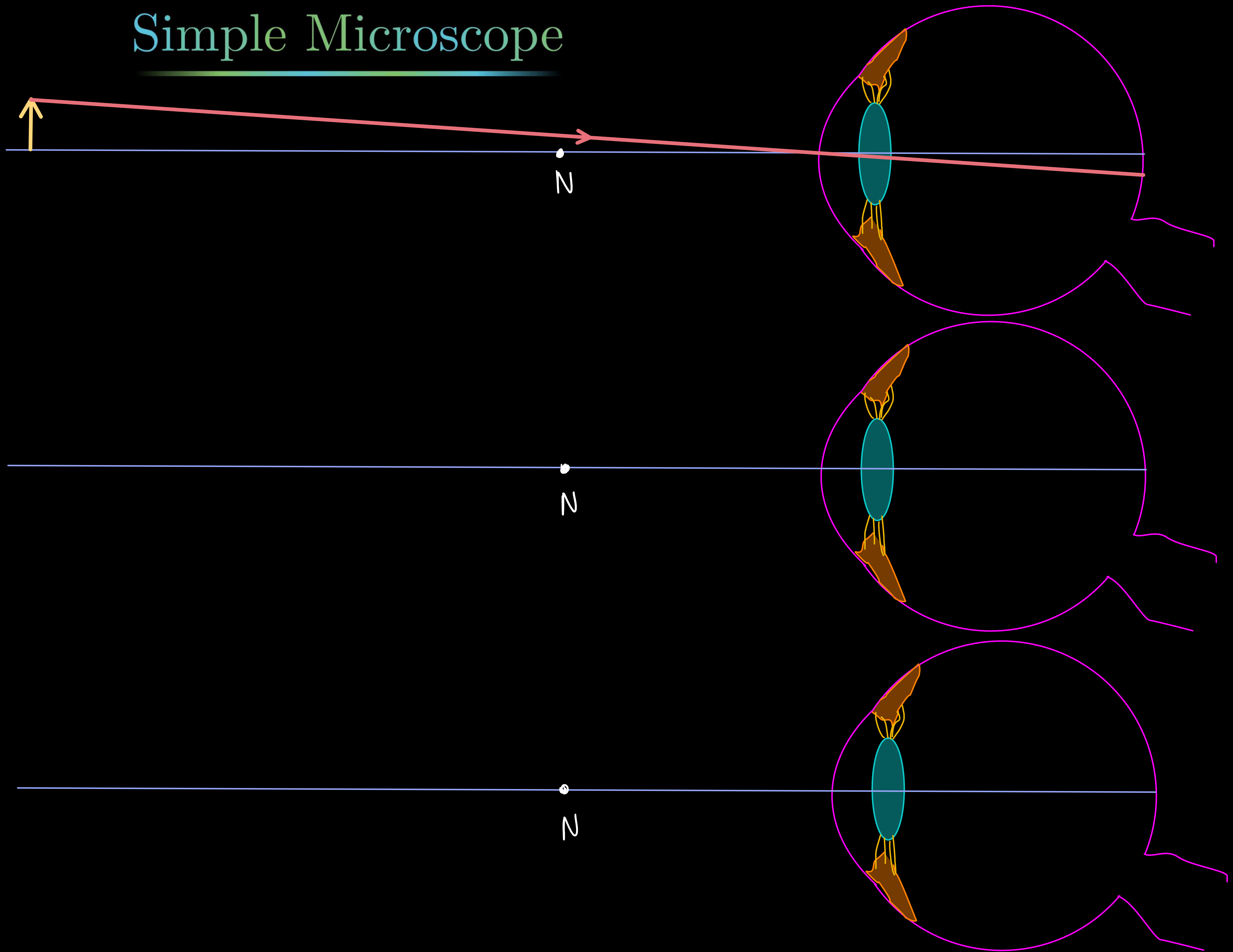
Simple Microscope



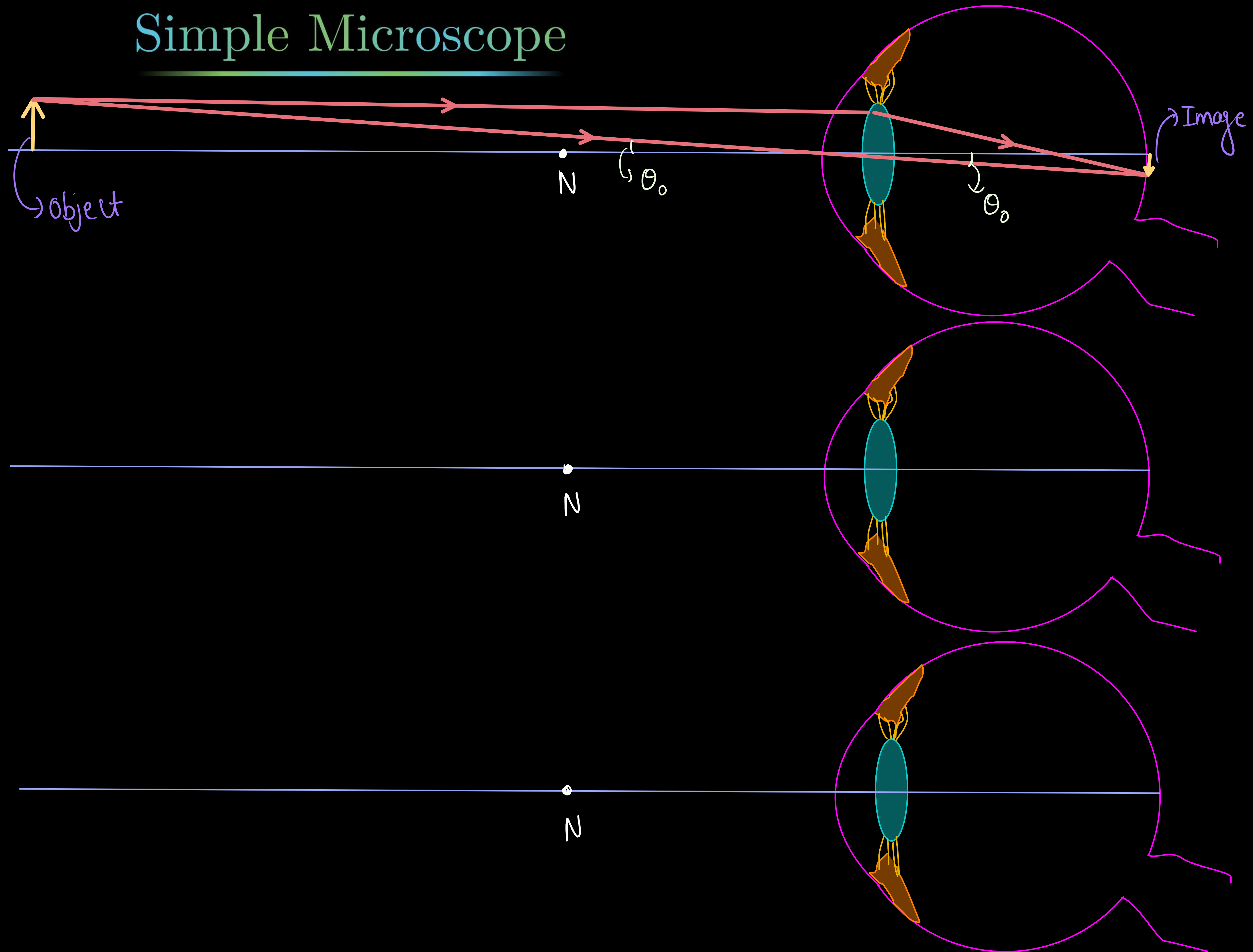
Simple Microscope



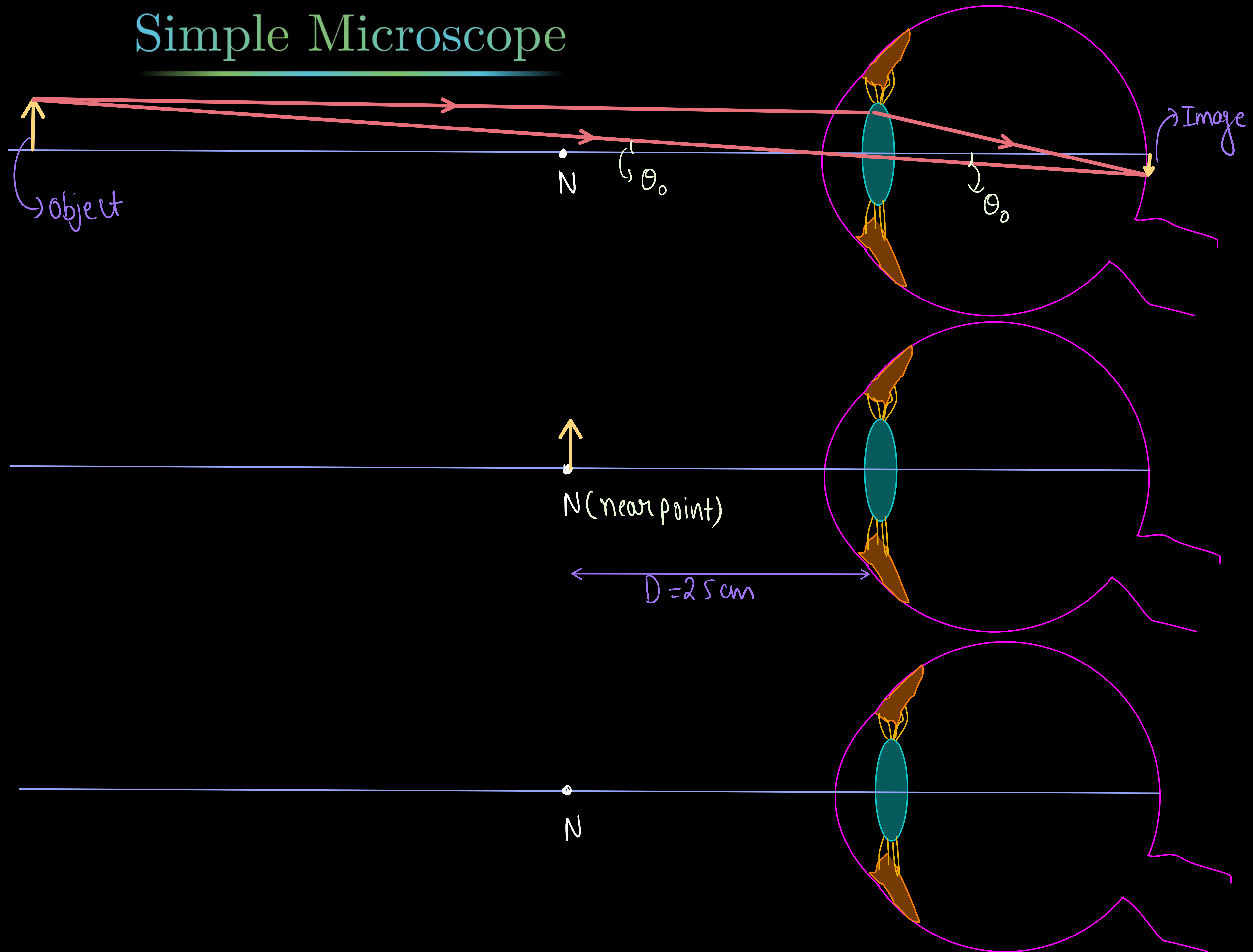
Simple Microscope



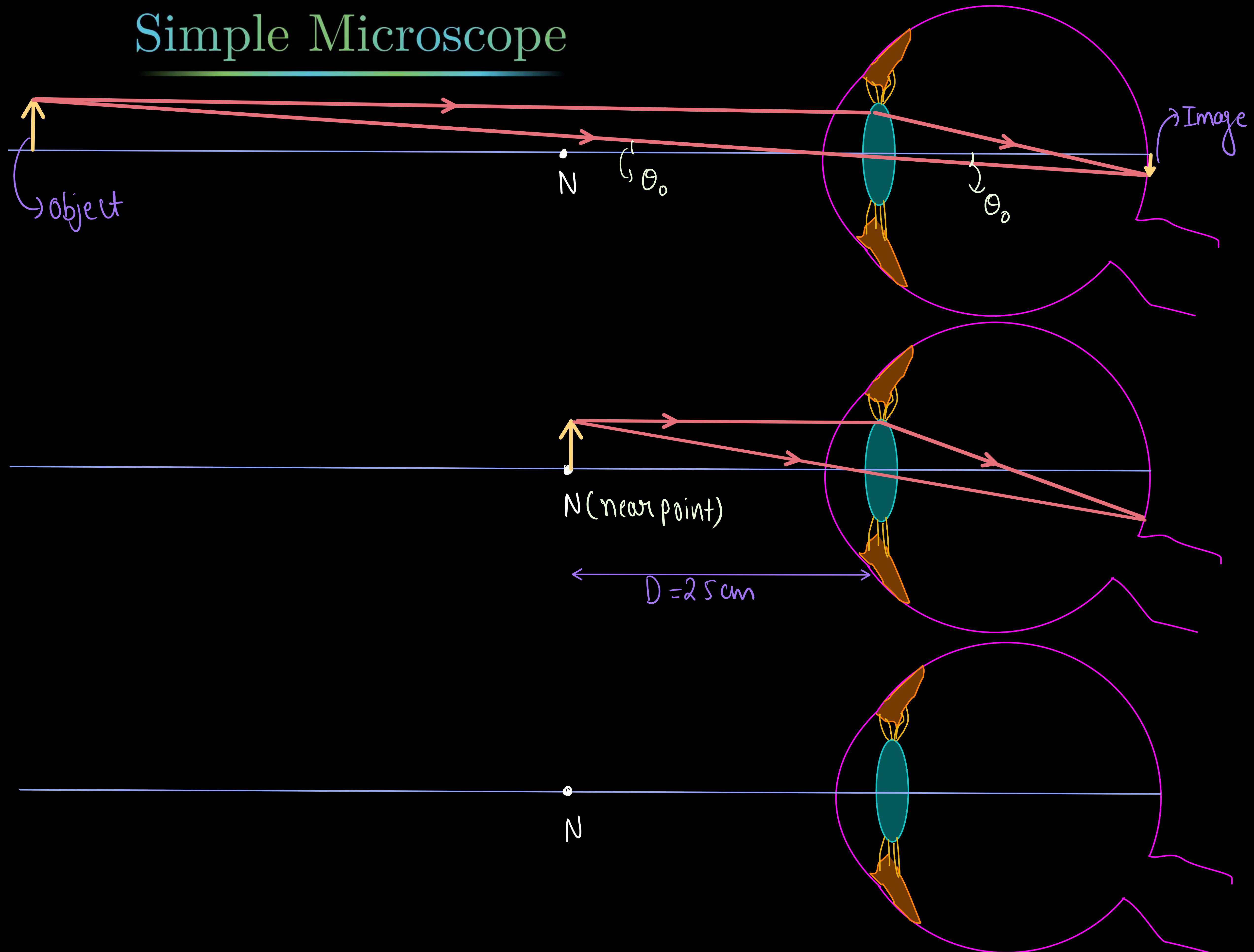
Simple Microscope



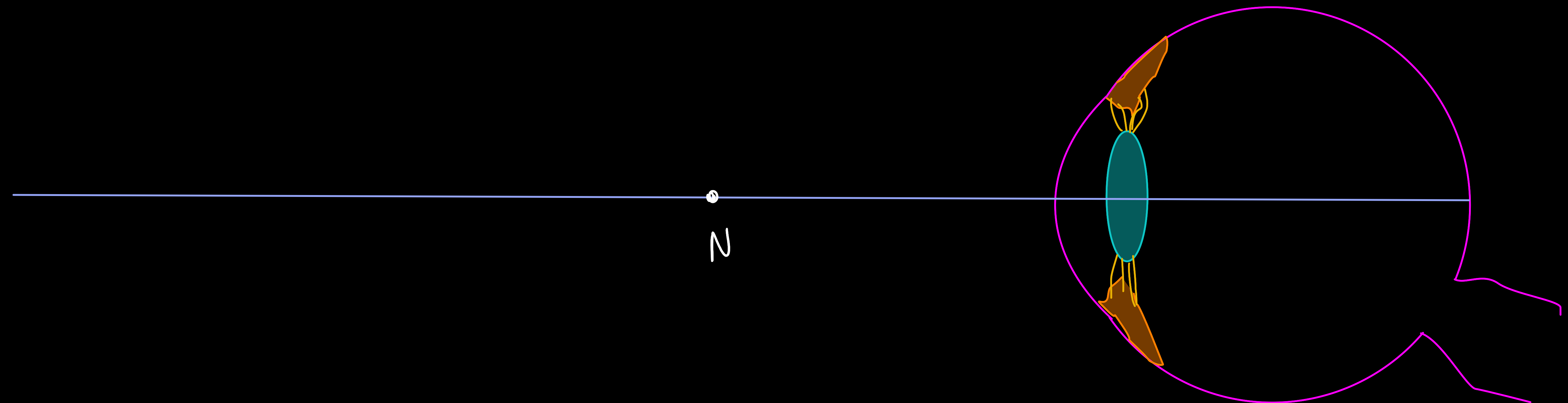
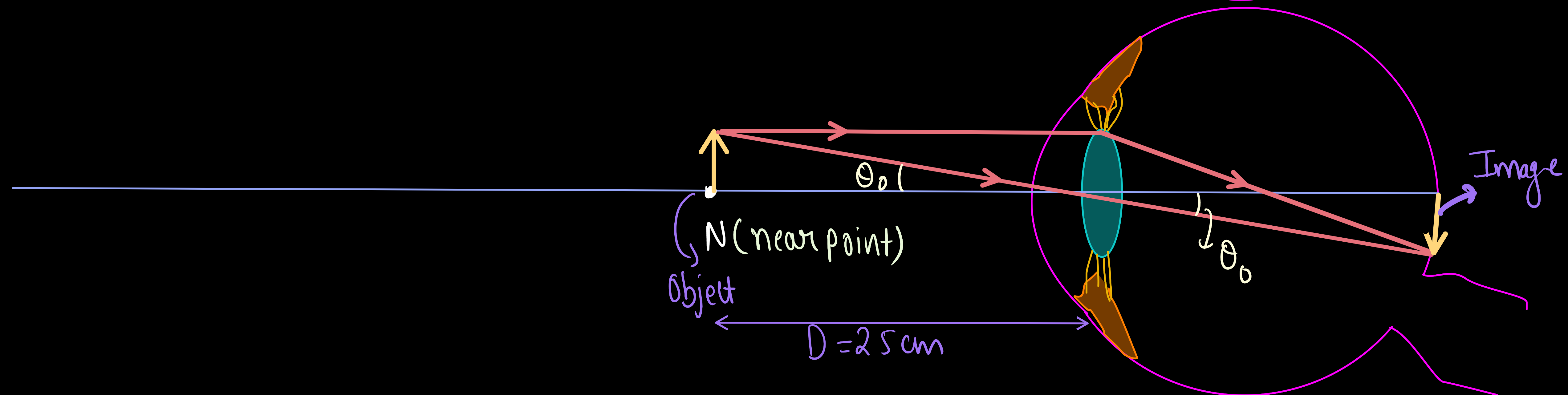
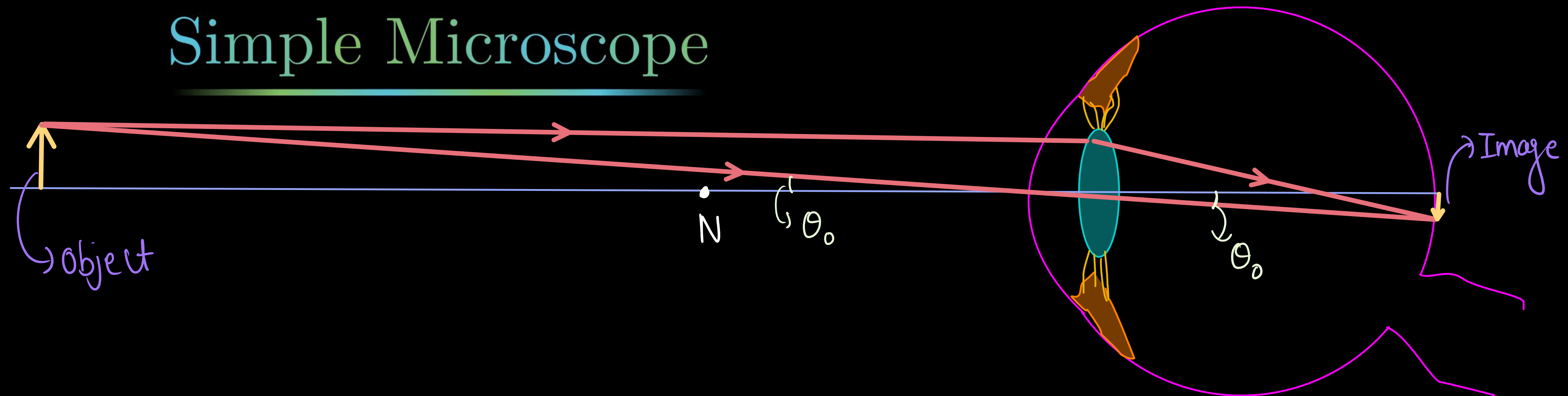
Simple Microscope



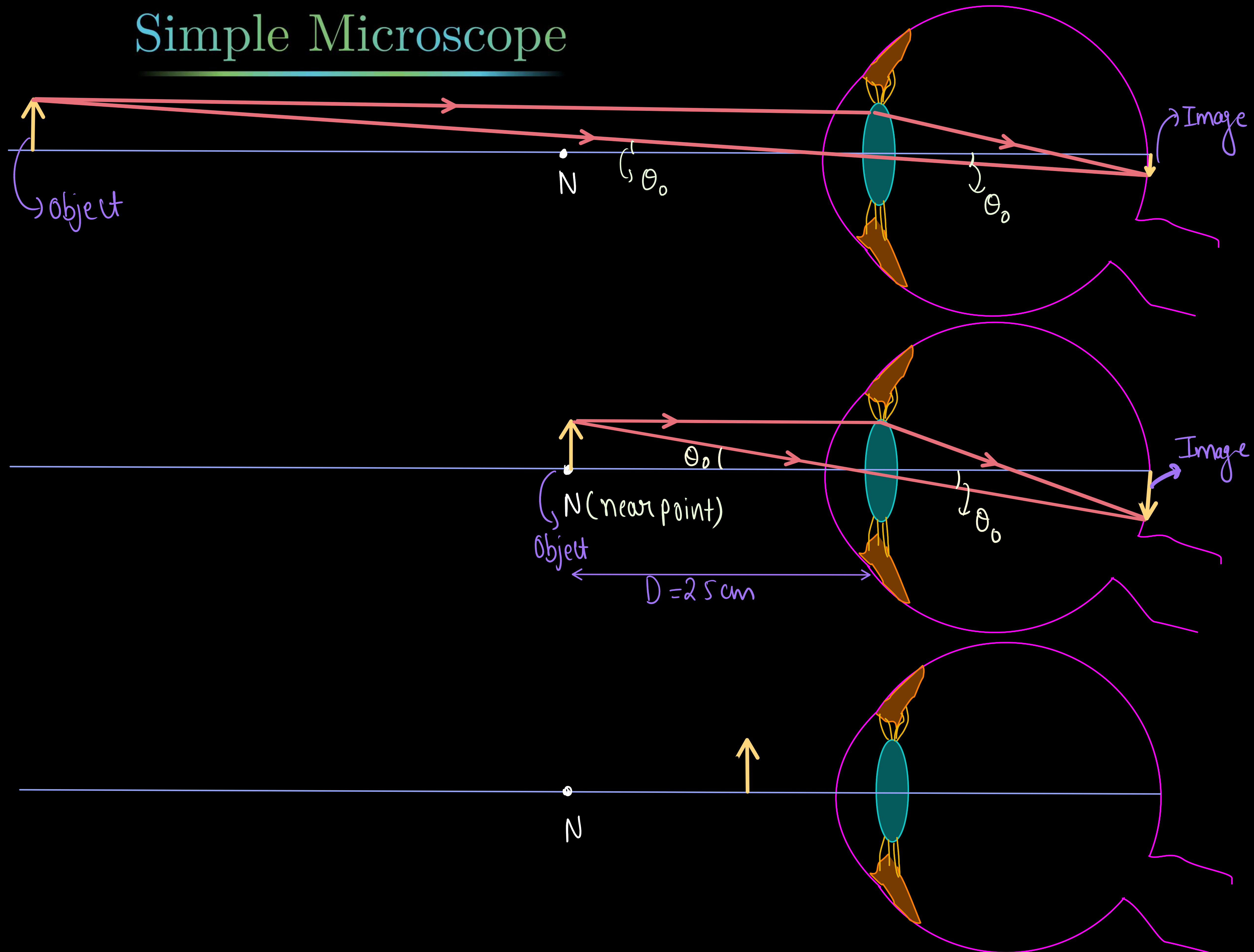
Simple Microscope



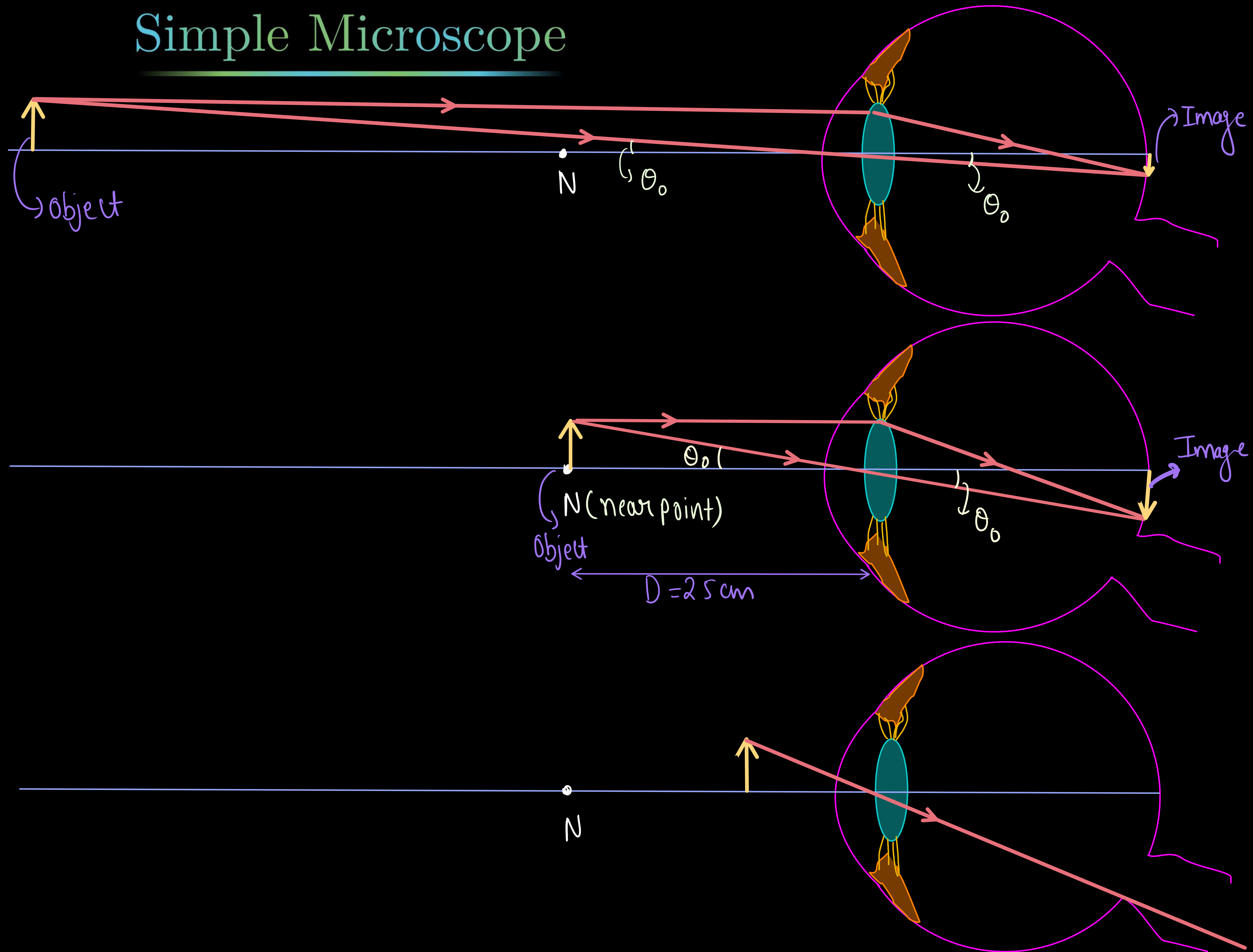
Simple Microscope



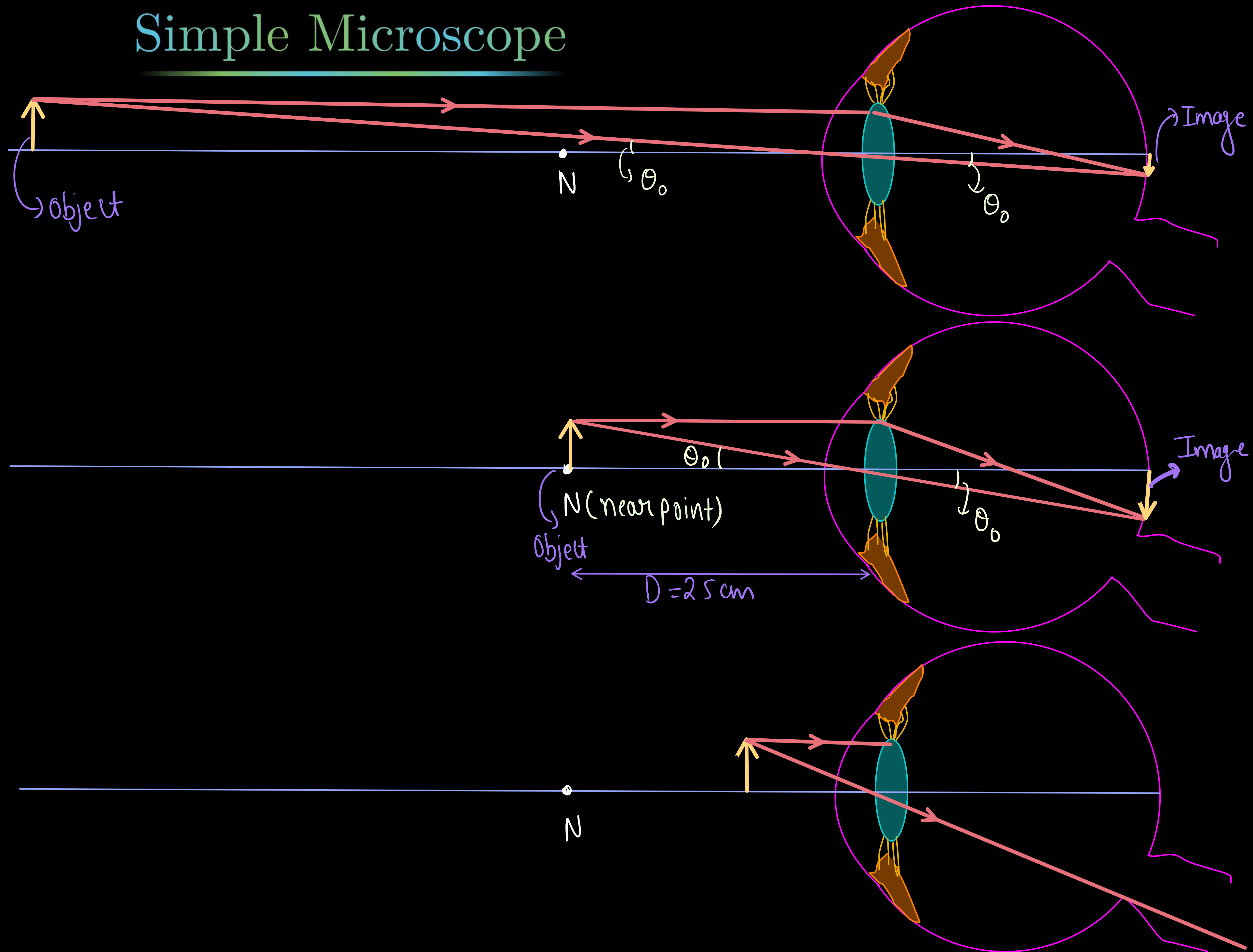
Simple Microscope



Simple Microscope



Simple Microscope



Simple Microscope

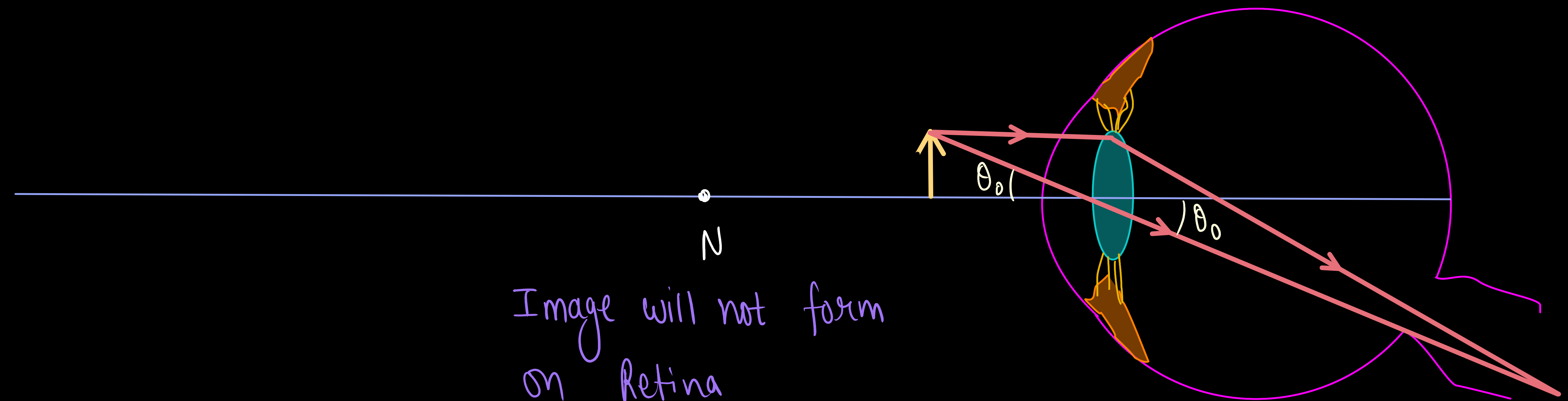
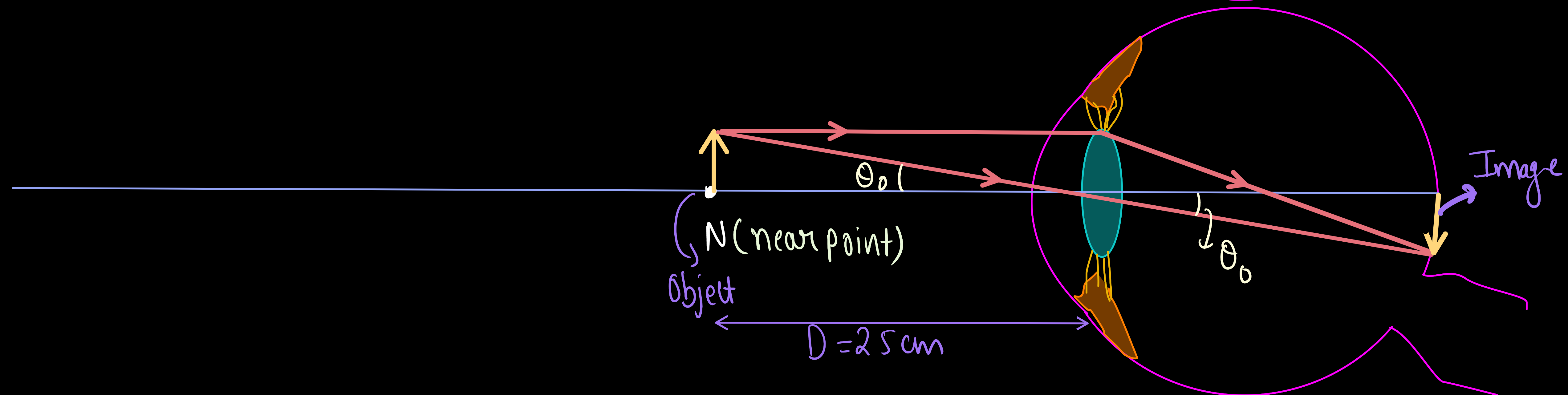
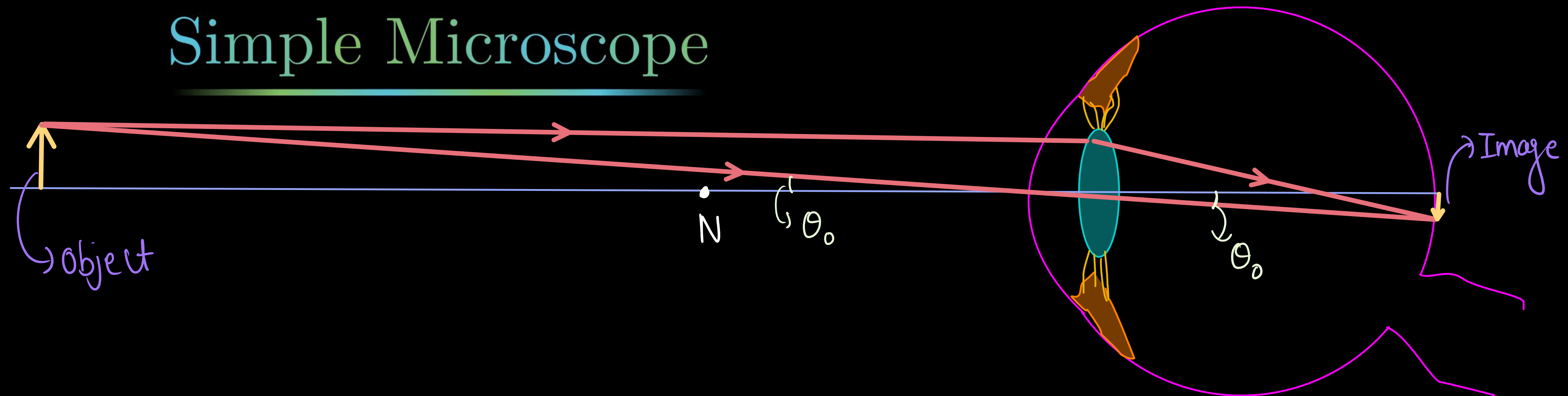
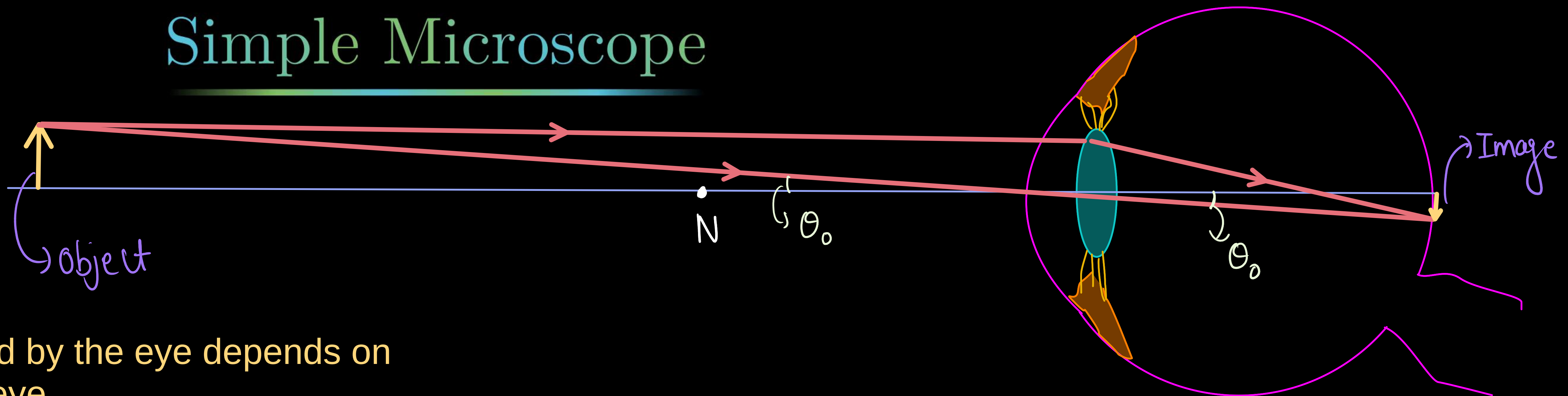
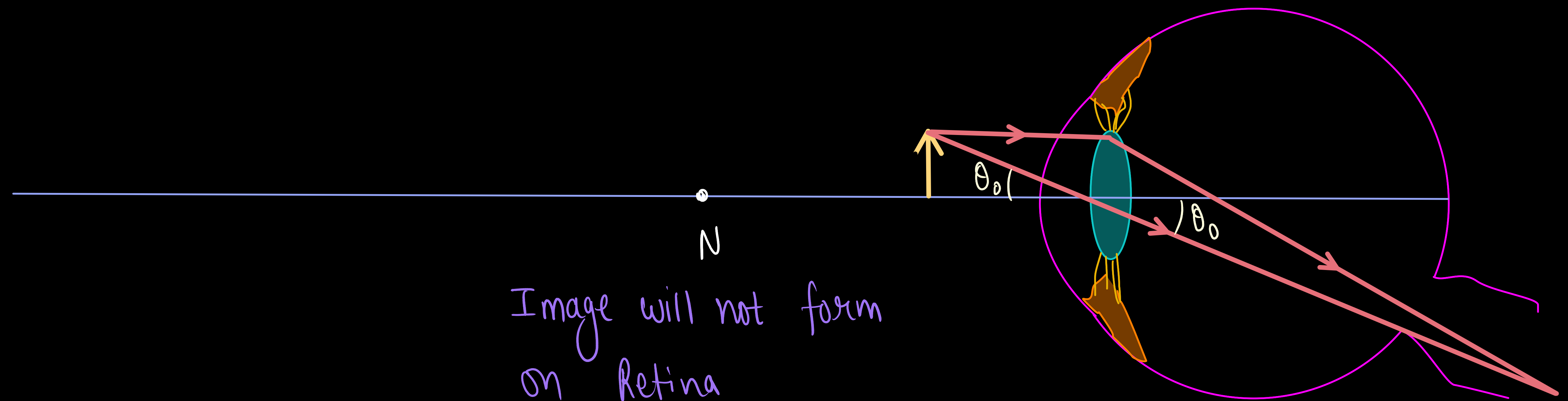
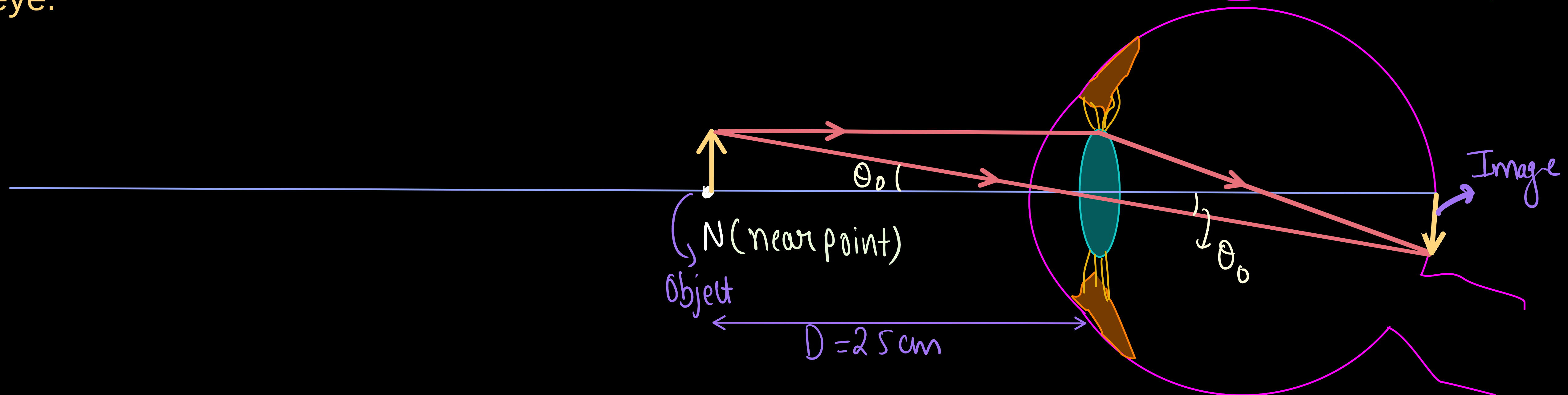


Image will not form
on Retina

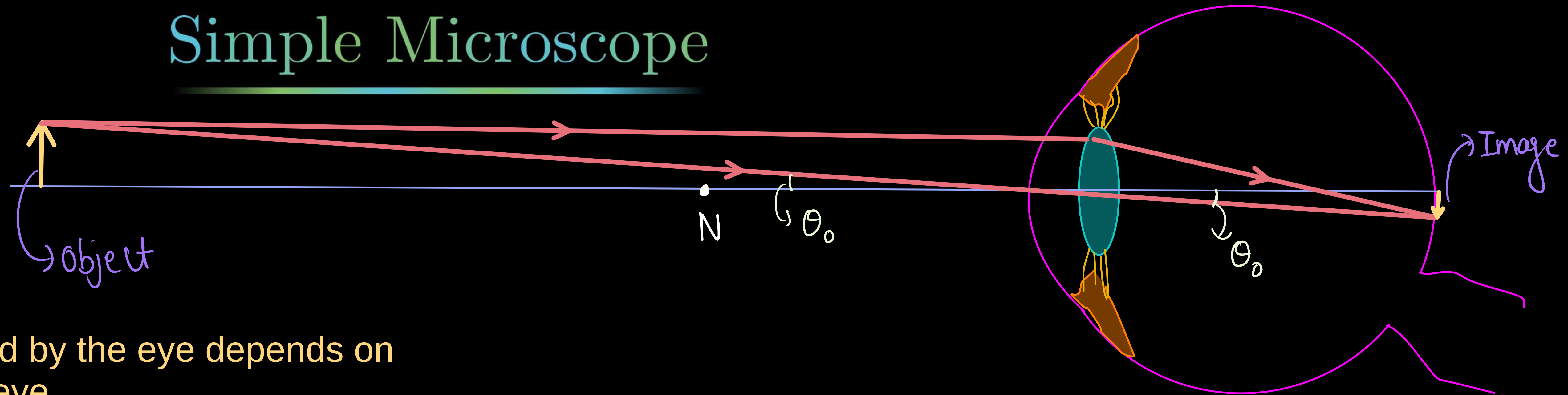
Simple Microscope



The apparent size of an object perceived by the eye depends on the angle the object subtends from the eye.

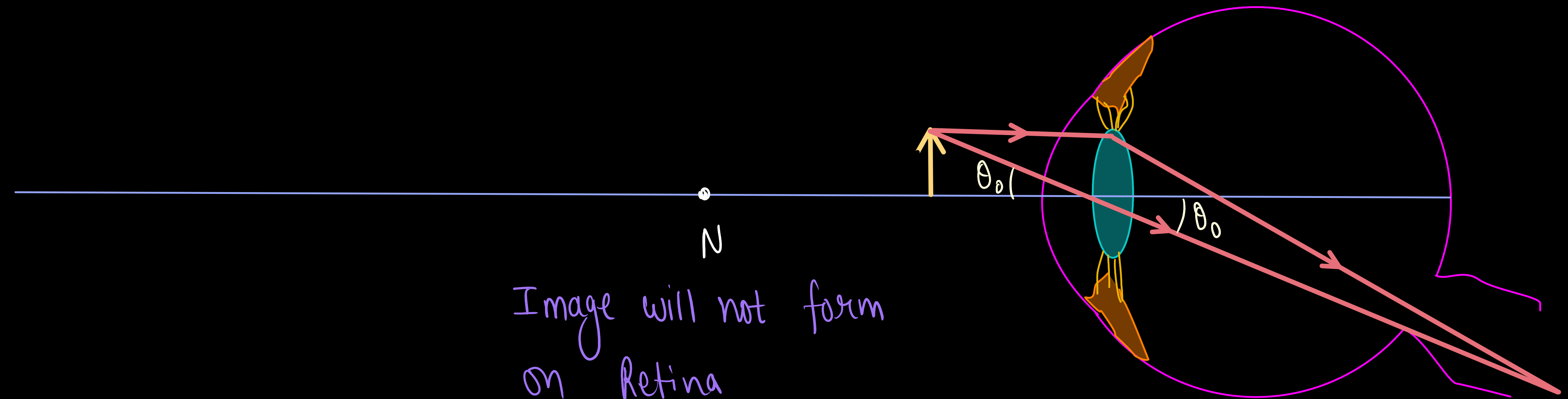
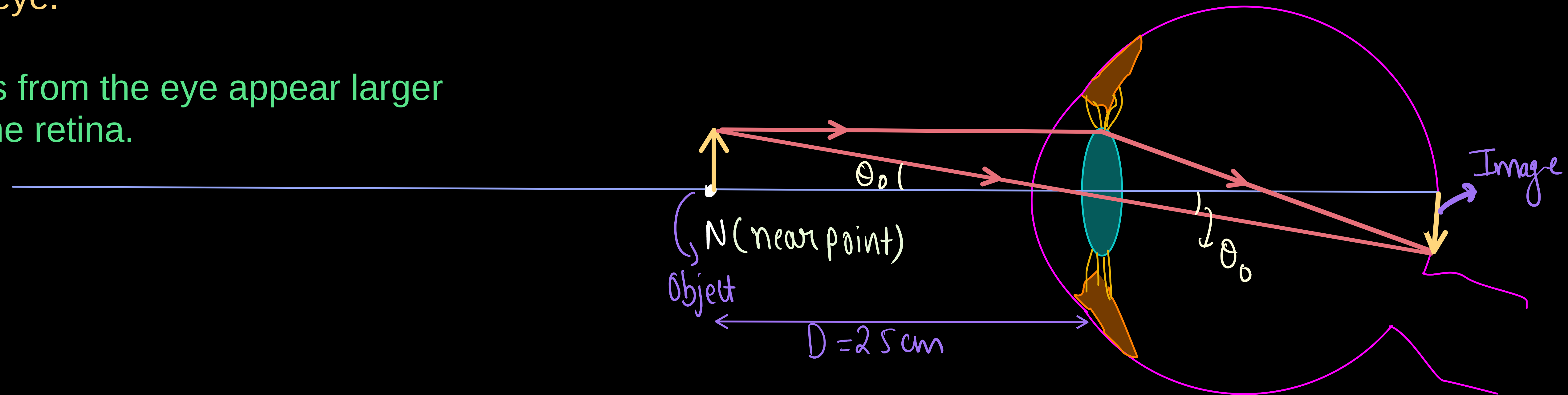


Simple Microscope



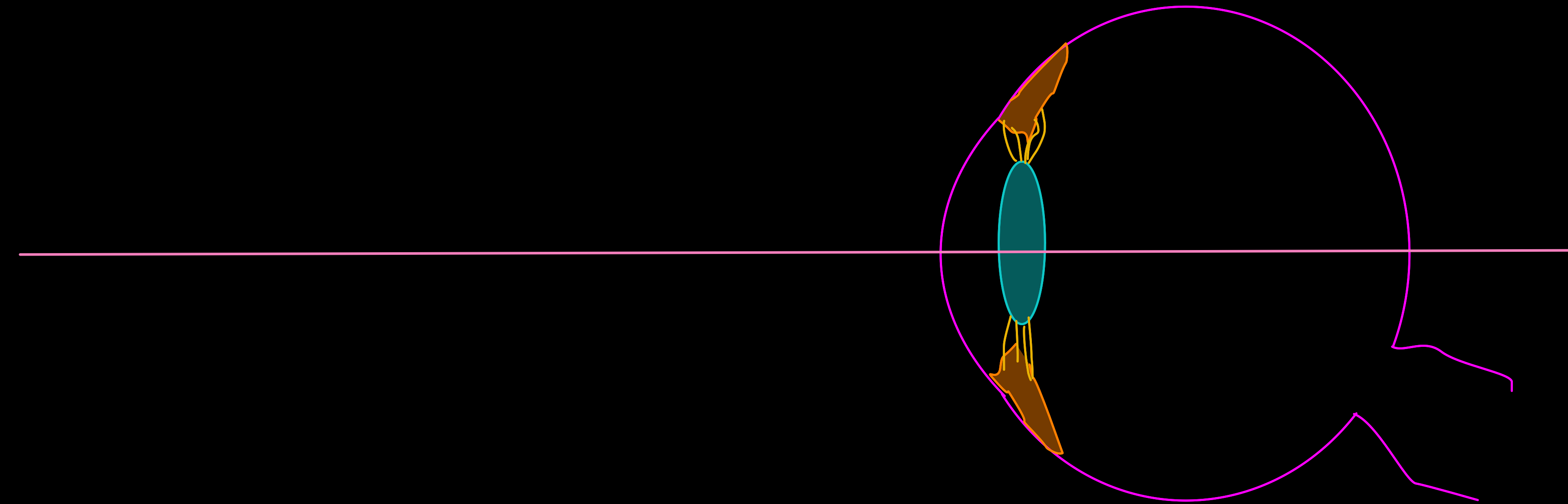
The apparent size of an object perceived by the eye depends on the angle the object subtends from the eye.

Thus, objects that subtend large angles from the eye appear larger because they form larger images on the retina.

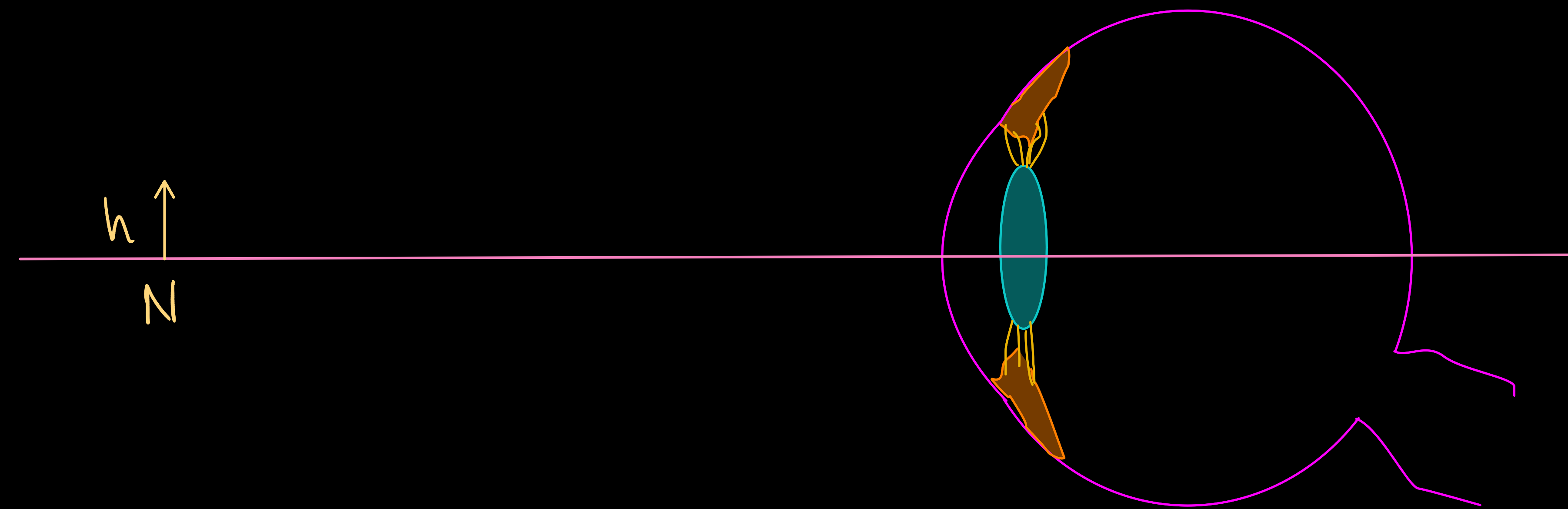


Simple Microscope

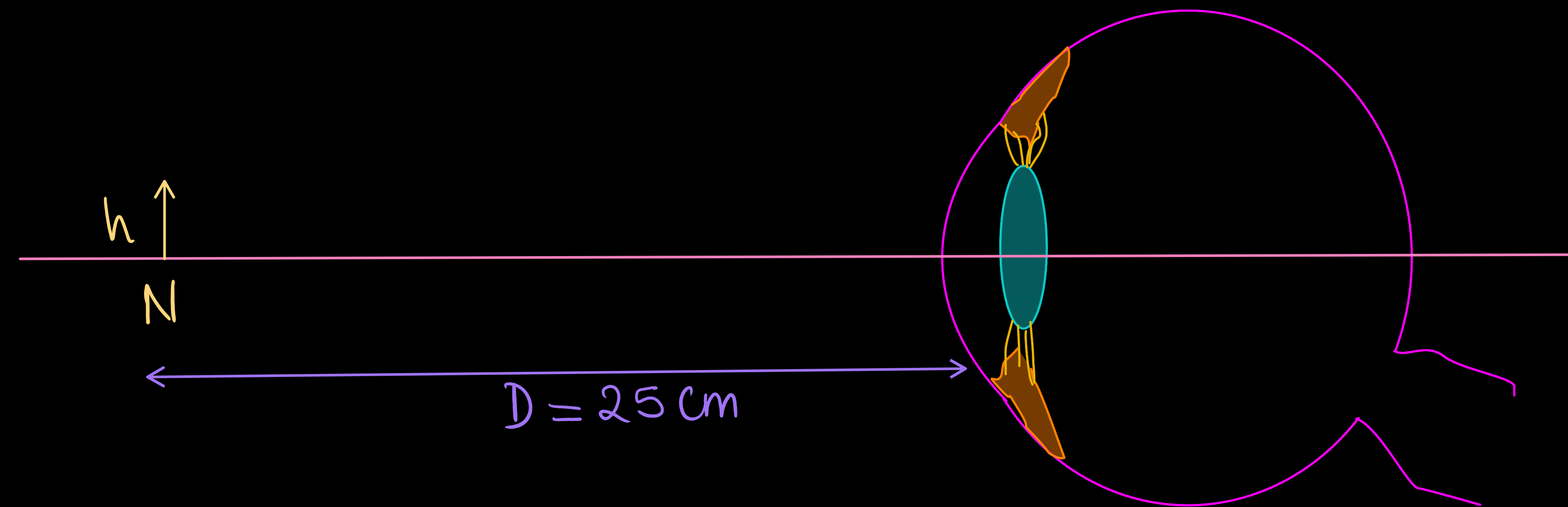
Simple Microscope



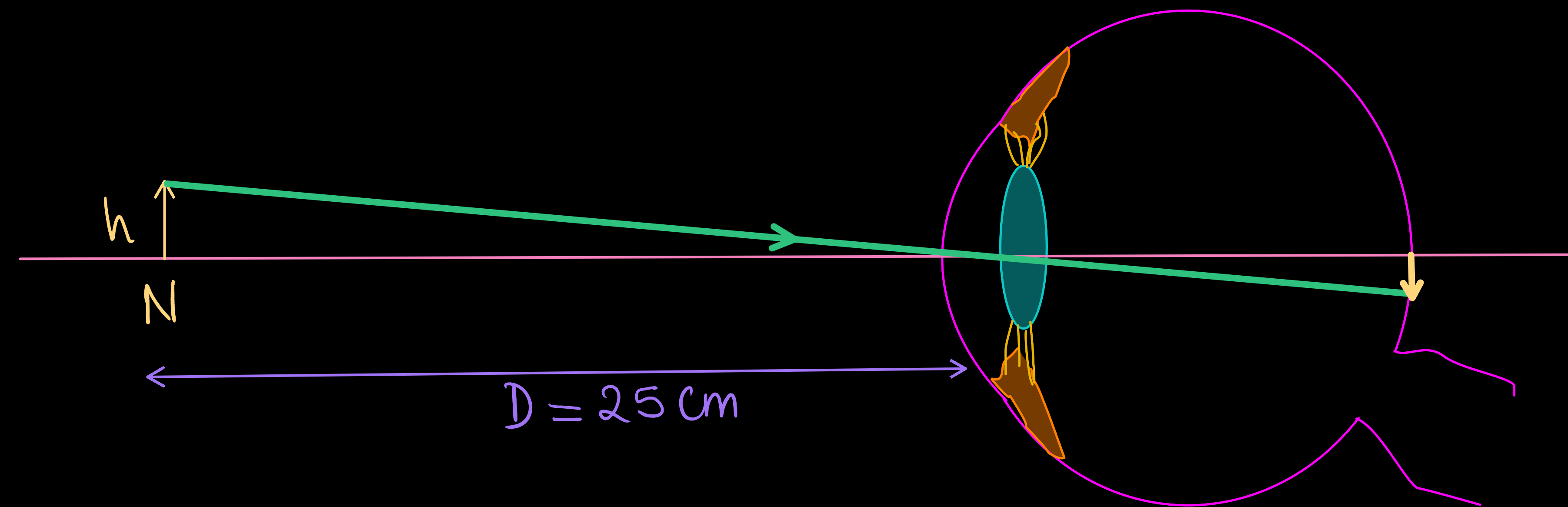
Simple Microscope



Simple Microscope

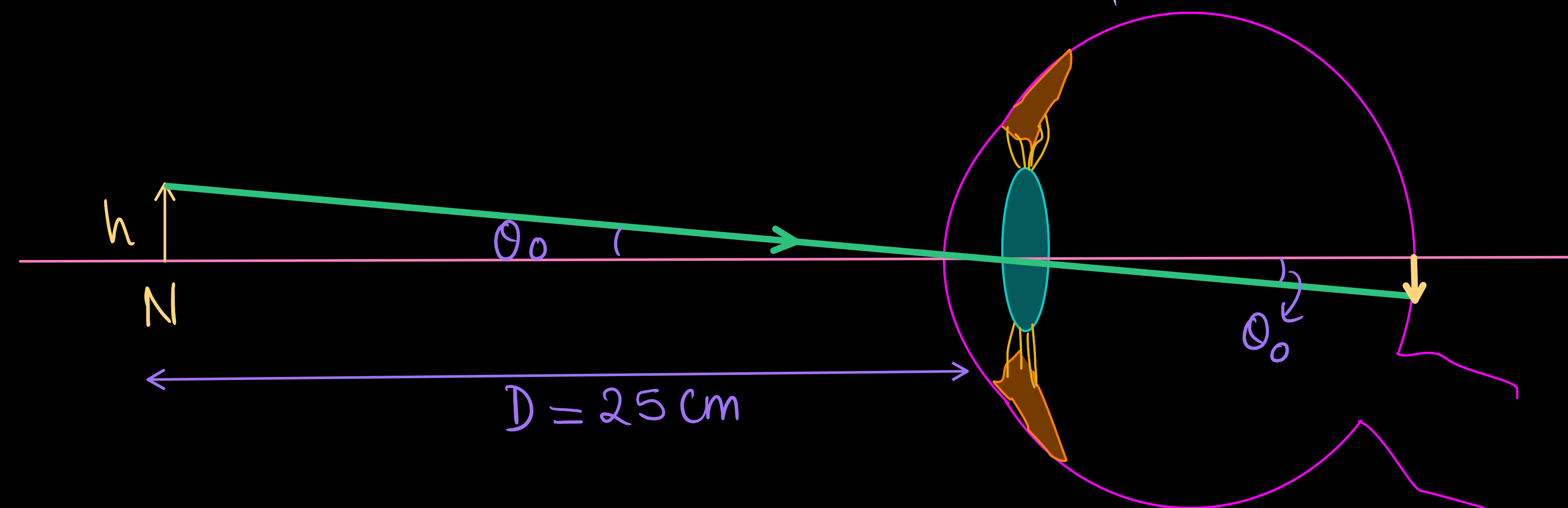


Simple Microscope



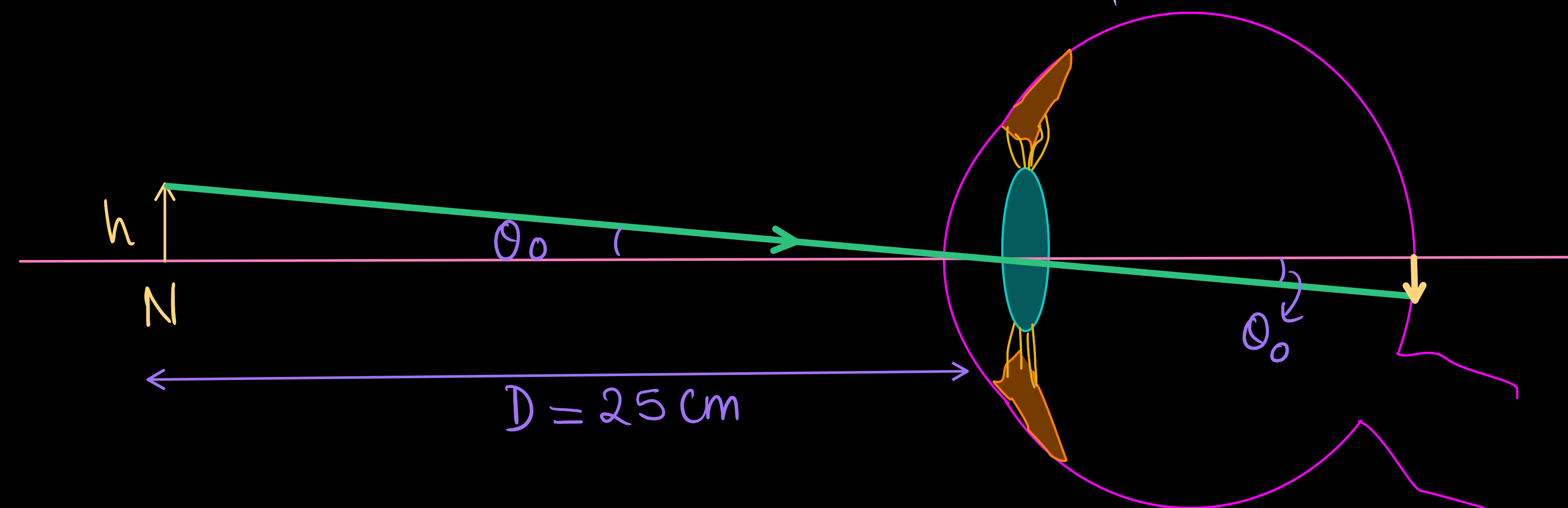
Simple Microscope

for naked eye -
The angular size (θ_0) of image formed at retina is maximum when object is at near point of the eye.



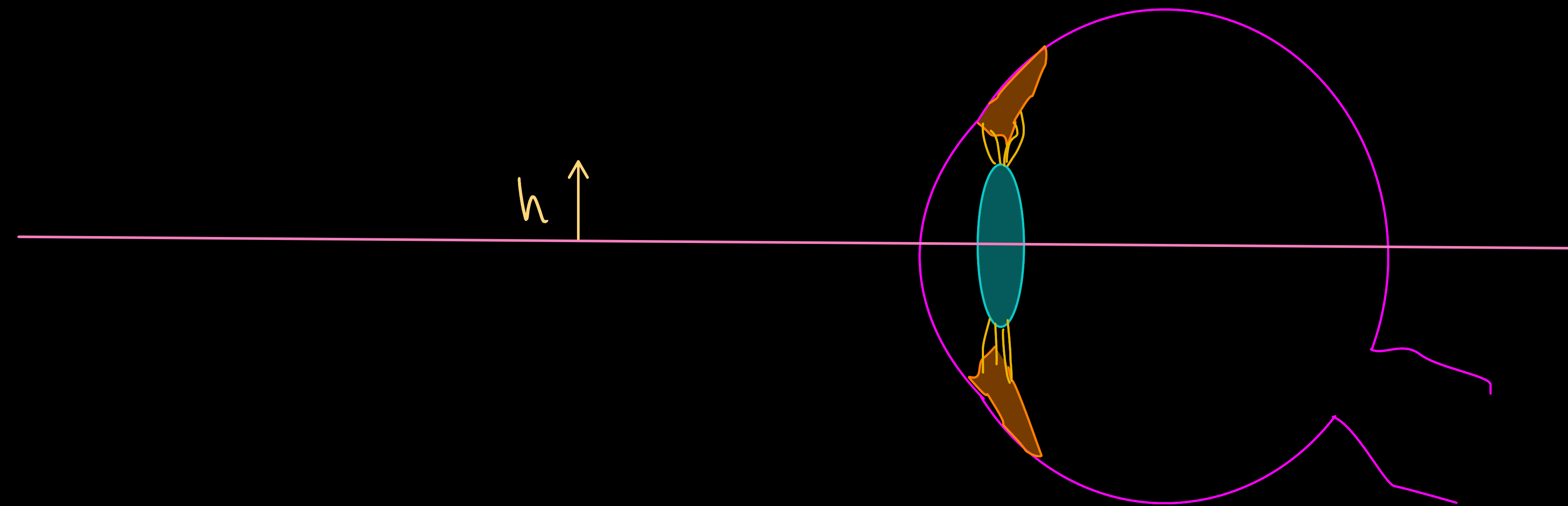
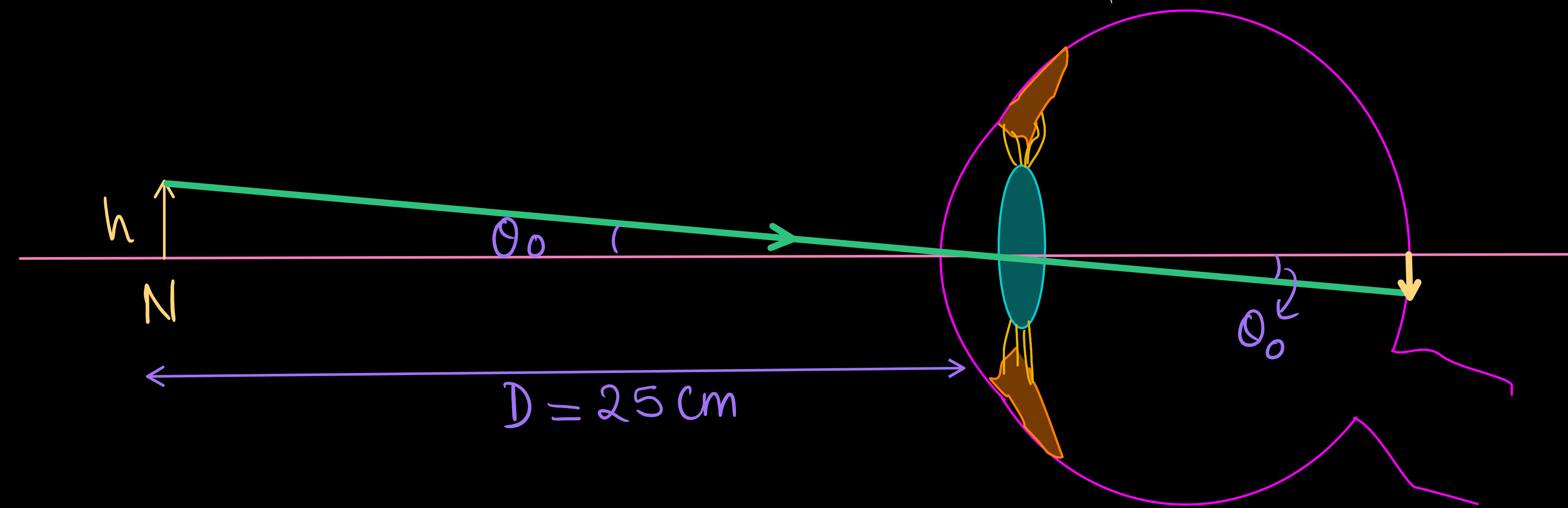
Simple Microscope

for naked eye -
The angular size (θ_o) of image formed at retina is maximum when object is at near point of the eye.



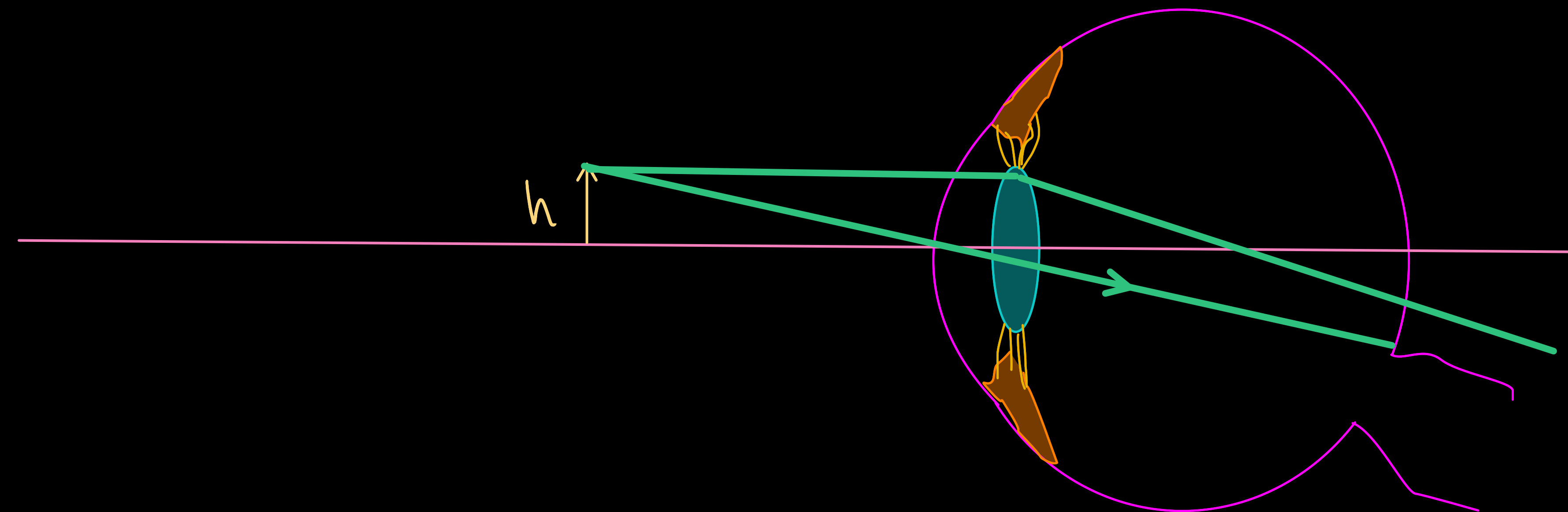
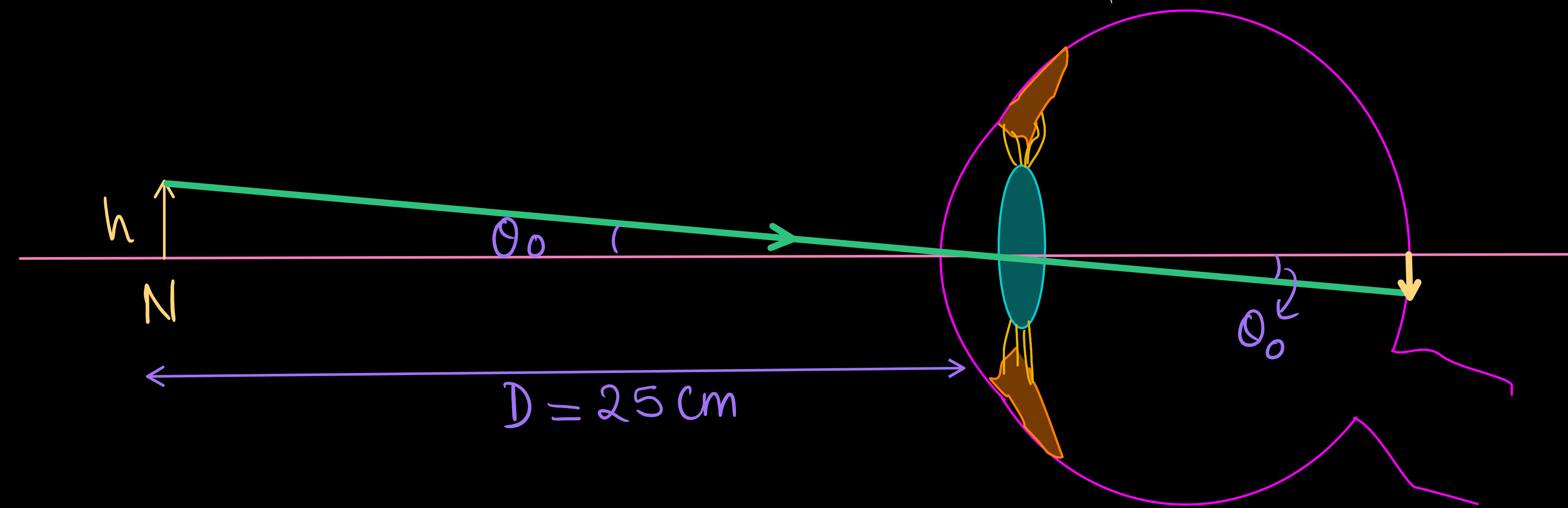
Simple Microscope

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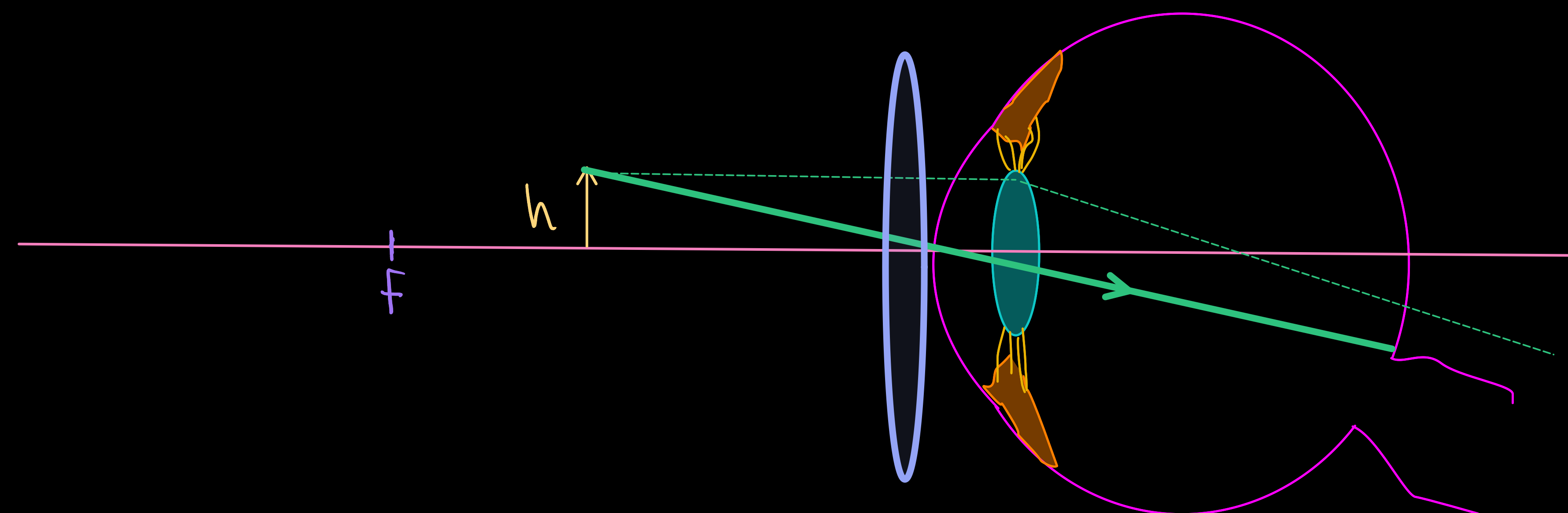
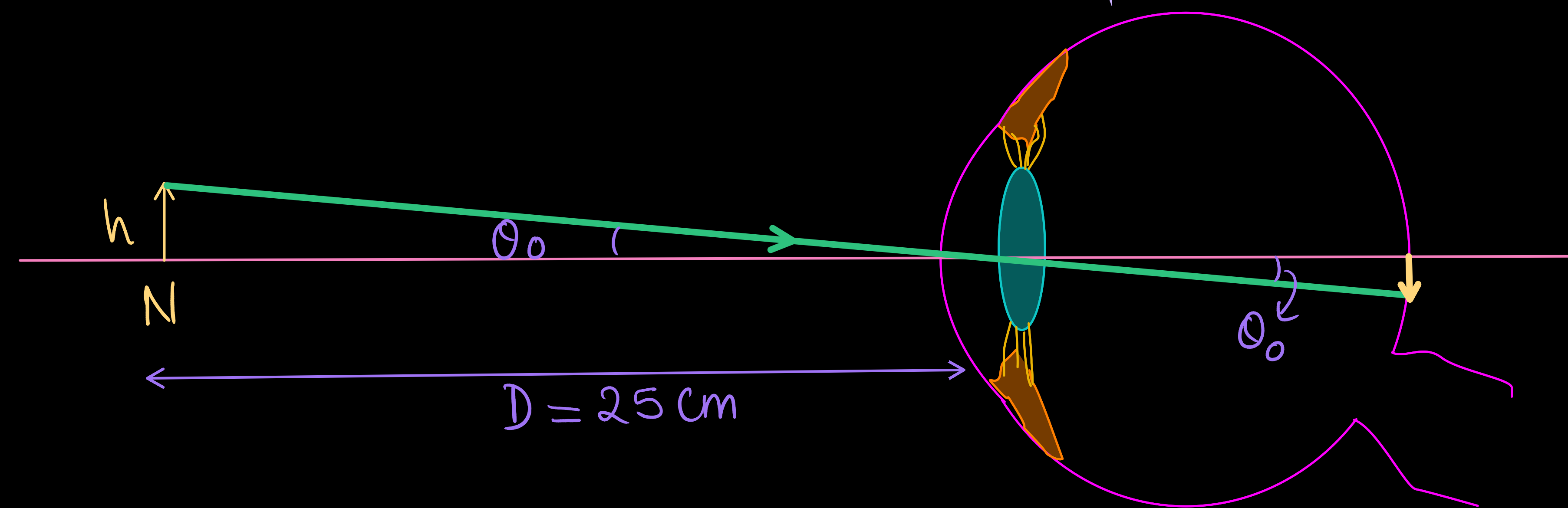
Simple Microscope

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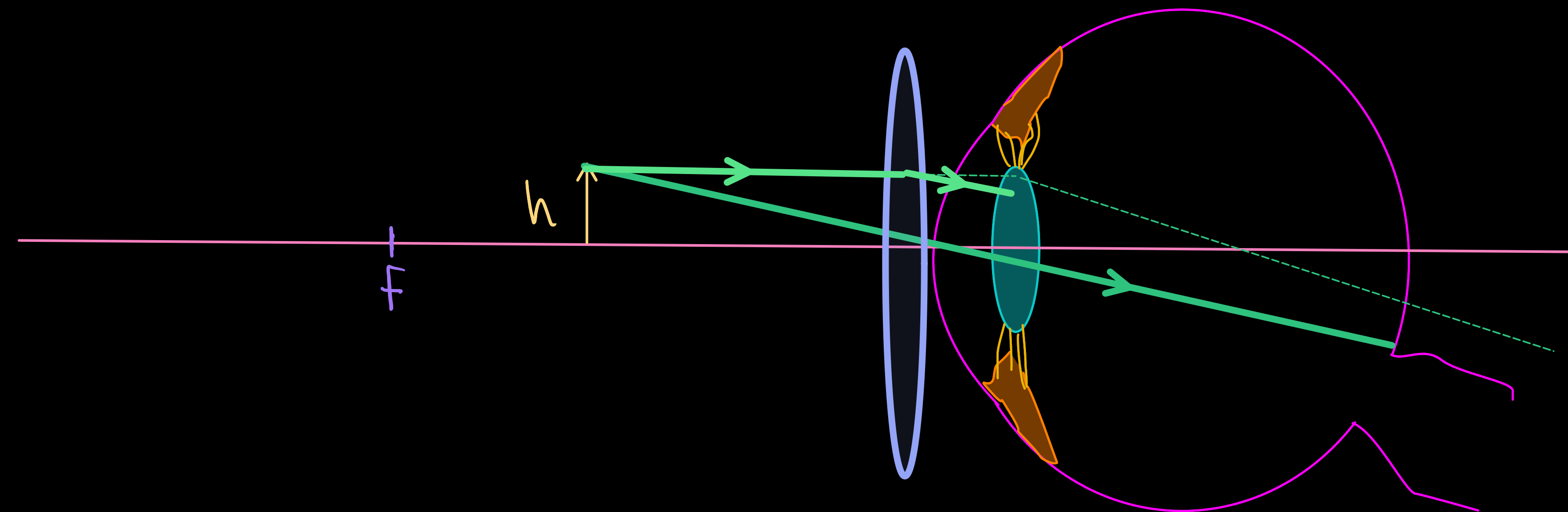
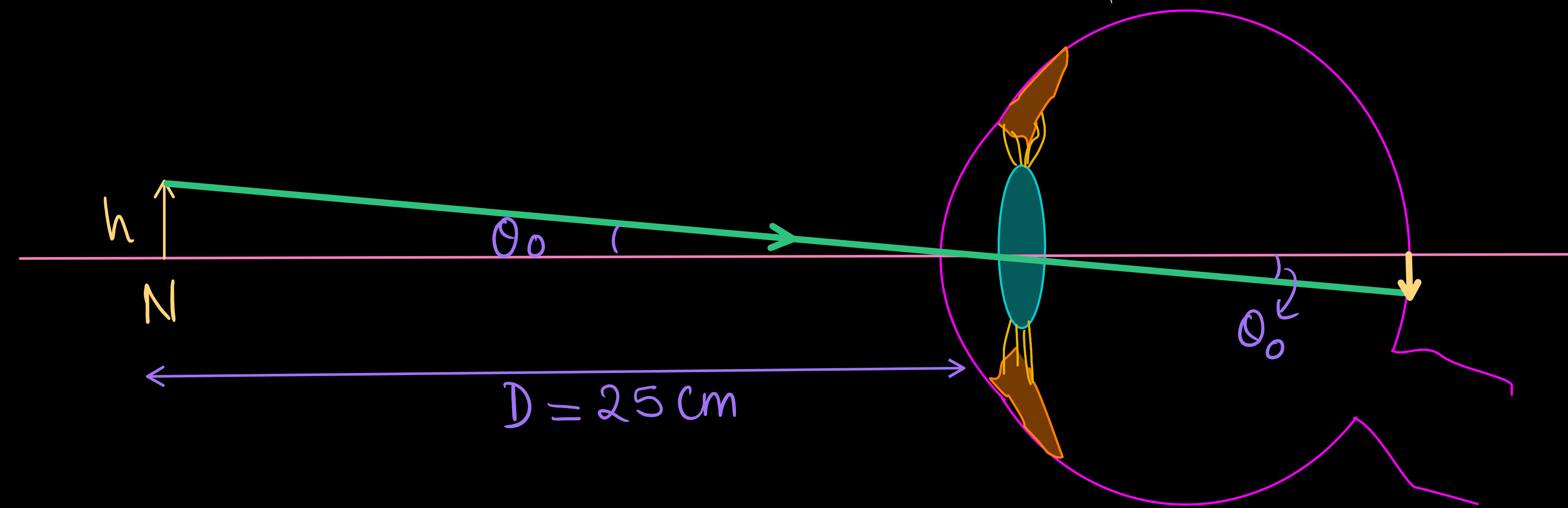
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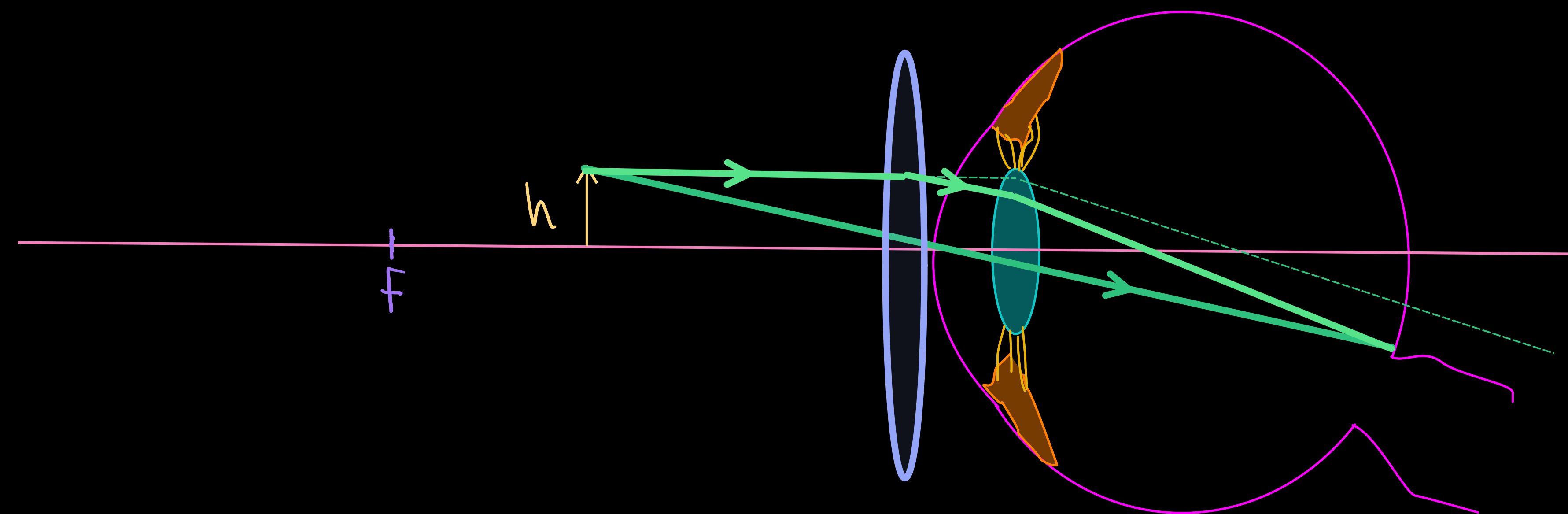
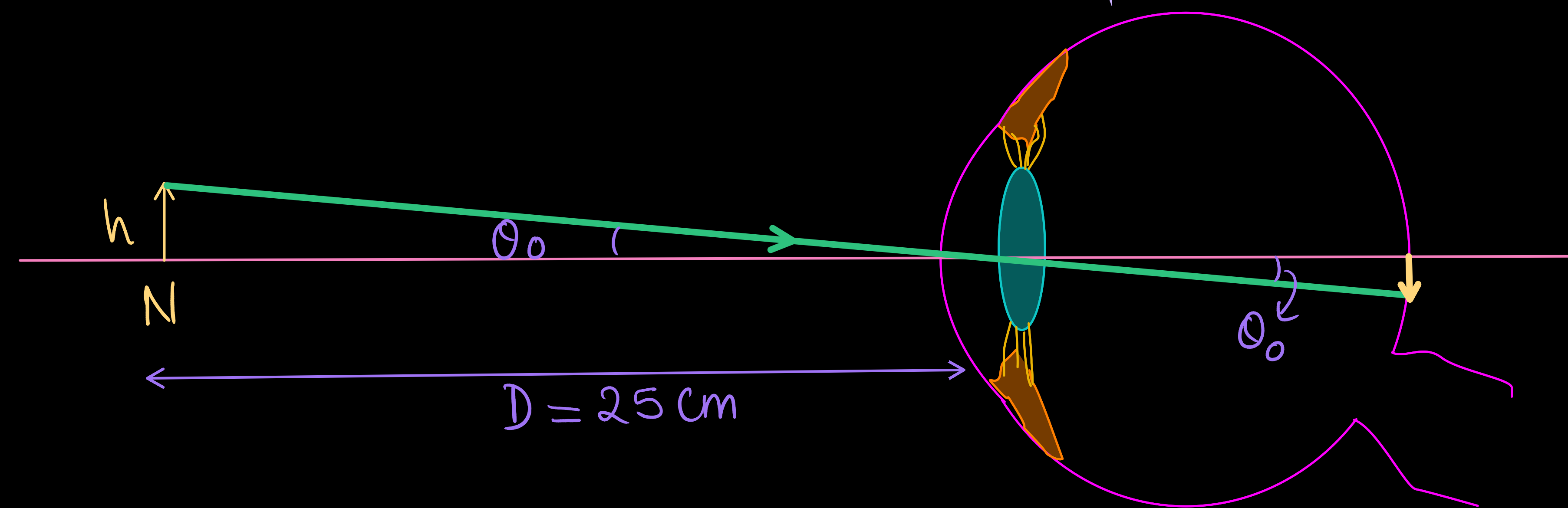
Simple Microscope

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The angular size (θ_0) of image formed at retina is maximum when object is at near point of the eye.



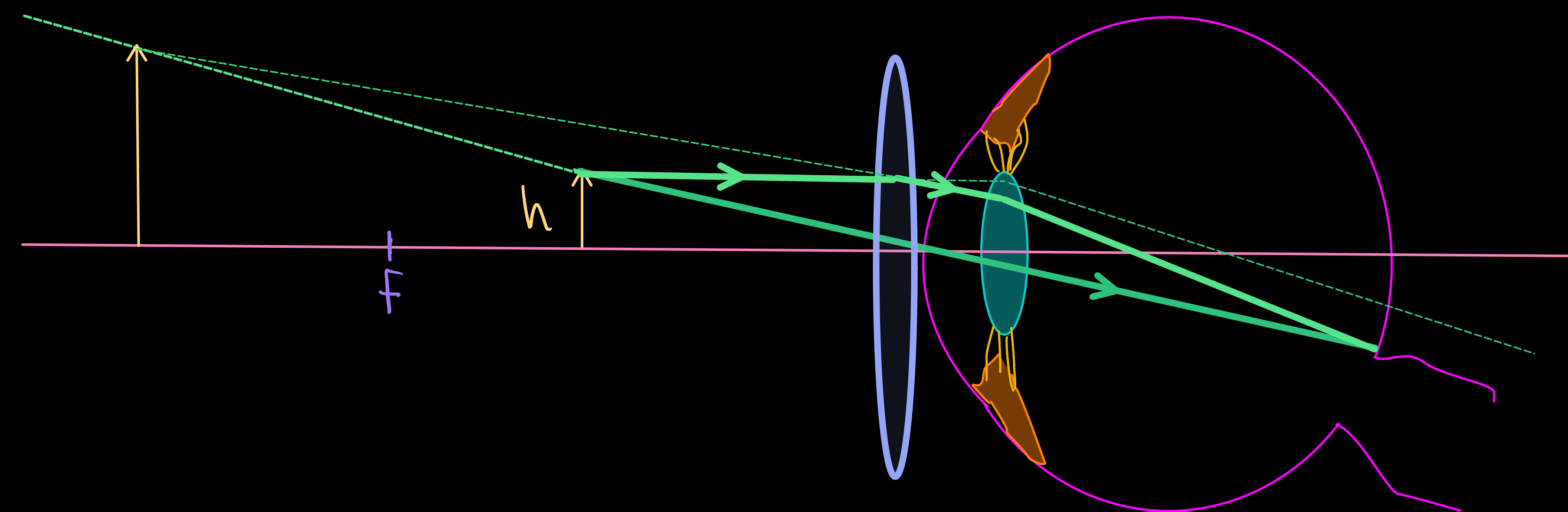
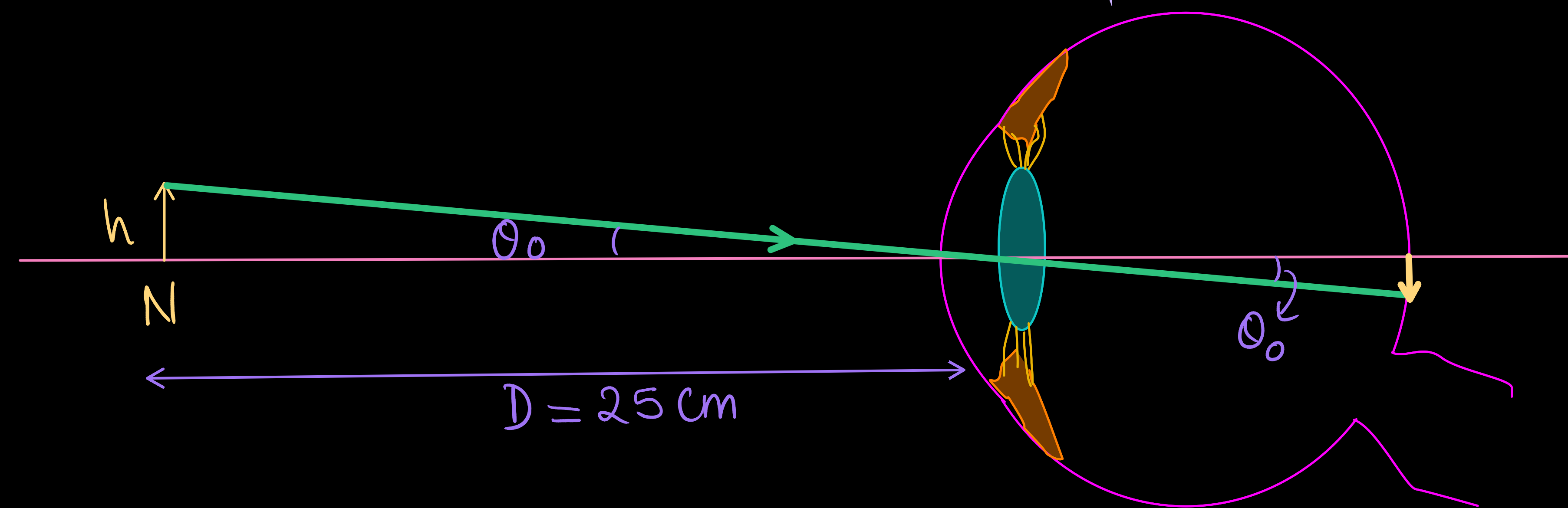
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for naked eye -
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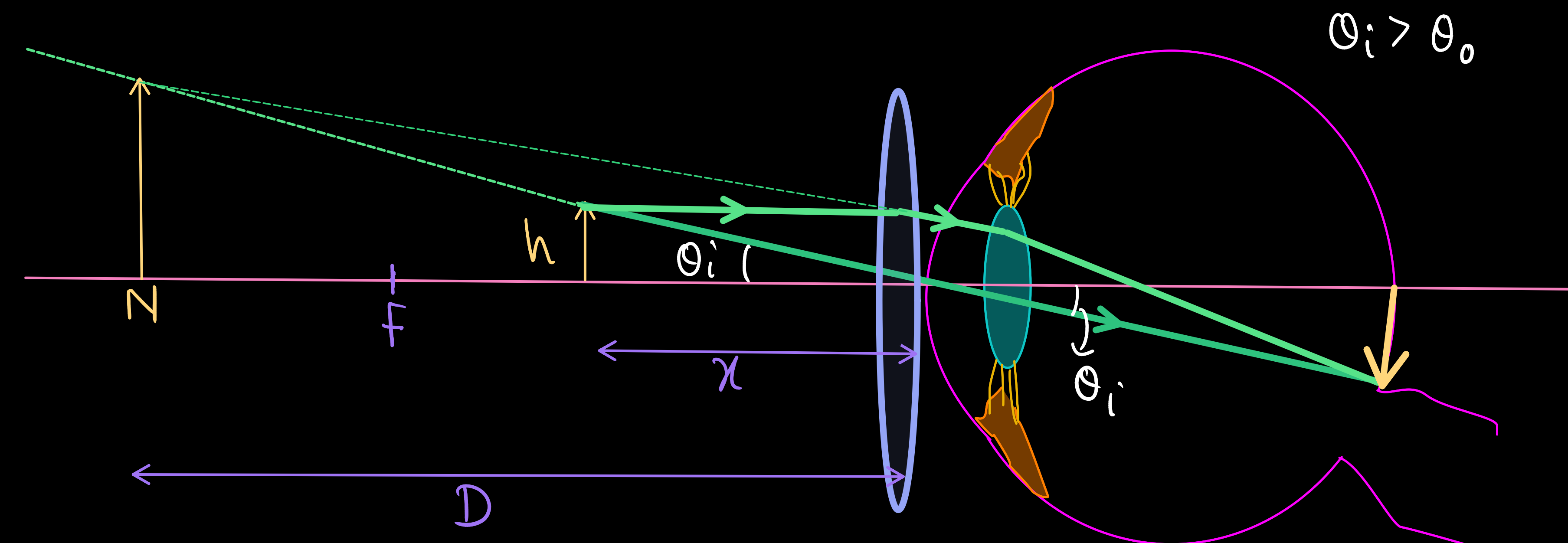
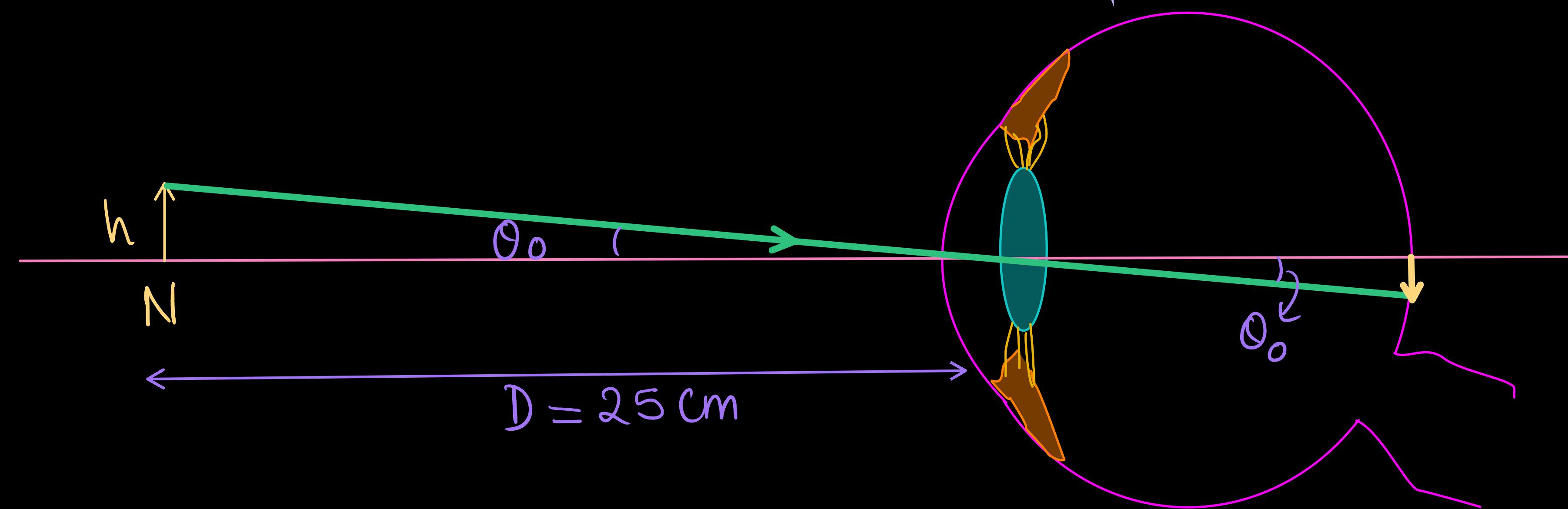
Simple Microscope

for naked eye -
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Simple Microscope

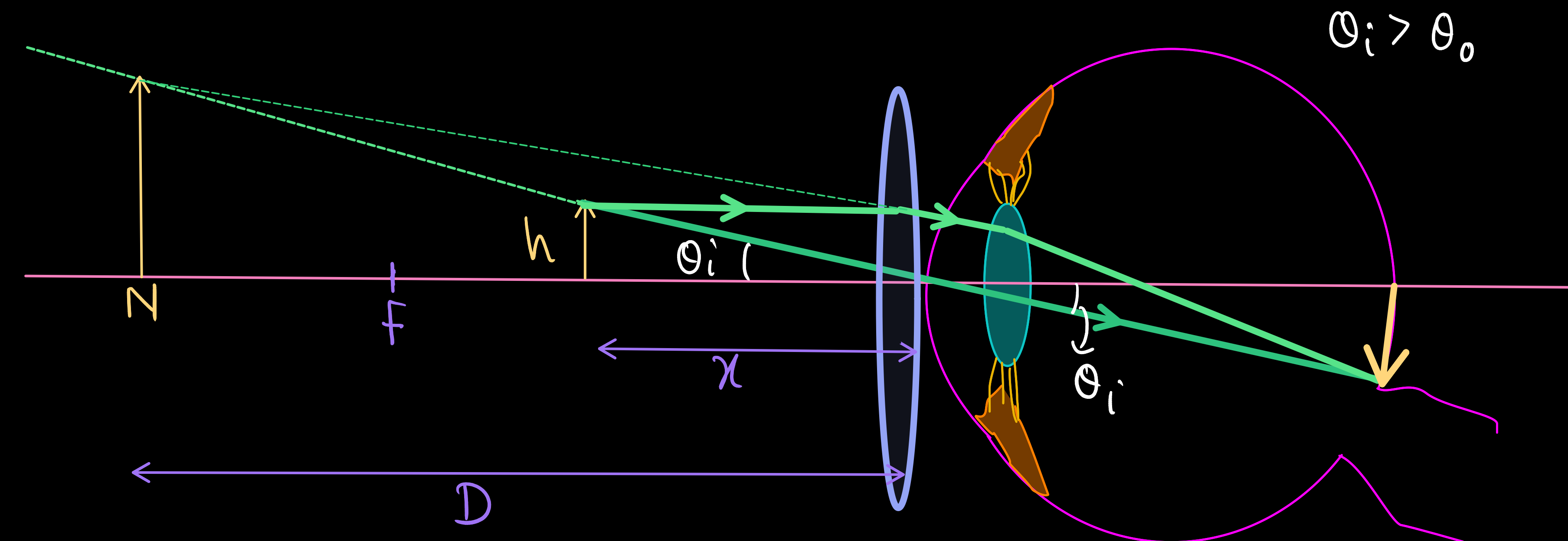
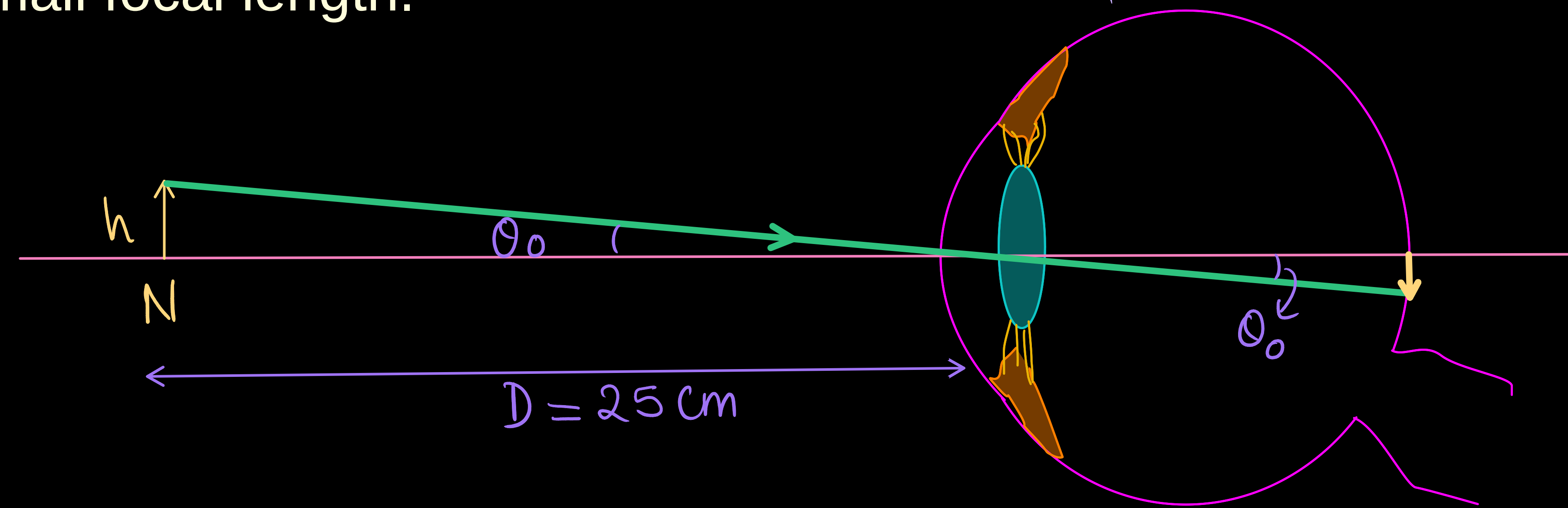
for naked eye -
The angular size (θ_o) of image formed at retina is maximum when object is at near point of the eye.



Simple Microscope

* A simple magnifier or microscope is a converging lens of small focal length.

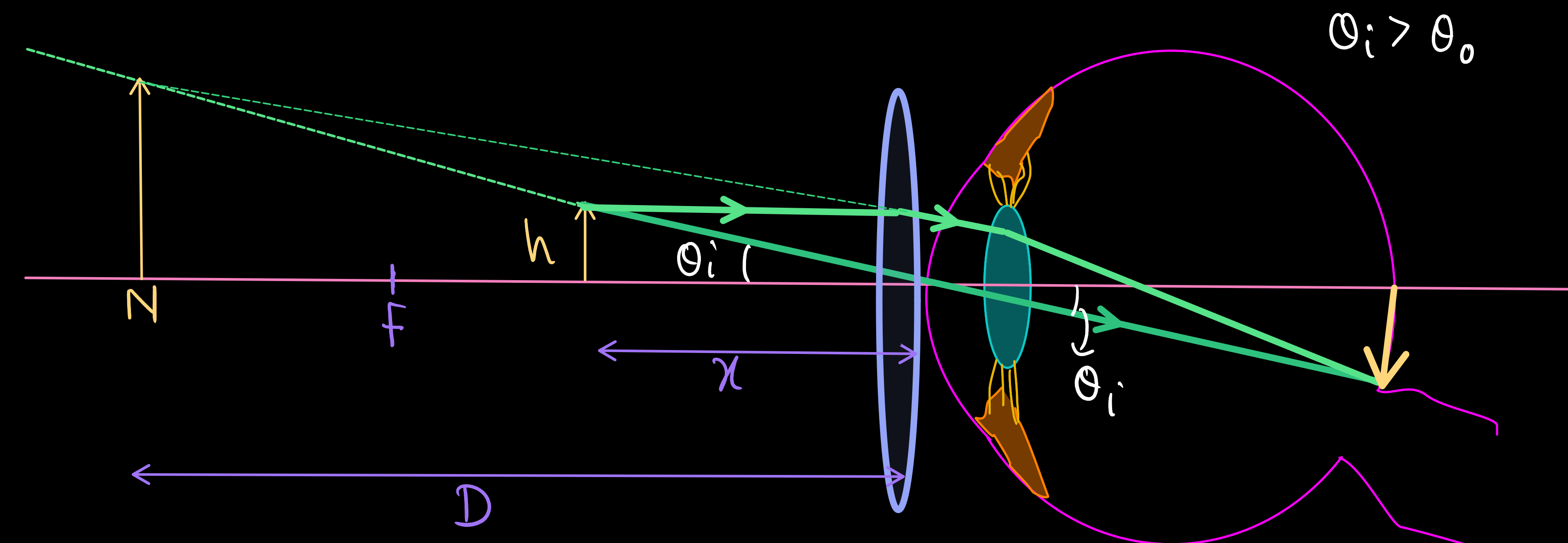
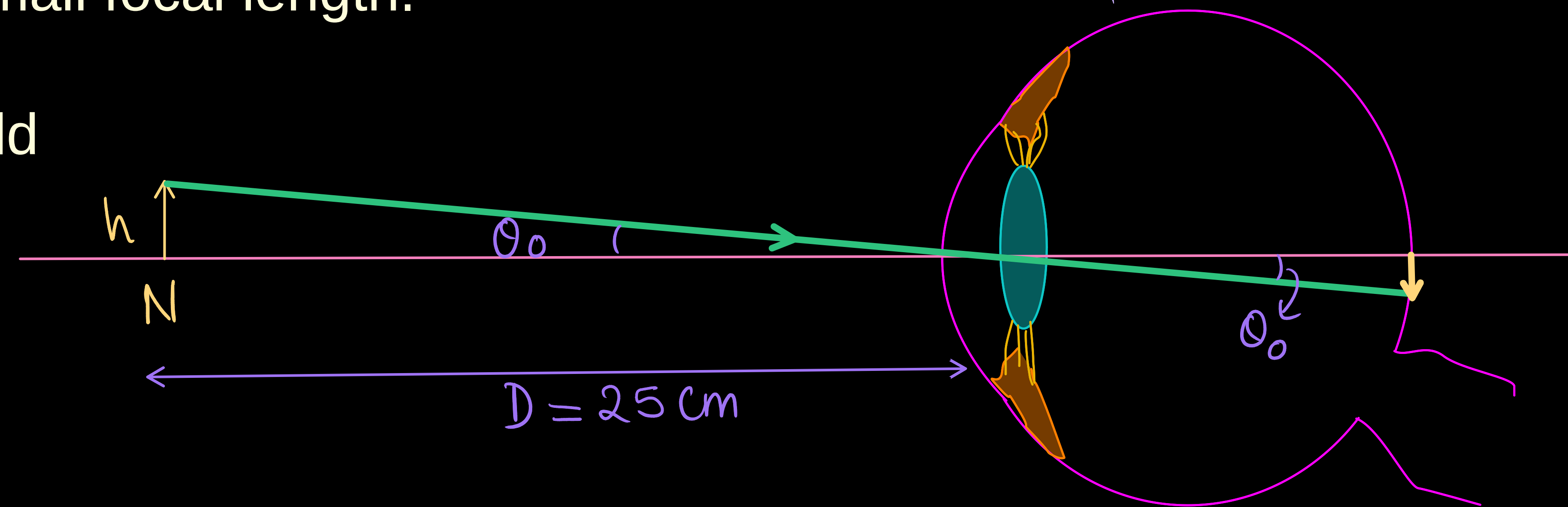
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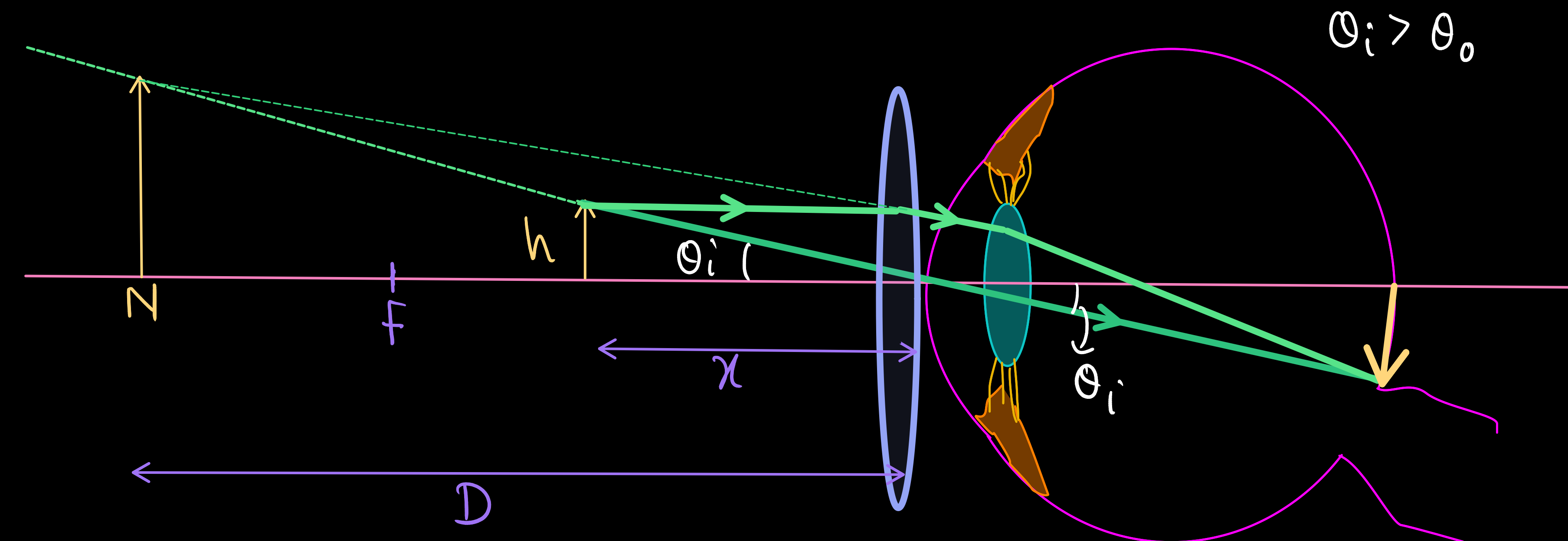
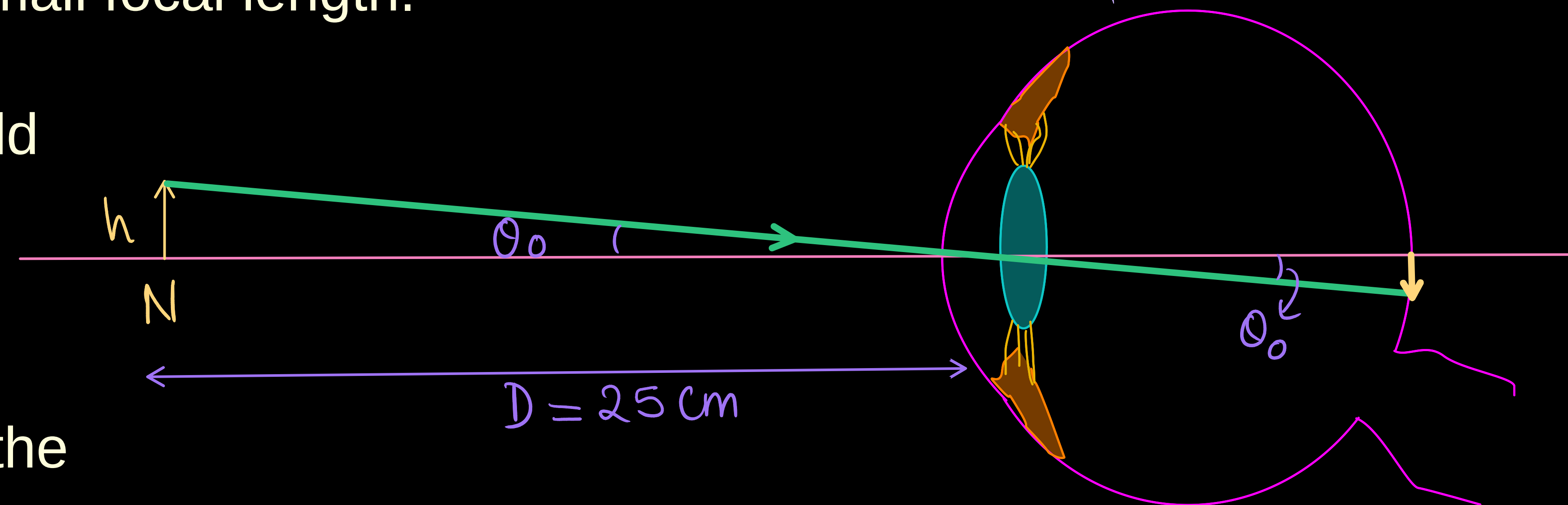
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- * The idea is to get an erect, magnified and virtual image of the object at a distance so that it can be viewed comfortably, i.e., at 25 cm or more.

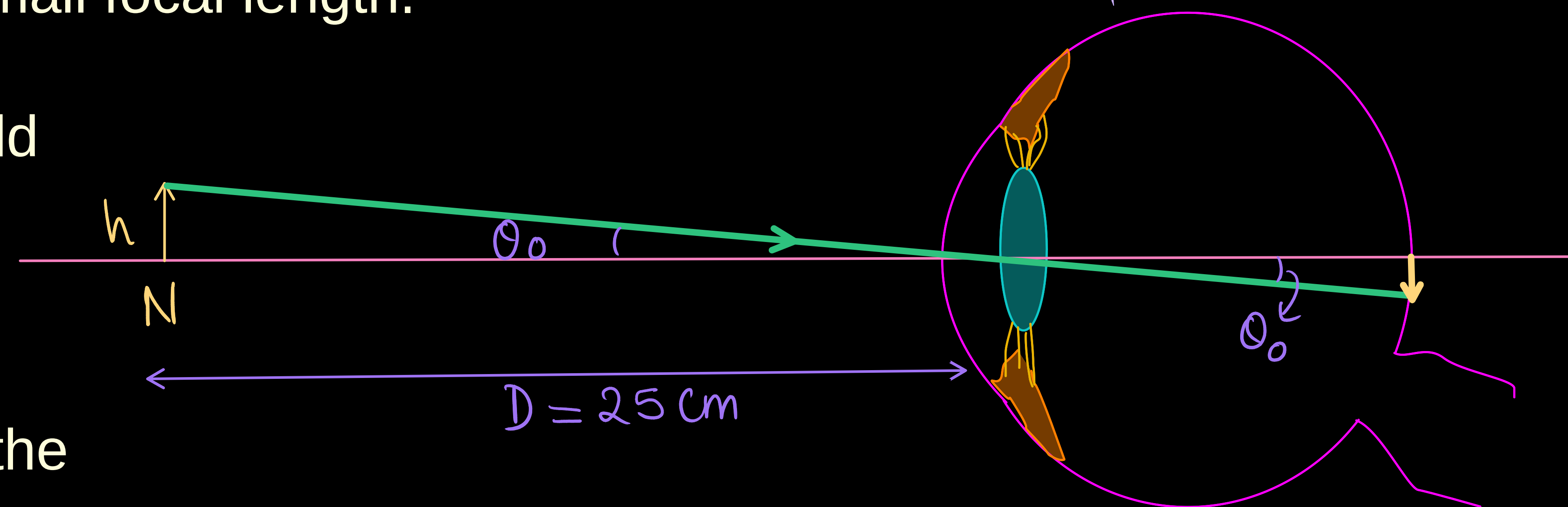
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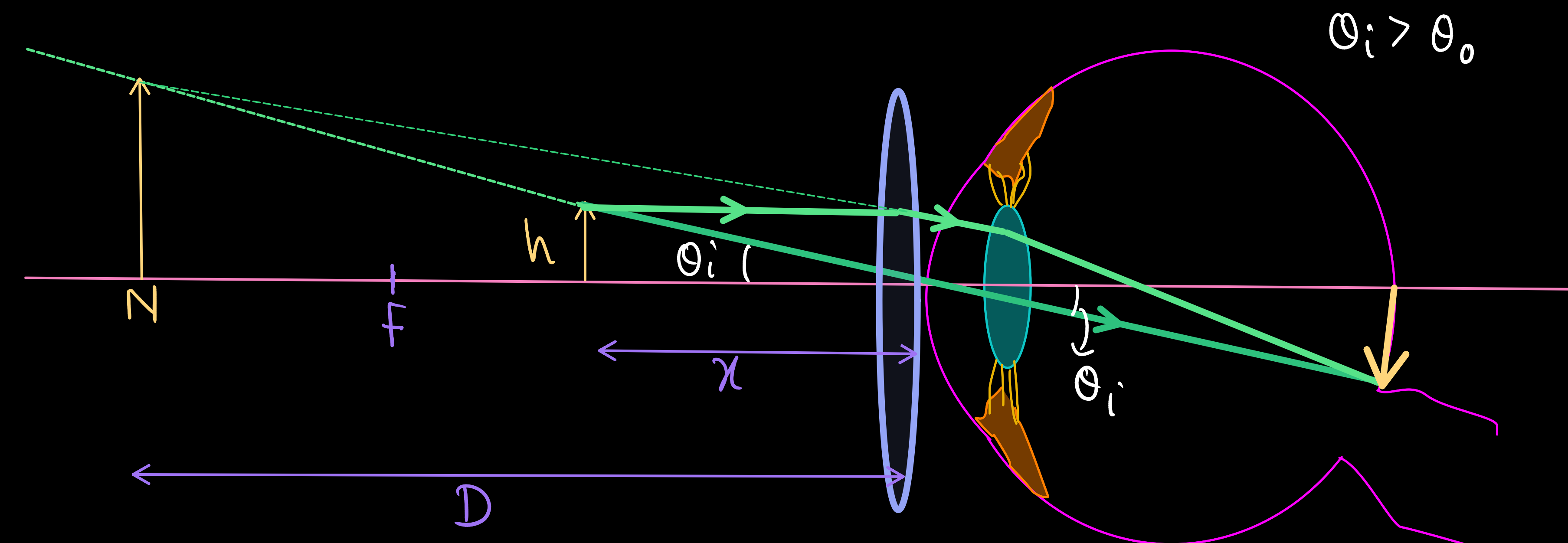
for naked eye -
The angular size (θ_0) of image formed at retina is maximum when object is at near point of the eye.



Case I -> When image is formed at Near point ($D = 25 \text{ cm}$)

The max. angle subtended by the object at eye and can be clearly visible (without lens)

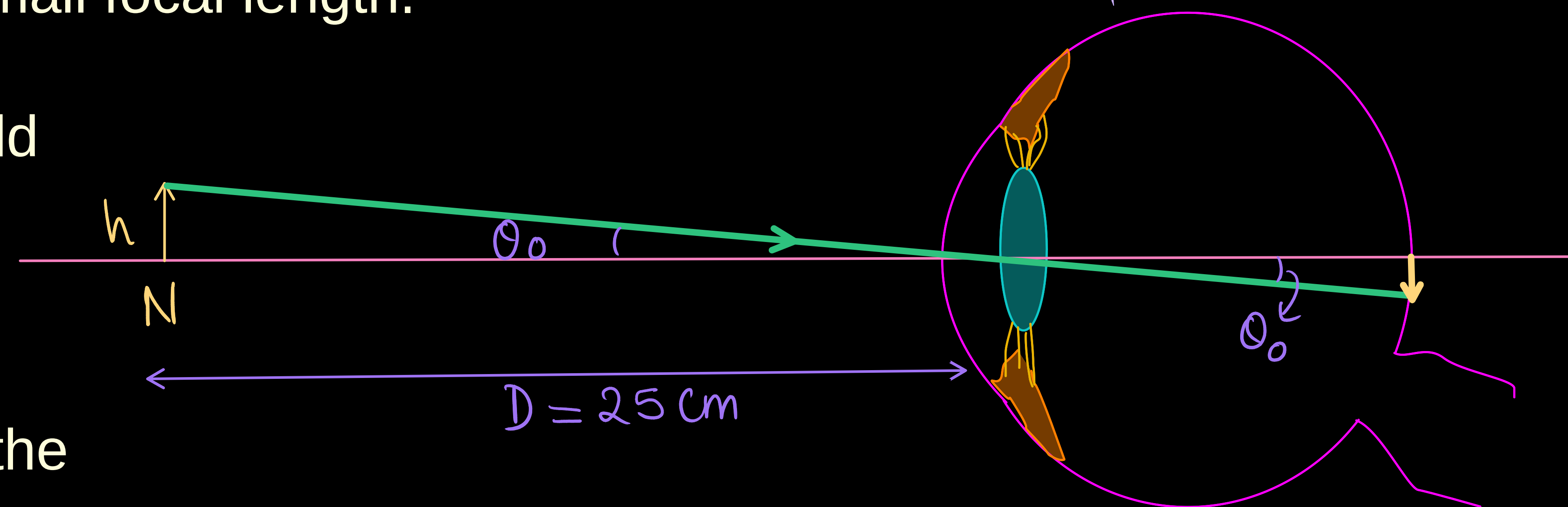
$$\theta_0 = \frac{h}{D} - \textcircled{1}$$



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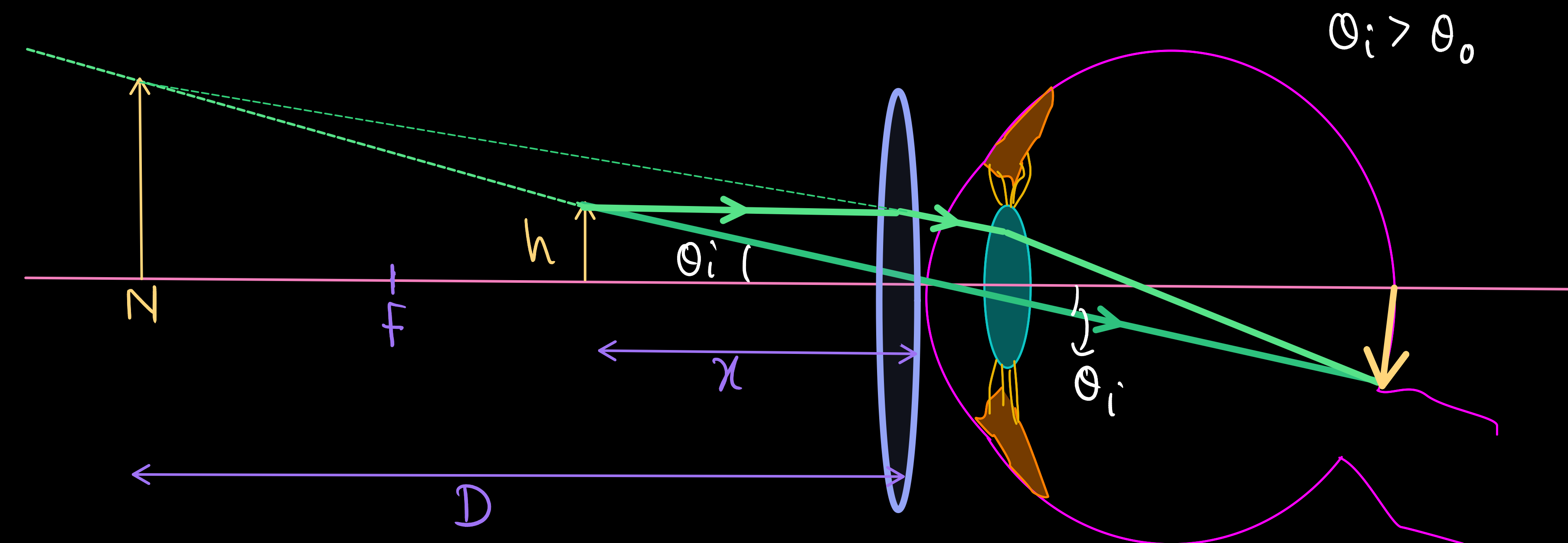
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$$\theta_0 = \frac{h}{D} \quad \text{--- (1)}$$

Now, the angle subtended by image at eye.

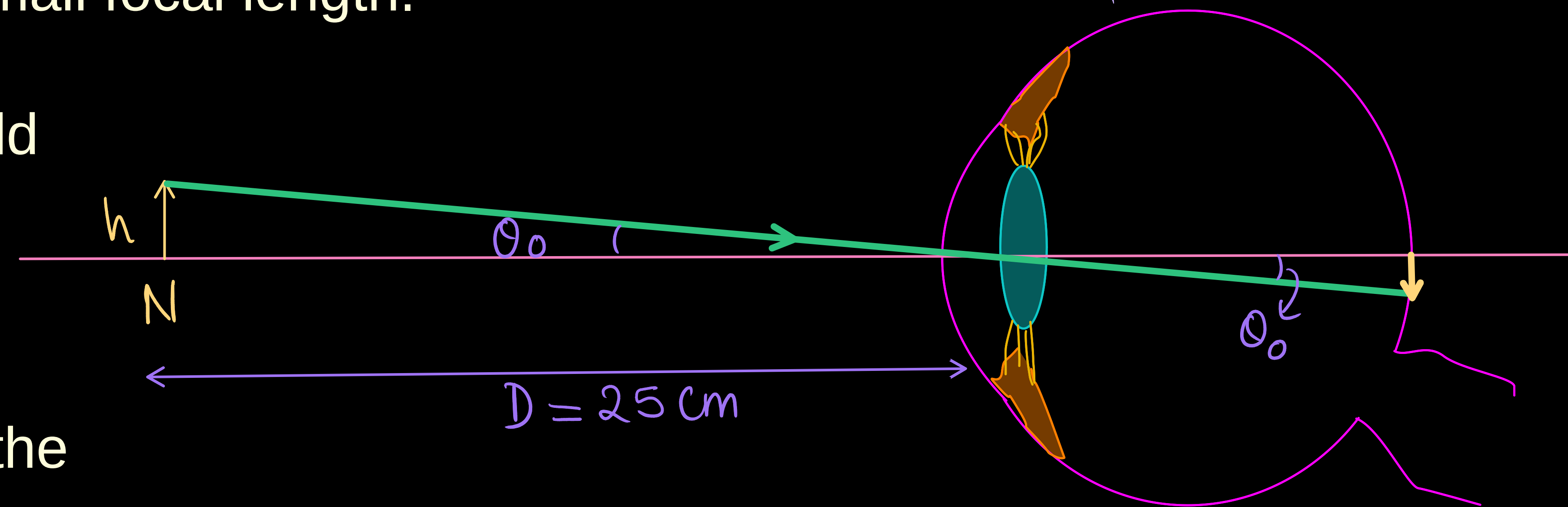
$$\theta_i = \frac{h}{x} \quad \text{--- (2)}$$



Simple Microscope

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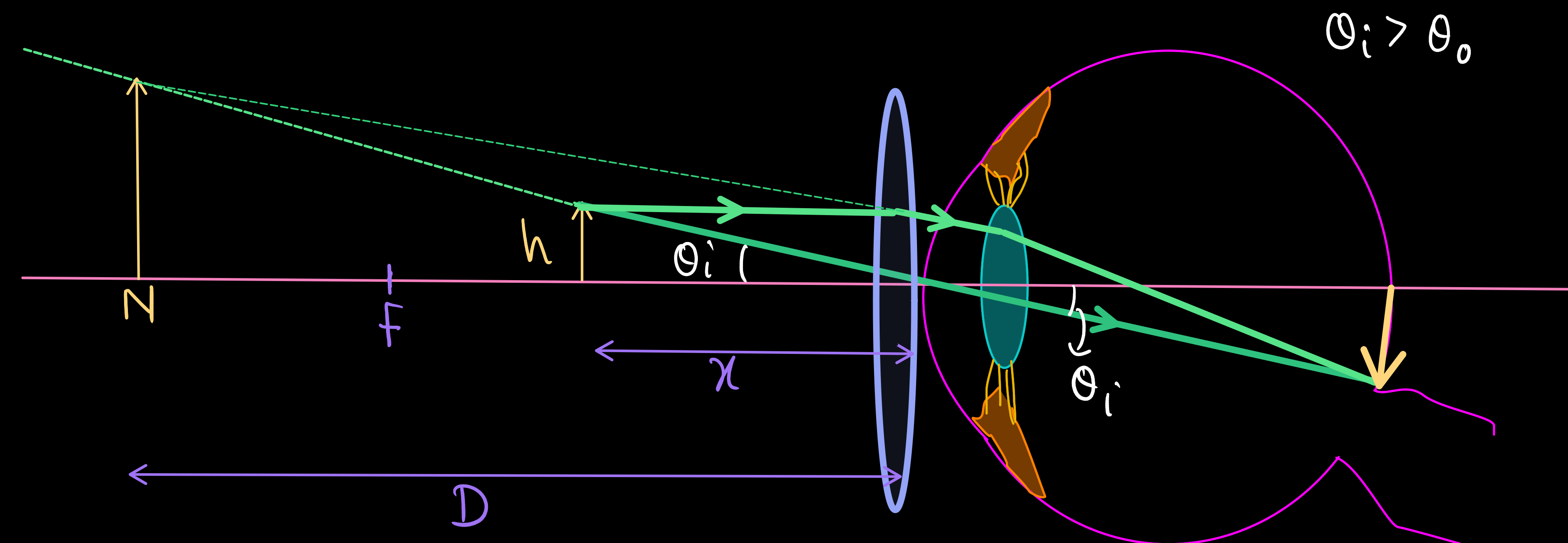
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Now, the angle subtended by image at eye.

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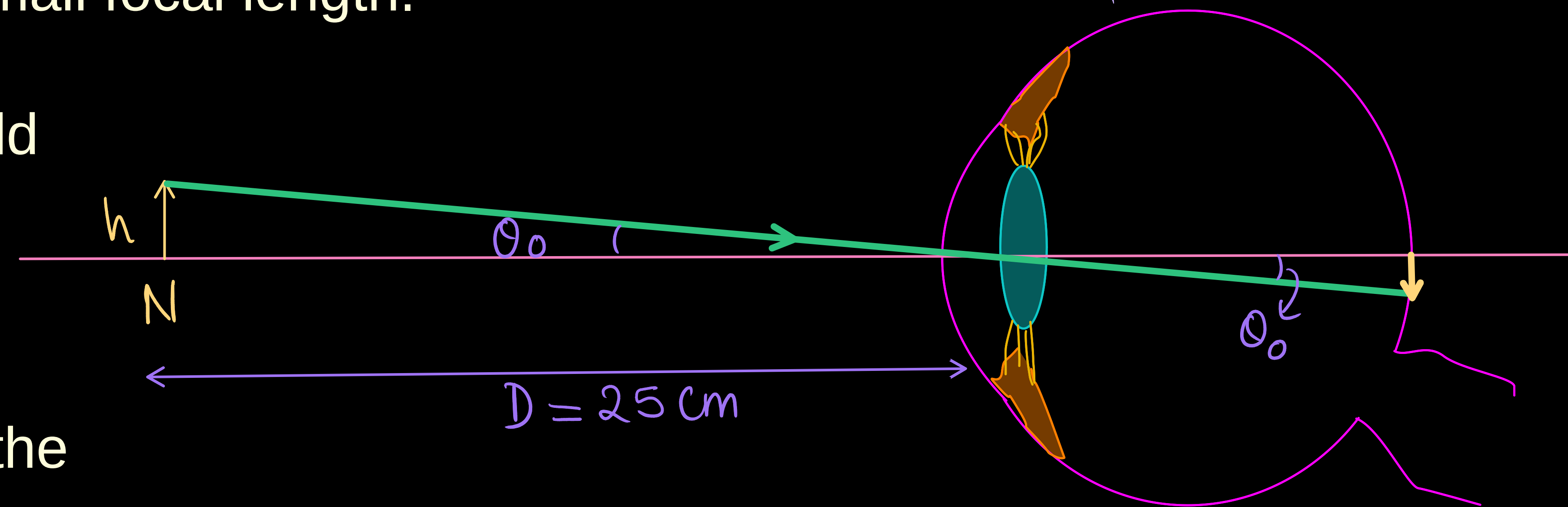
using lens formula $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$



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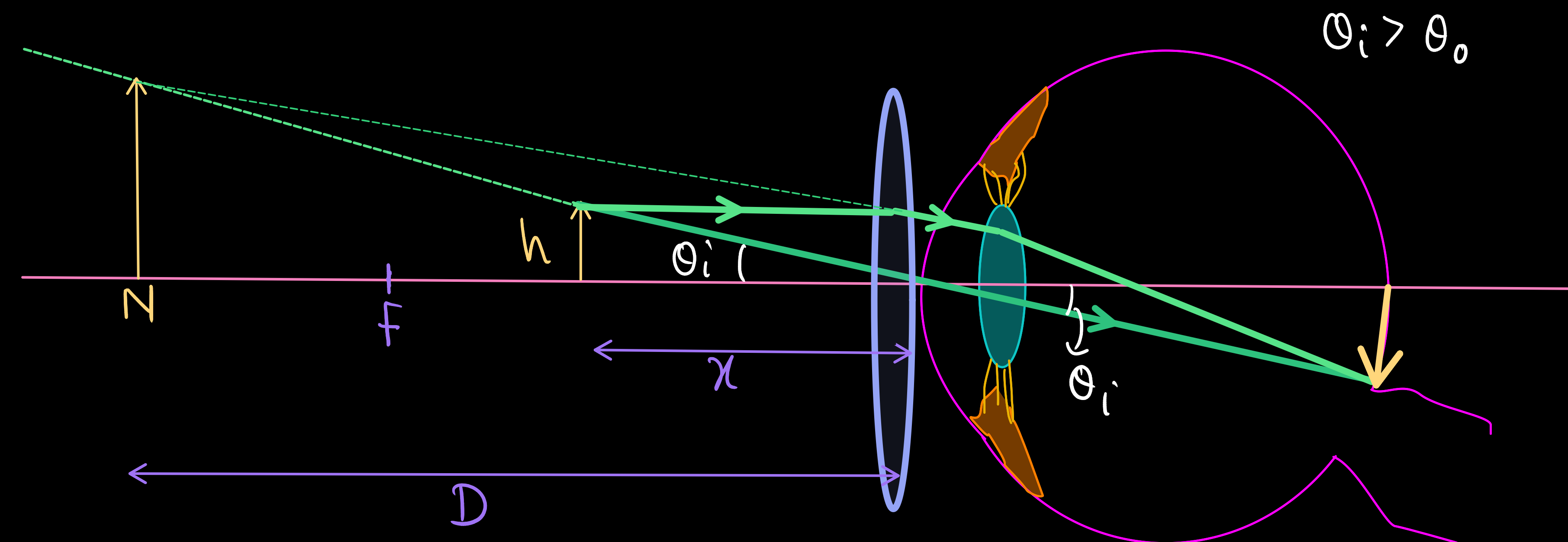
$$\theta_i = \frac{h}{x} \quad \text{--- (2)}$$

using lens formula $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$$v = -D, u = -x$$

$$\frac{1}{f} = \frac{1}{-D} - \frac{1}{-x}$$

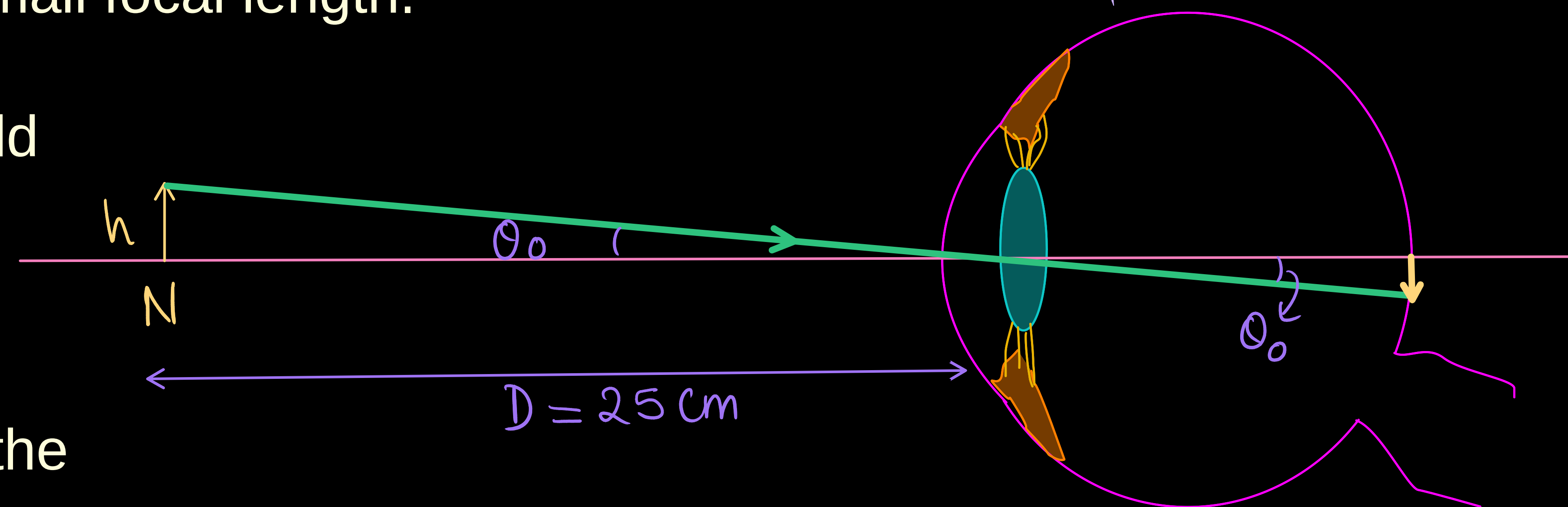
$$\boxed{\frac{1}{f} + \frac{1}{D} = \frac{1}{x}} \quad \text{--- (3)}$$



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The max. angle subtended by the object at eye and can be clearly visible (without lens)

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Now, the angle subtended by image at eye.

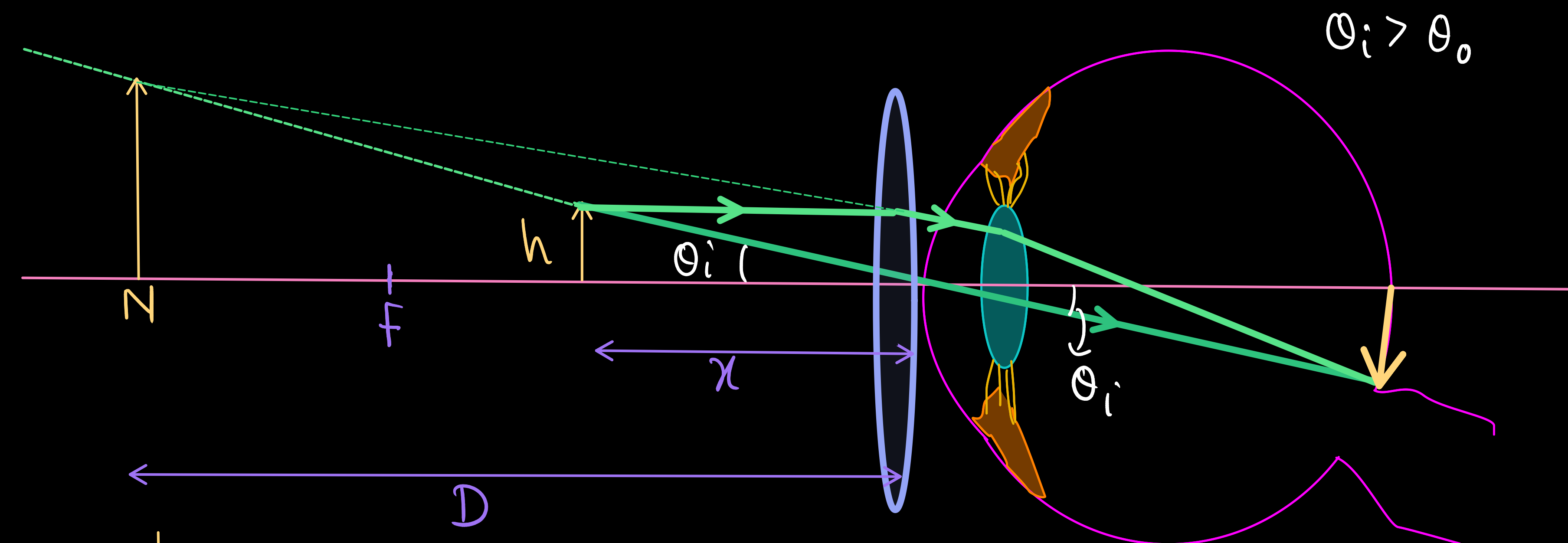
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$$\boxed{\frac{1}{f} + \frac{1}{D} = \frac{1}{x}} \quad \text{--- (3)}$$



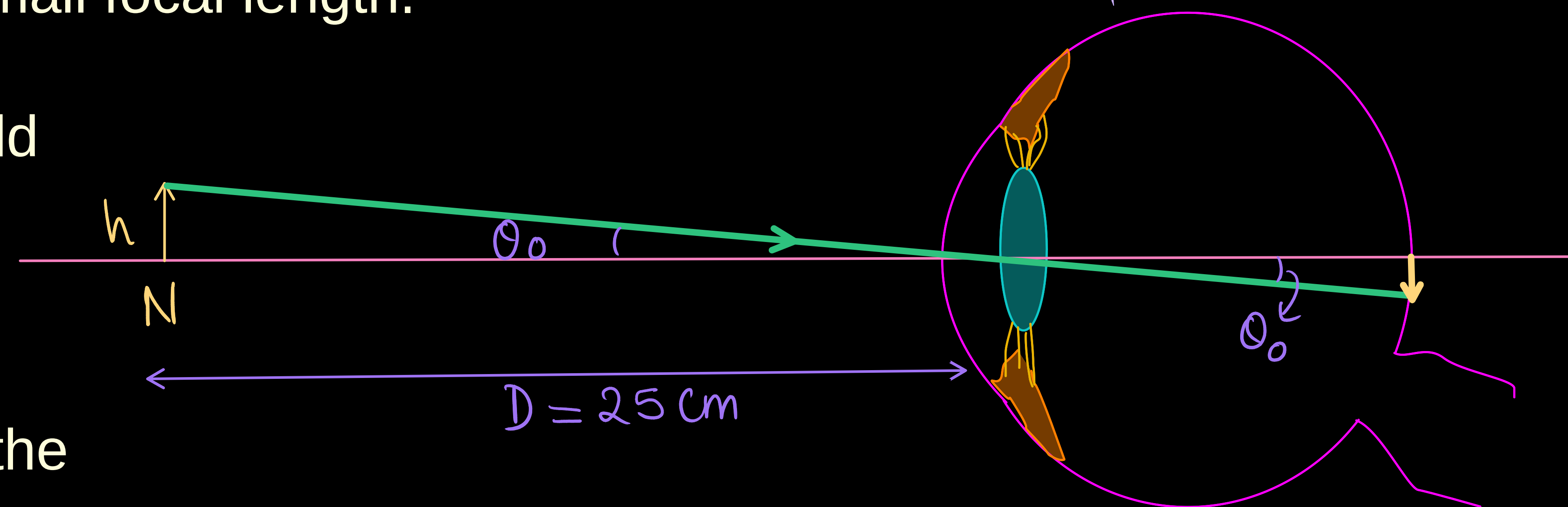
from eq (2) & (3)

$$\boxed{\theta_i = h \left(\frac{1}{f} + \frac{1}{D} \right)} \quad \text{--- (4)}$$

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Now, the angle subtended by image at eye.

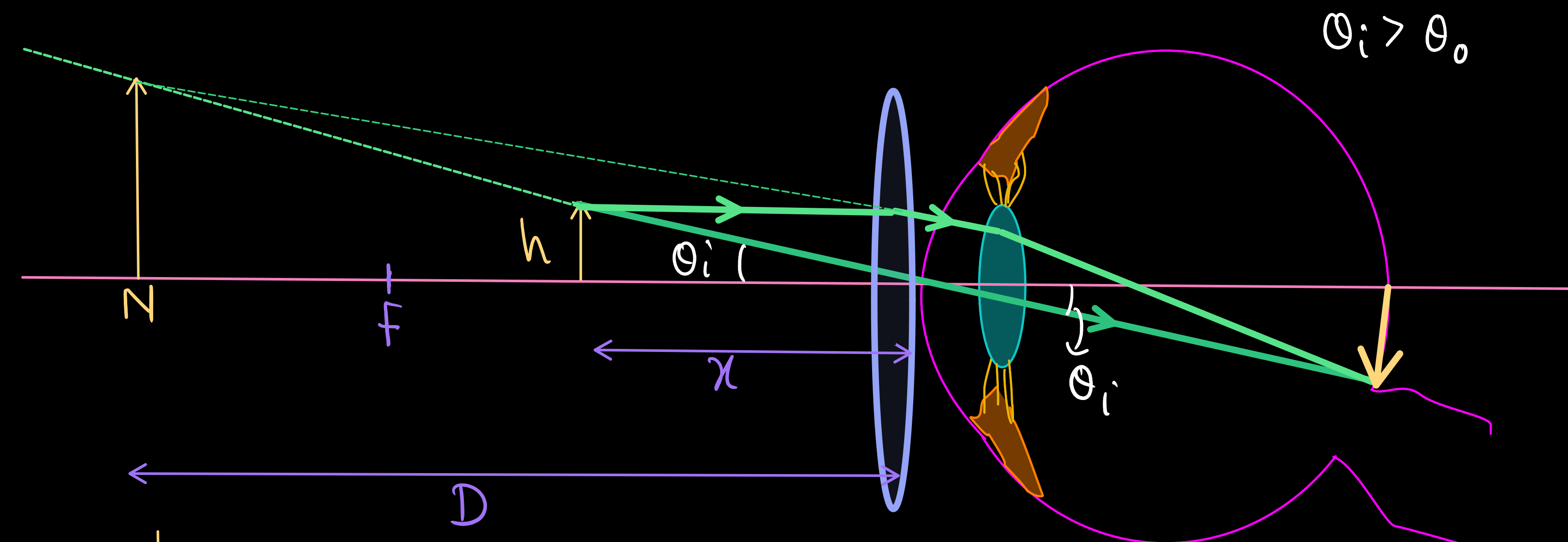
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Simple Microscope

Case I → When image is formed at Near point ($D = 25\text{ cm}$)

The max. angle subtended by the object at eye and can be clearly visible (without lens)

$$\theta_o = \frac{h}{D} \quad \text{--- (1)}$$

Now, the angle subtended by image at eye

$$\theta_i = \frac{h}{x} \quad \text{--- (2)}$$

using lens formula $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

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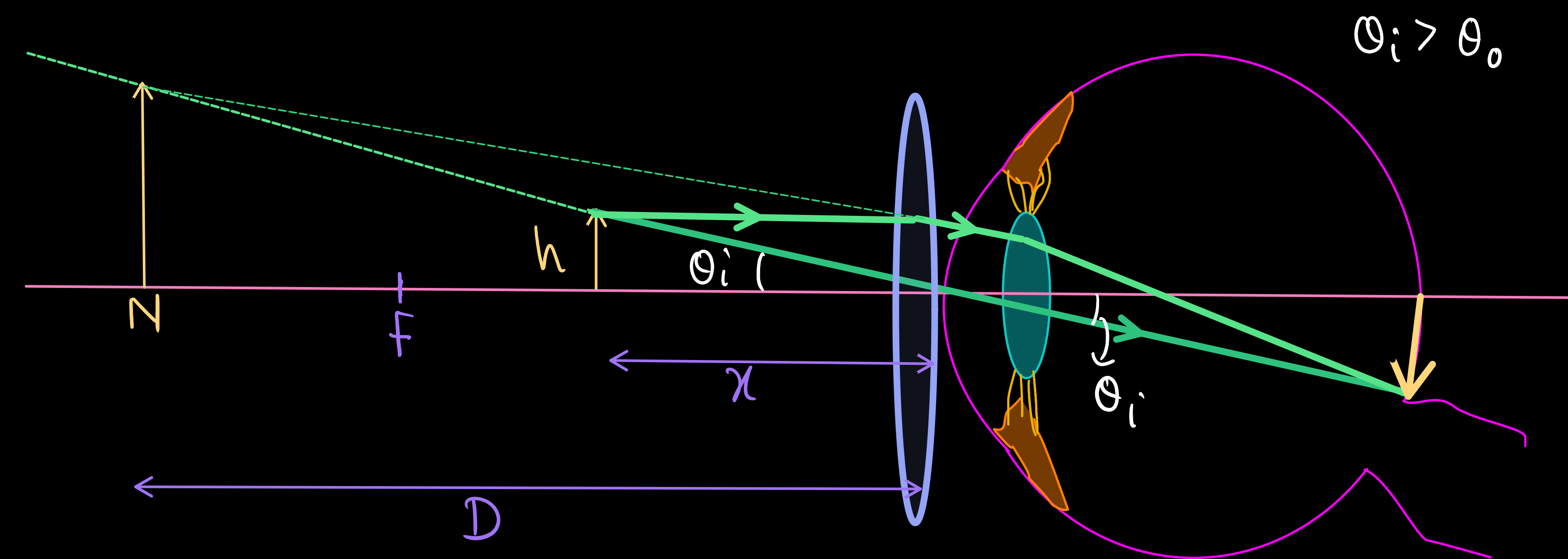
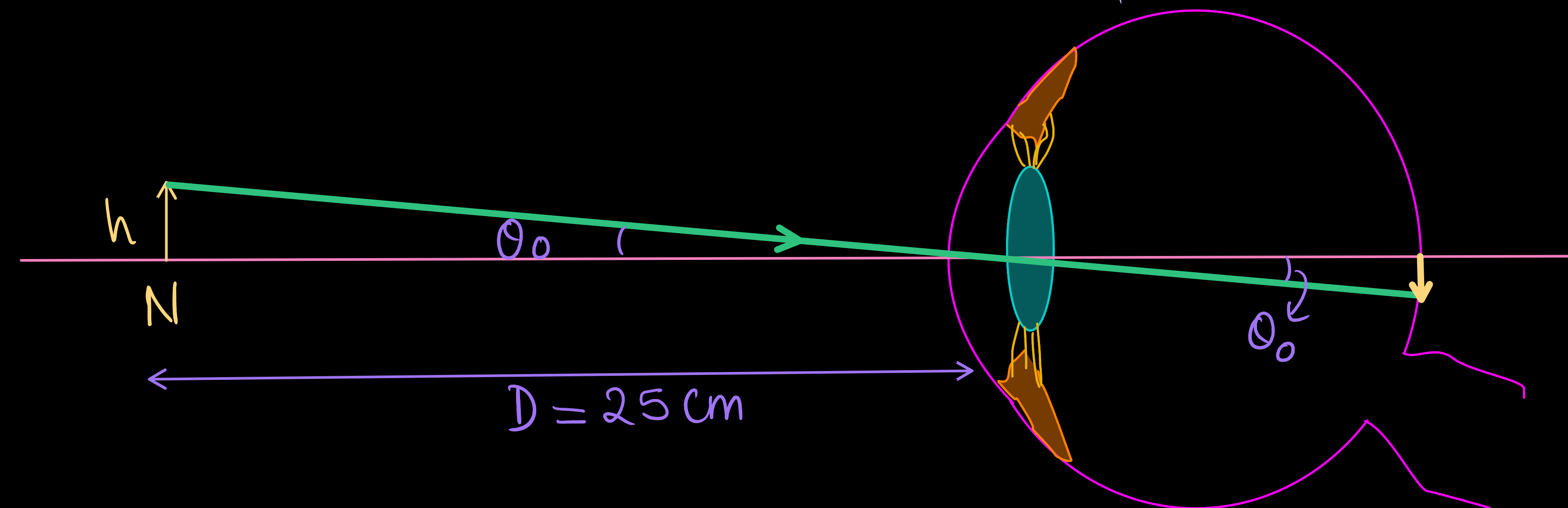
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$$v = -D, u = -x$$

$$\frac{1}{f} = \frac{1}{-D} - \frac{1}{-x}$$

$$\boxed{\frac{1}{f} + \frac{1}{D} = \frac{1}{x}} \quad \text{--- (3)}$$

from eq (2) & (3)

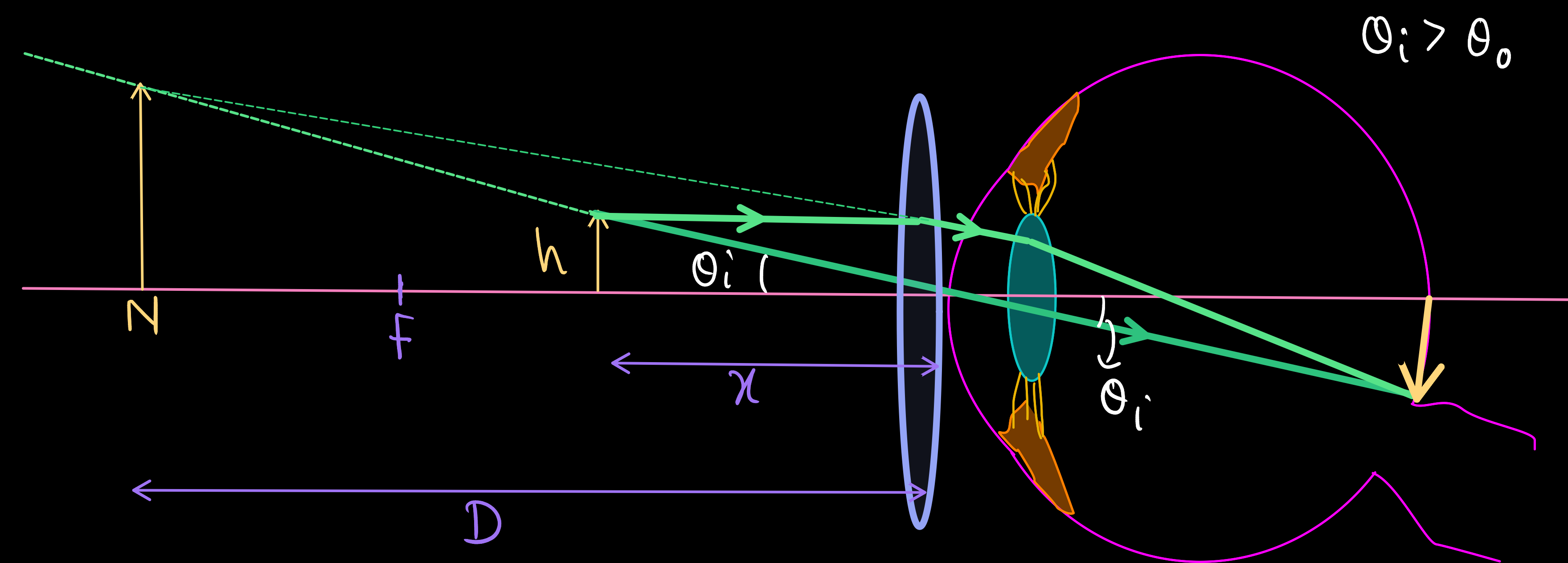
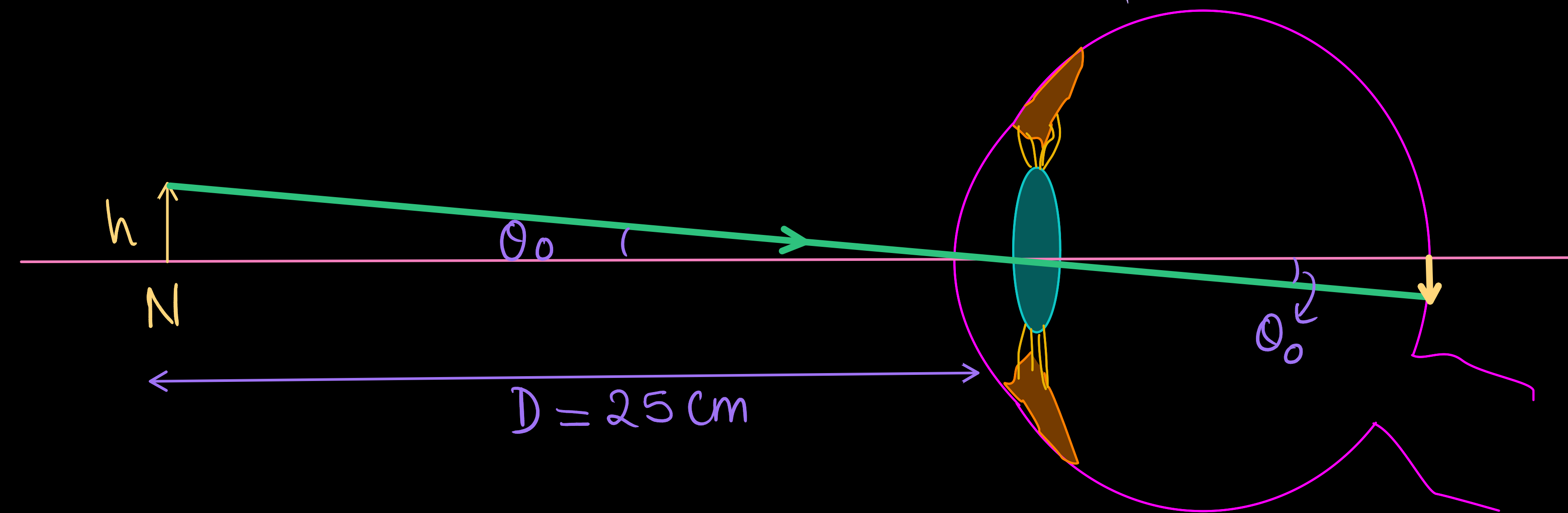
$$\boxed{\theta_i = h \left(\frac{1}{f} + \frac{1}{D} \right)} \quad \text{--- (4)}$$

Now, Angular Magnification

$$m = \frac{\theta_i}{\theta_0}$$

for naked eye -

The angular size (θ_0) of image formed at retina is maximum when object is at near point of the eye.



Simple Microscope

Case I → When image is formed at Near point ($D = 25\text{ cm}$)

The max. angle subtended by the object at eye and can be clearly visible (without lens)

$$\theta_0 = \frac{h}{D} \quad \text{--- (1)}$$

Now, the angle subtended by image at eye

$$\theta_i = \frac{h}{x} \quad \text{--- (2)}$$

using lens formula $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$$v = -D, u = -x$$

$$\frac{1}{f} = \frac{1}{-D} - \frac{1}{-x}$$

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from eq (2) & (3)

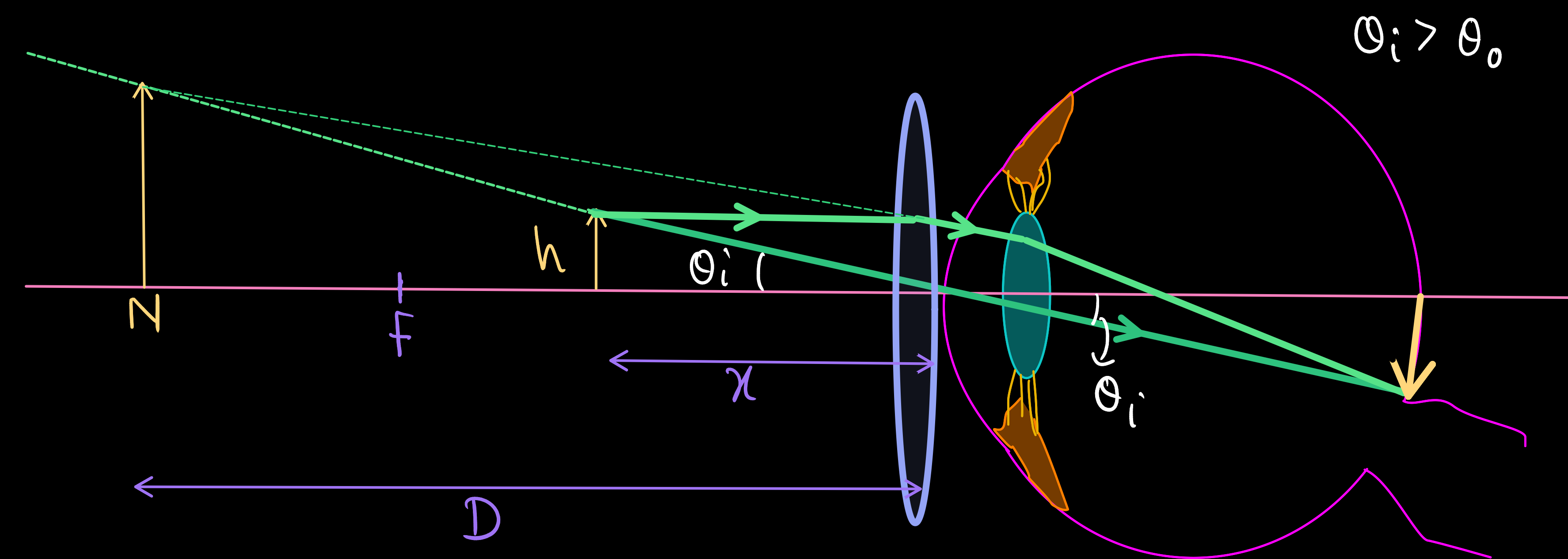
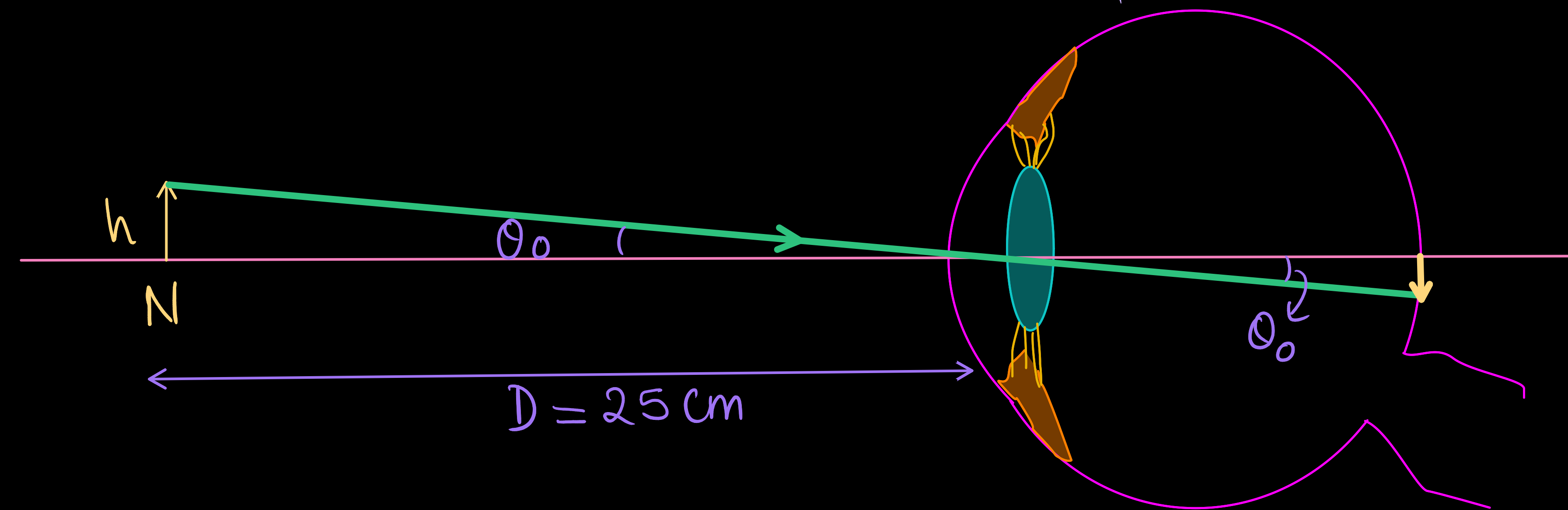
$$\boxed{\theta_i = h \left(\frac{1}{f} + \frac{1}{D} \right)} \quad \text{--- (4)}$$

Now, Angular Magnification

$$m = \frac{\theta_i}{\theta_0} = \frac{h \left(\frac{1}{f} + \frac{1}{D} \right)}{\frac{h}{D}}$$

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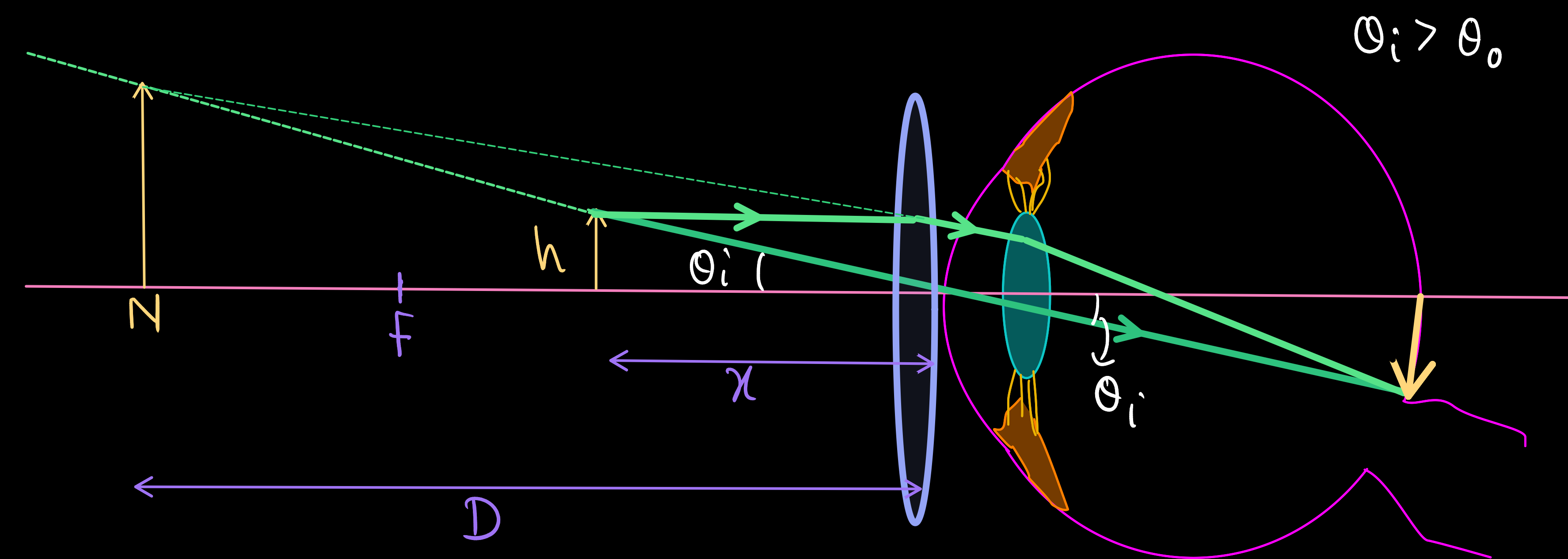
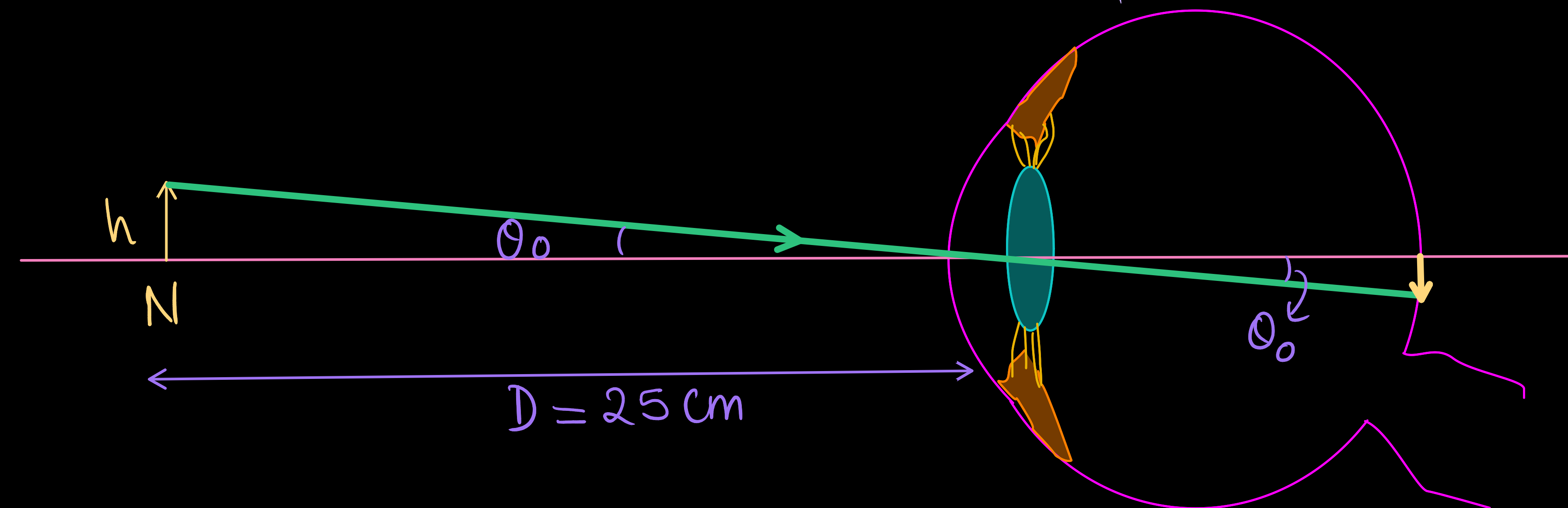
$$m = \frac{\theta_i}{\theta_0} = \frac{h \left(\frac{1}{f} + \frac{1}{D} \right)}{\frac{h}{D}}$$

$$m = D \left(\frac{1}{f} + \frac{1}{D} \right)$$

$$\boxed{m = 1 + \frac{D}{f}} \quad \text{--- (5)}$$

for naked eye -

The angular size (θ_0) of image formed at retina is maximum when object is at near point of the eye.



Simple Microscope

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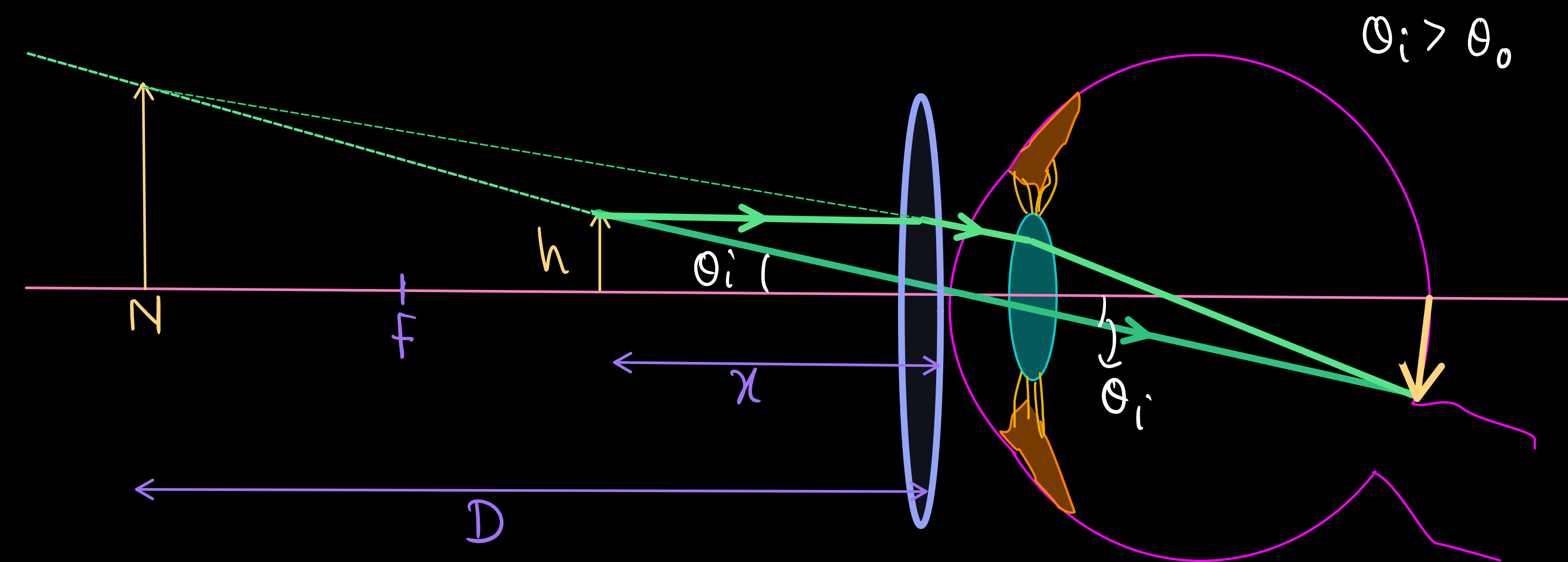
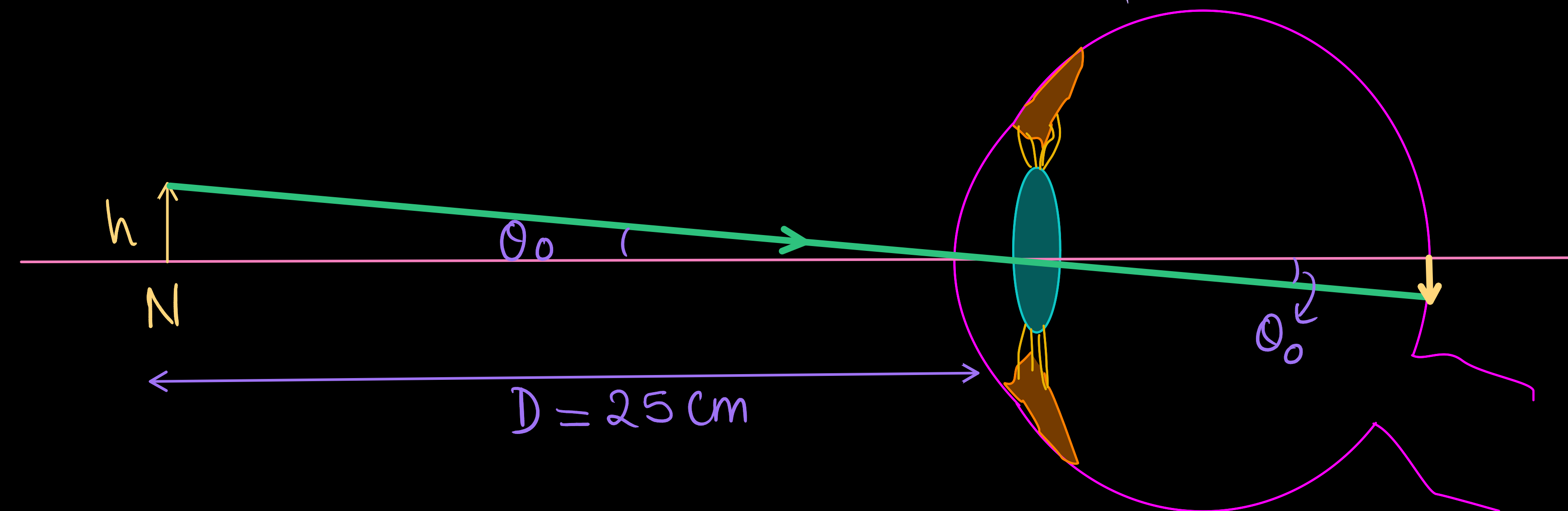
$$\boxed{m = 1 + \frac{D}{f}}$$

Magnification of simple microscope

when image is formed at Nearpoint (Least Distance of distinct vision)

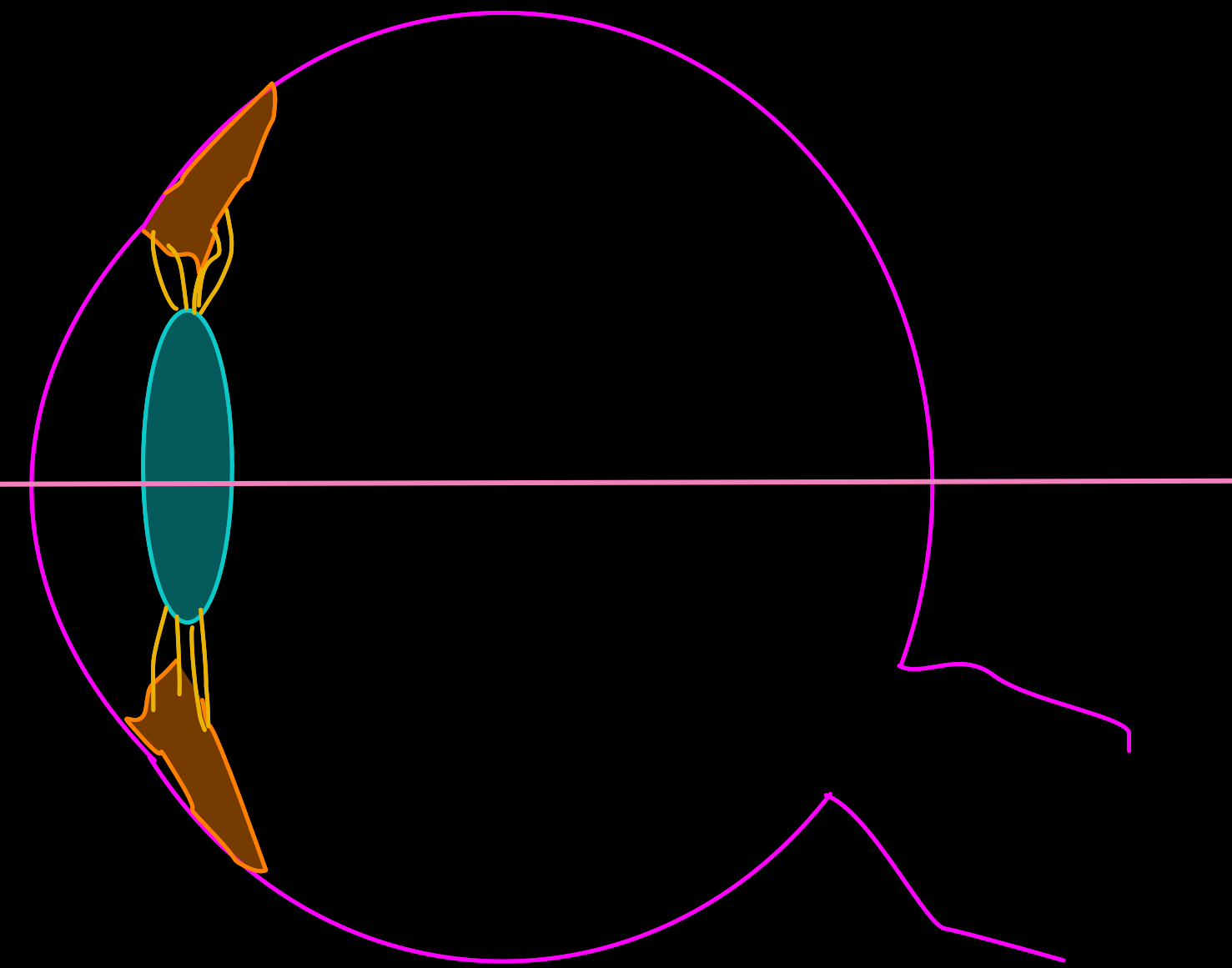
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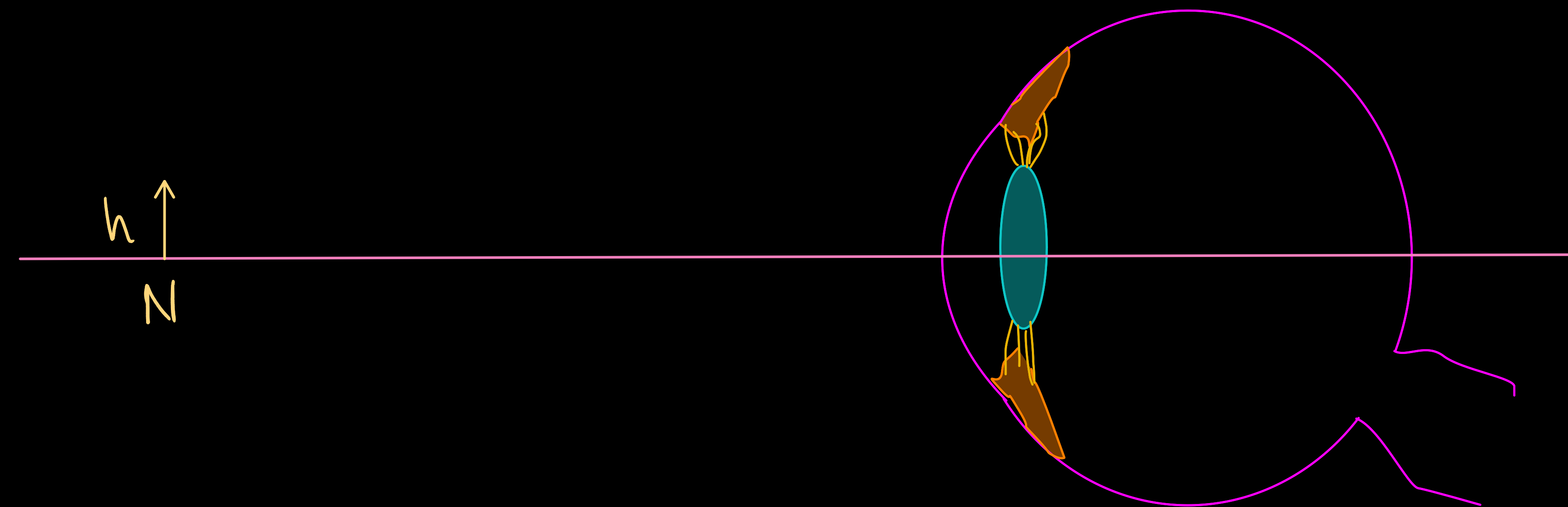


Simple Microscope

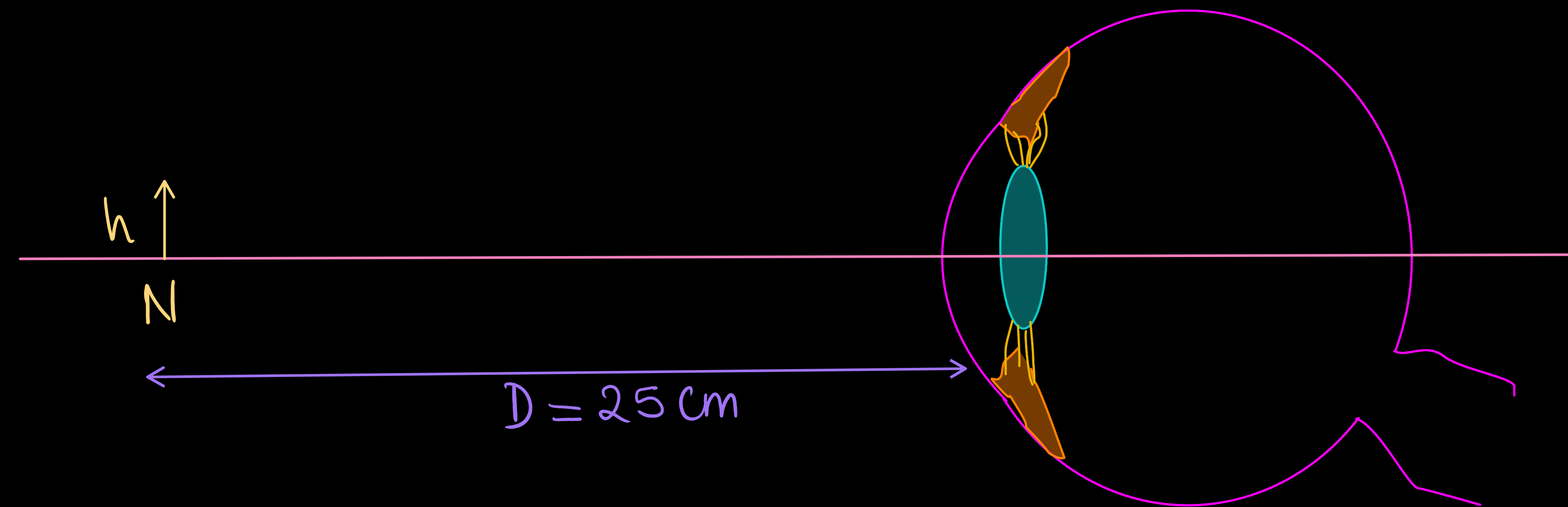
Simple Microscope



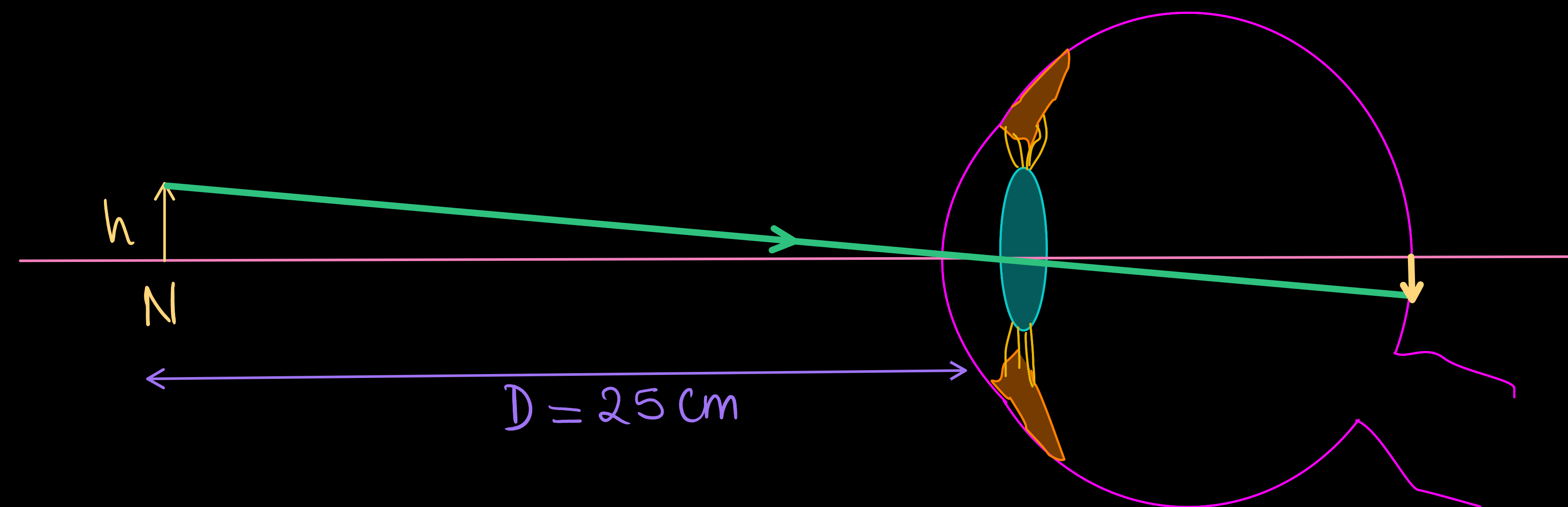
Simple Microscope



Simple Microscope

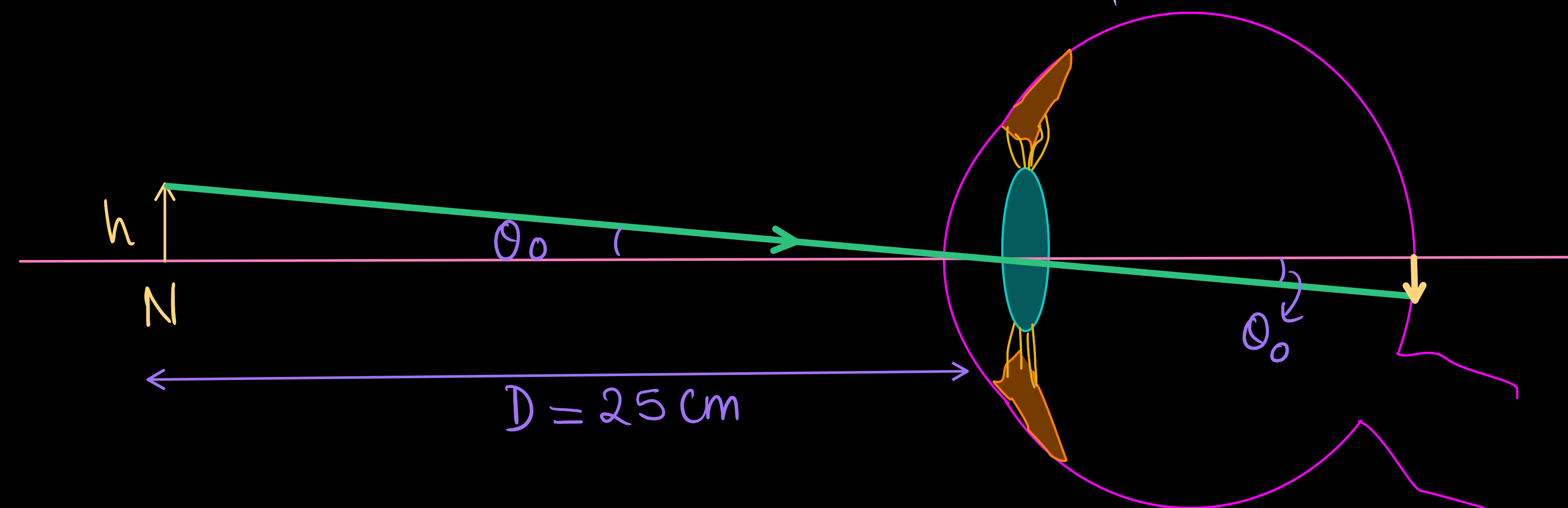


Simple Microscope



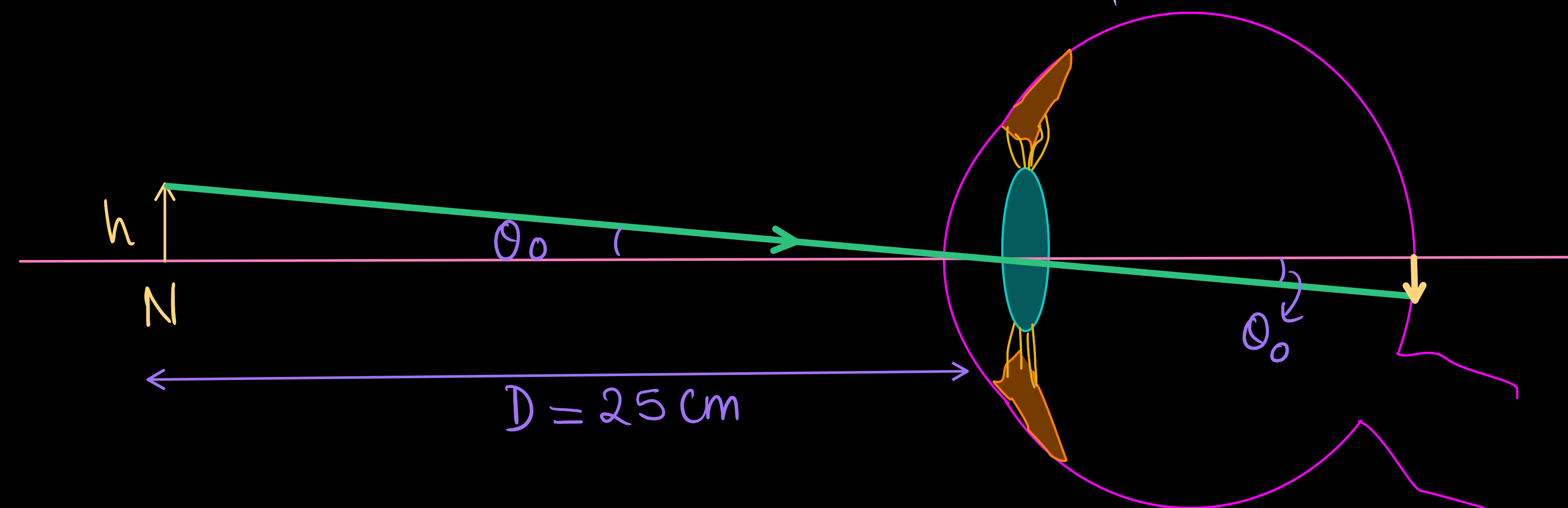
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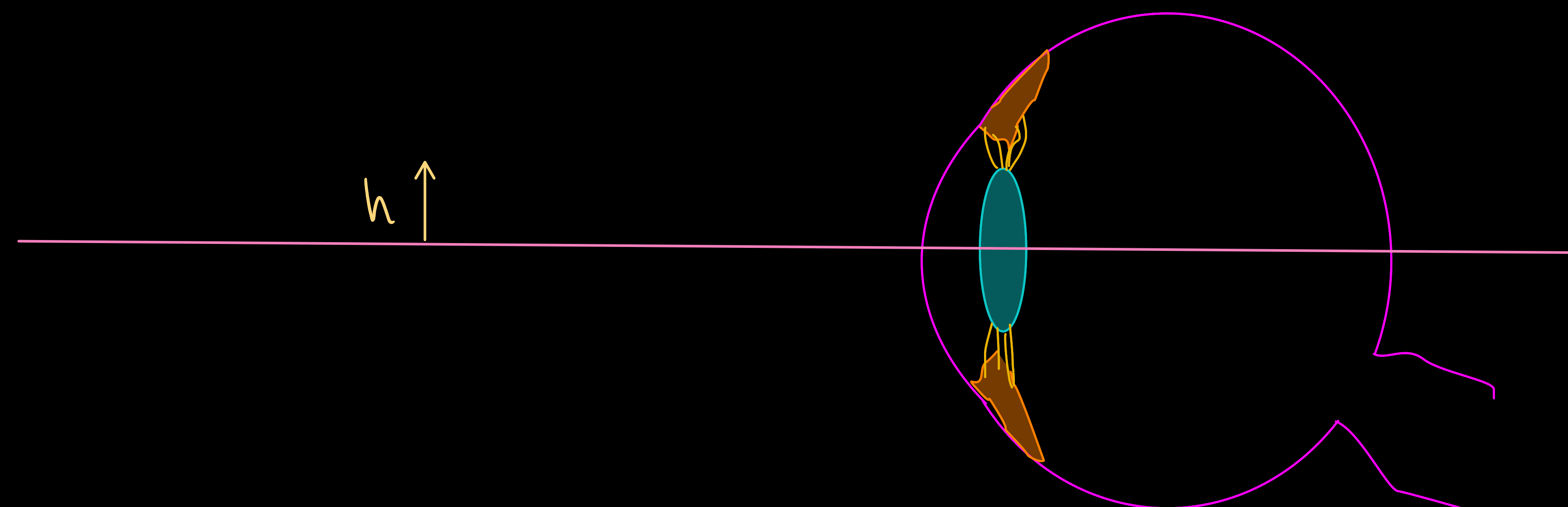
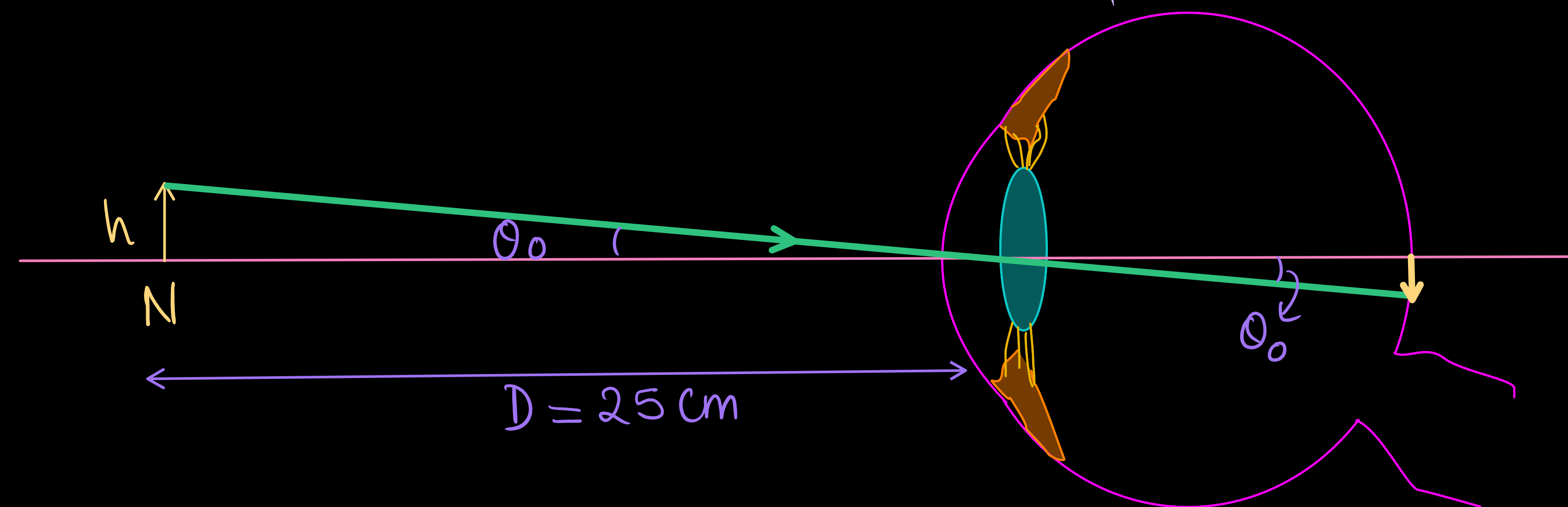
Simple Microscope

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Simple Microscope

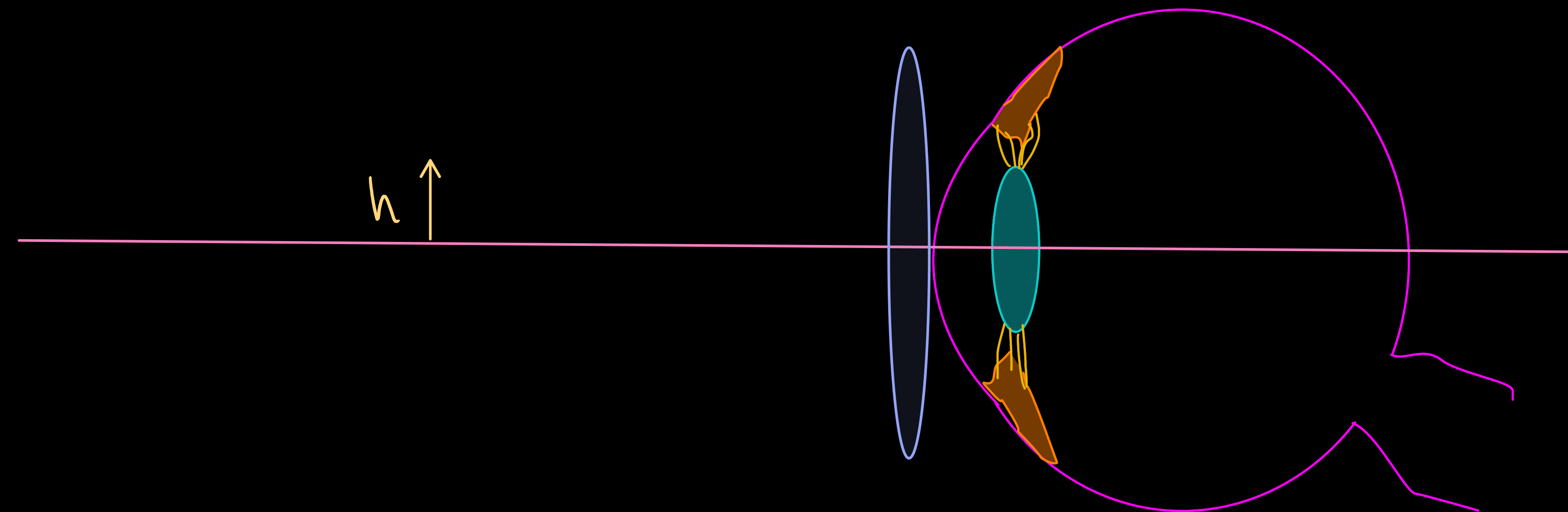
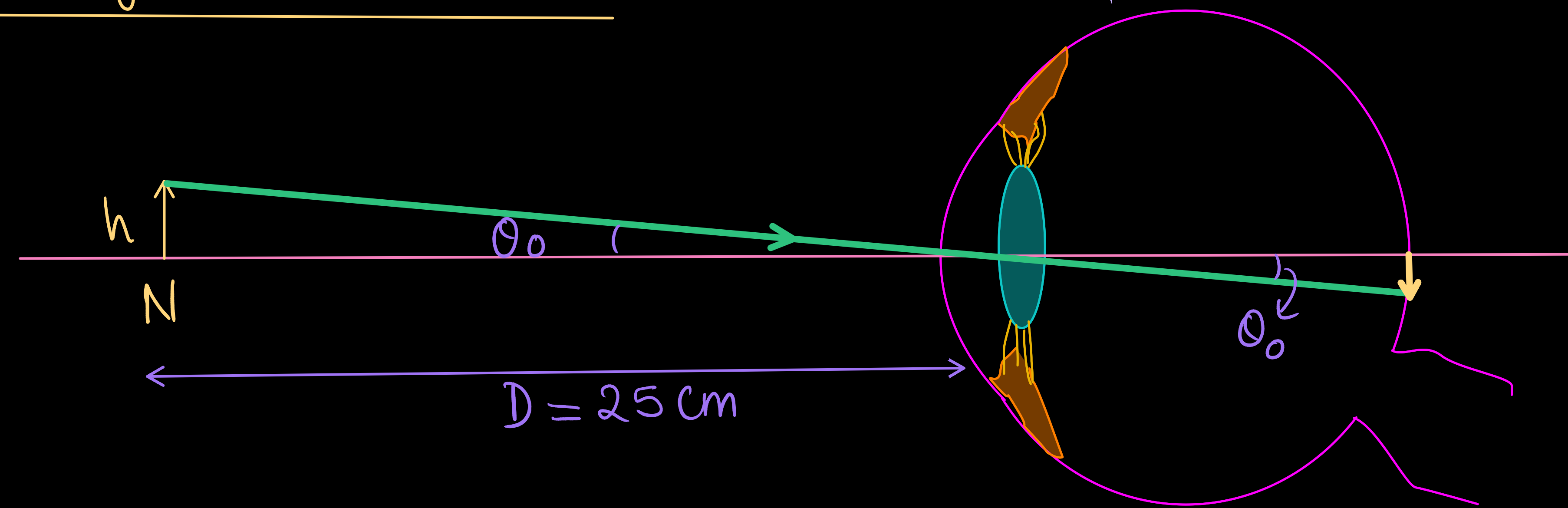
for naked eye -
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Simple Microscope

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Case II - When image is formed at infinity (eye is relaxed, viewing is more comfortable)

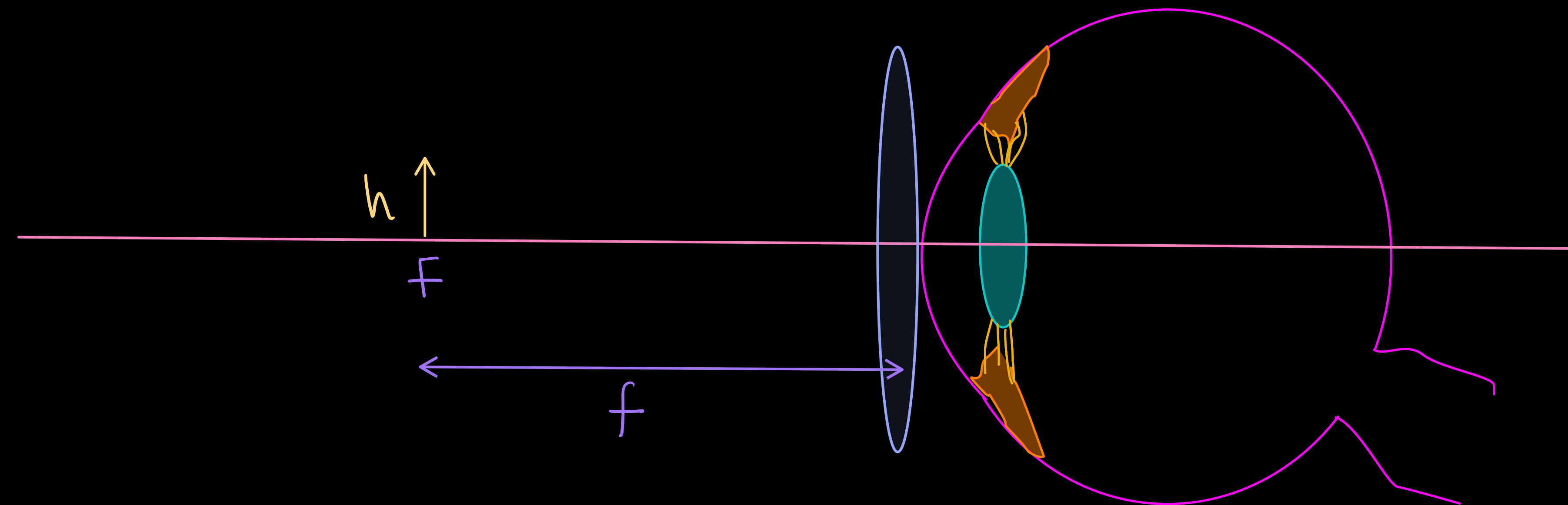
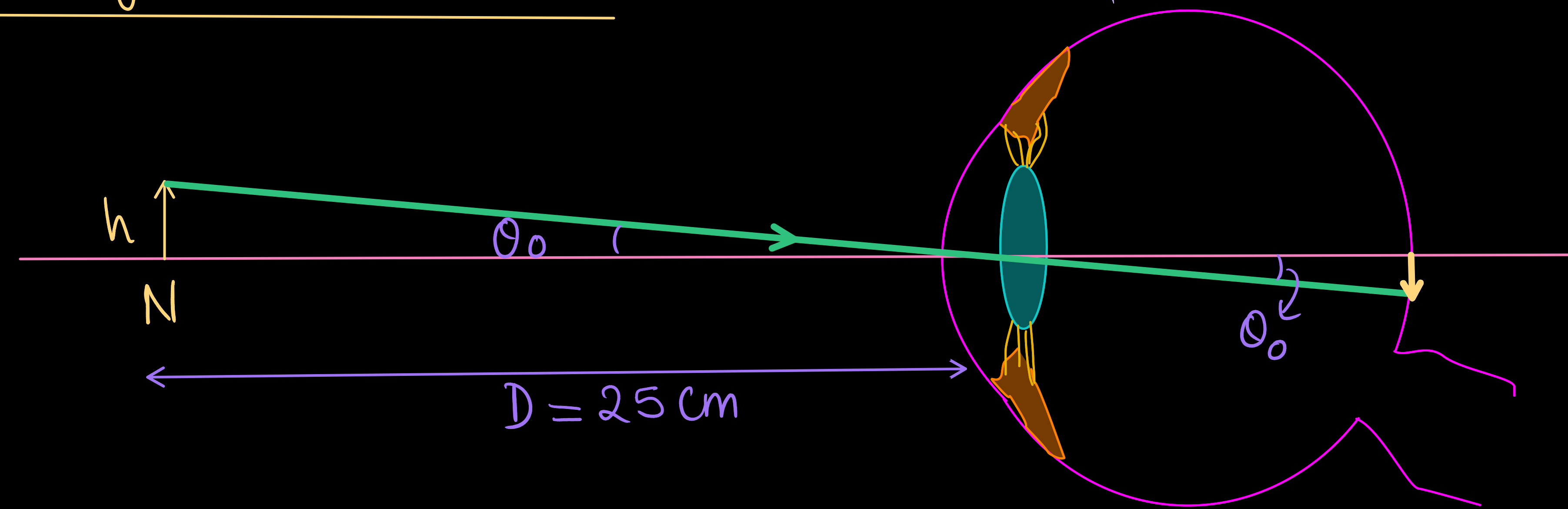


Simple Microscope

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Case II - When image is formed at infinity (eye is relaxed, viewing is more comfortable)

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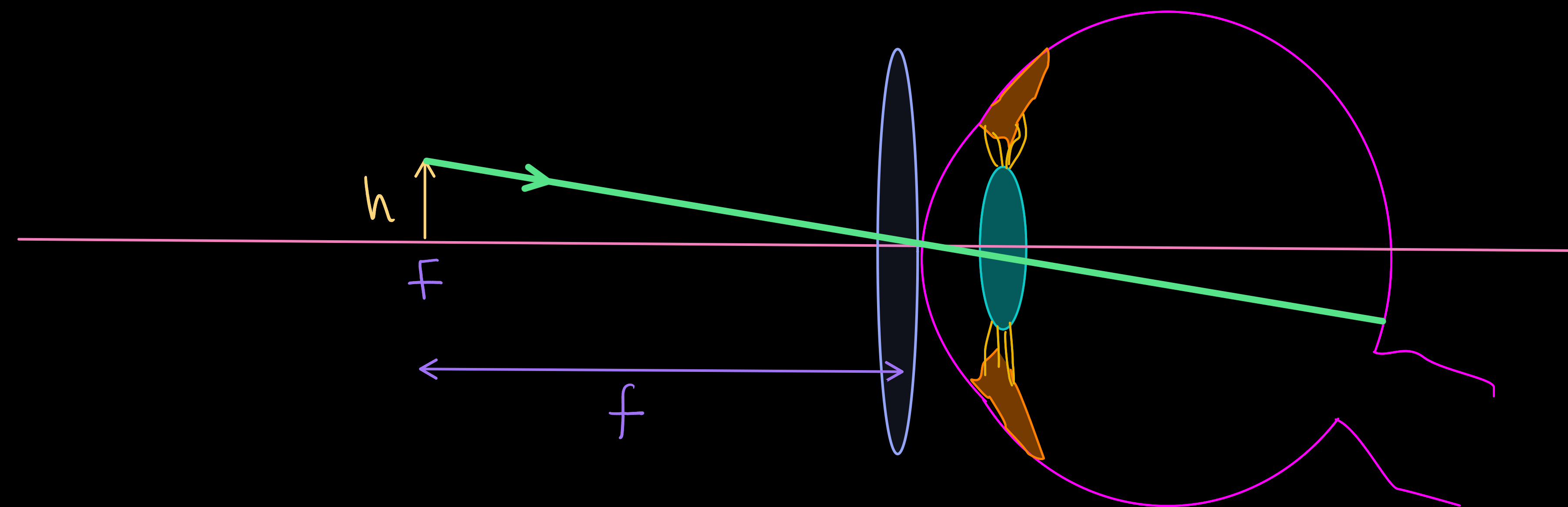
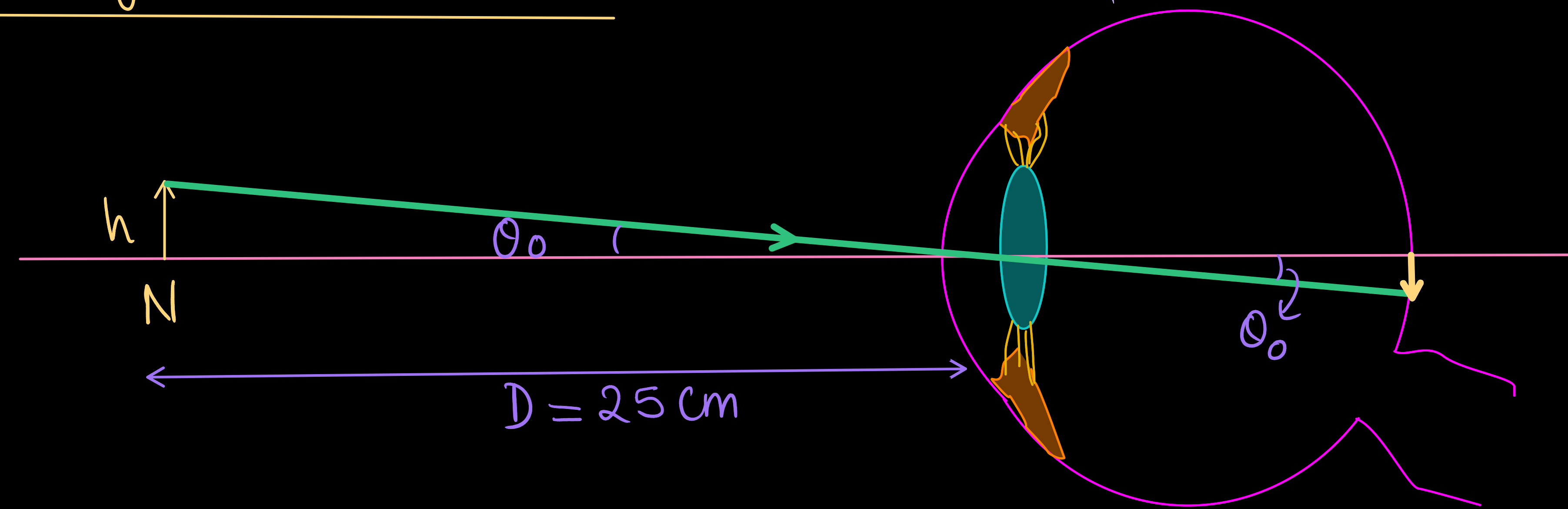


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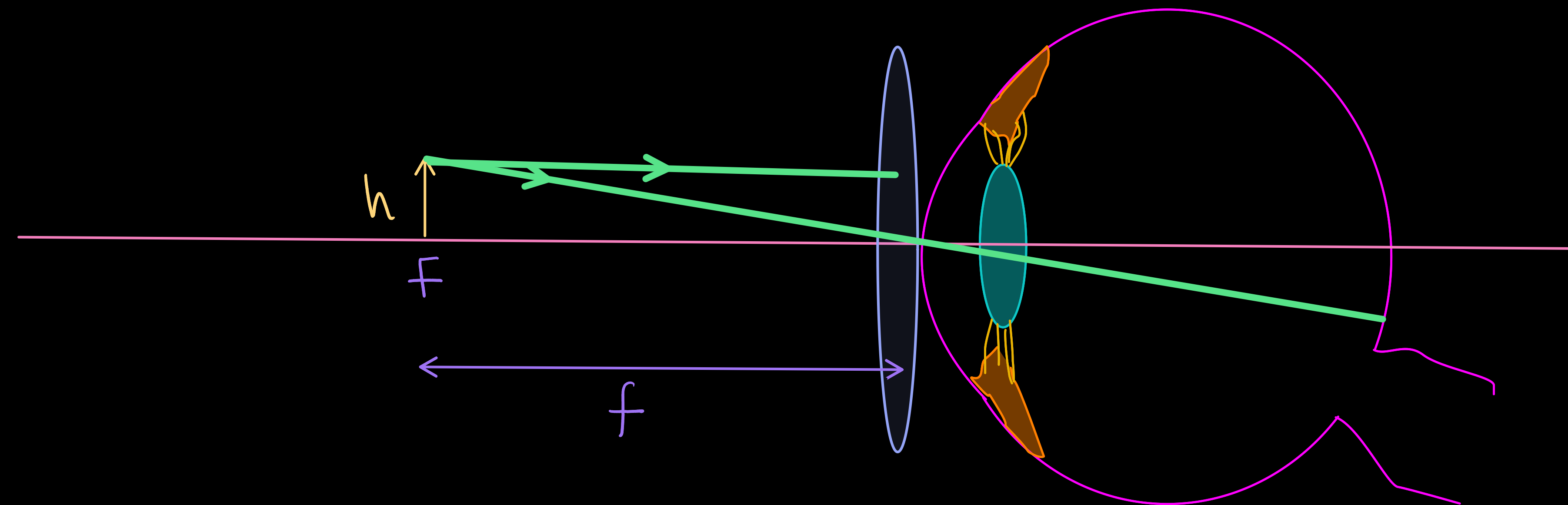
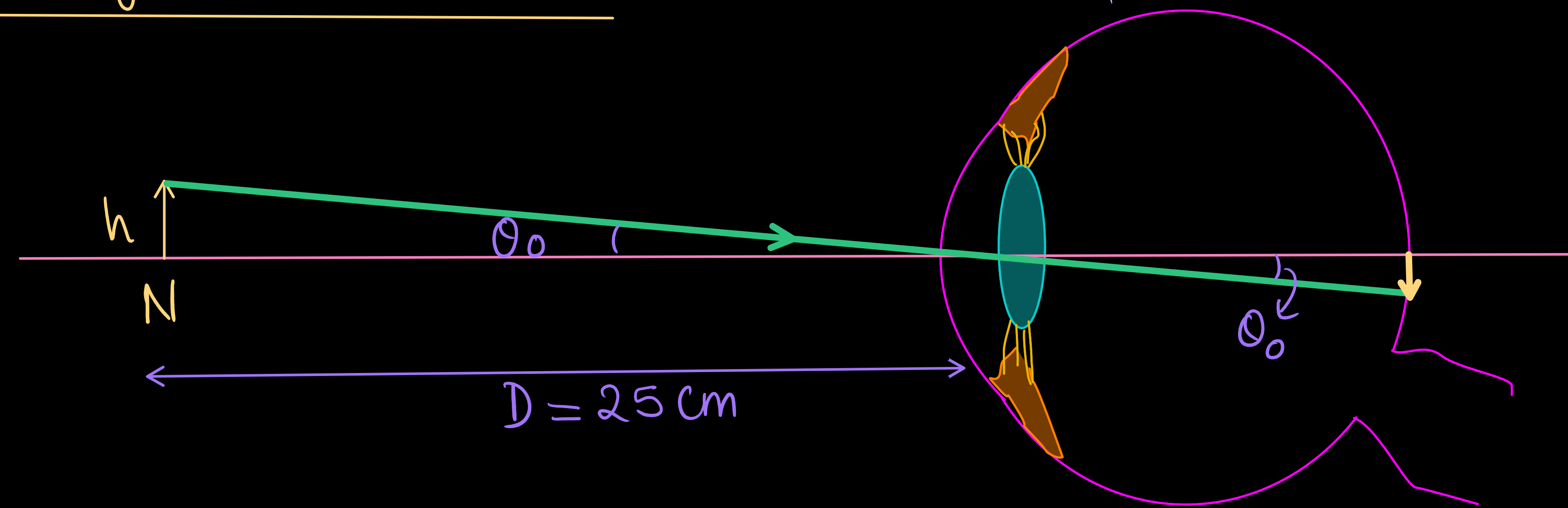


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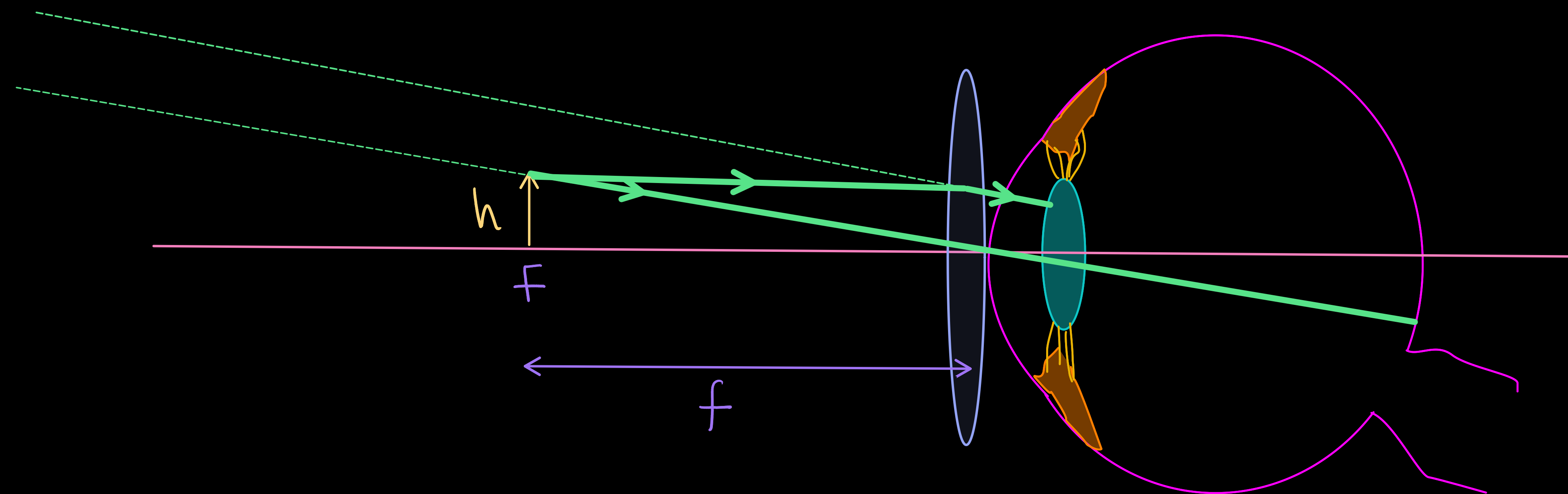
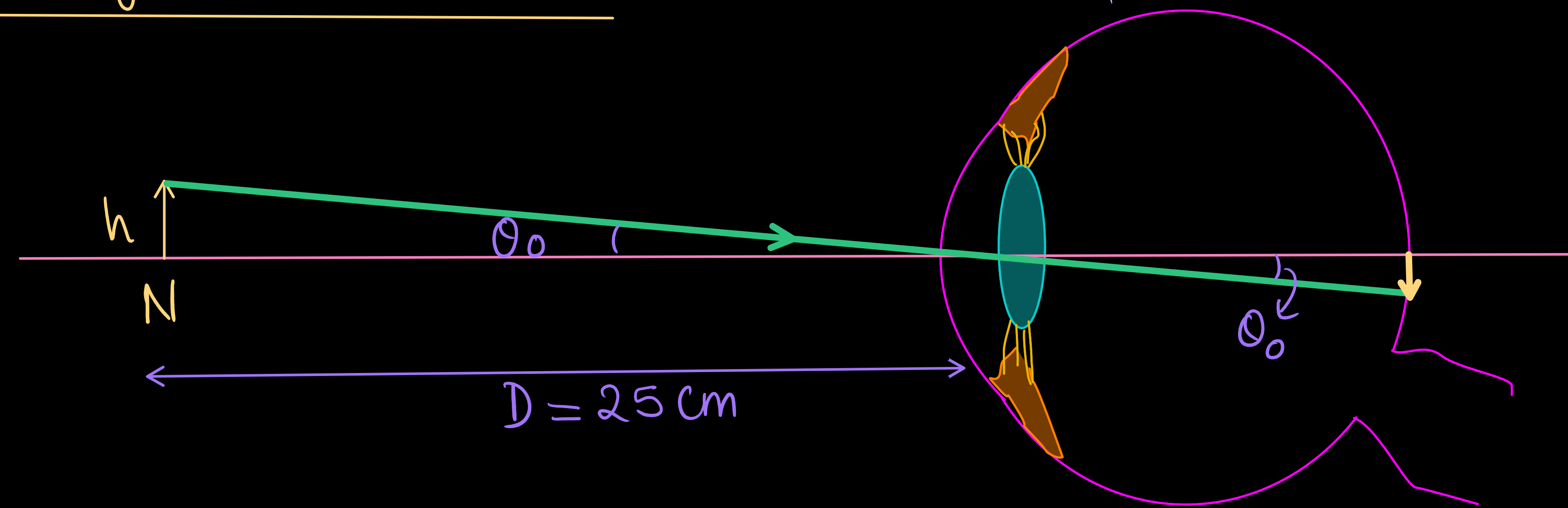


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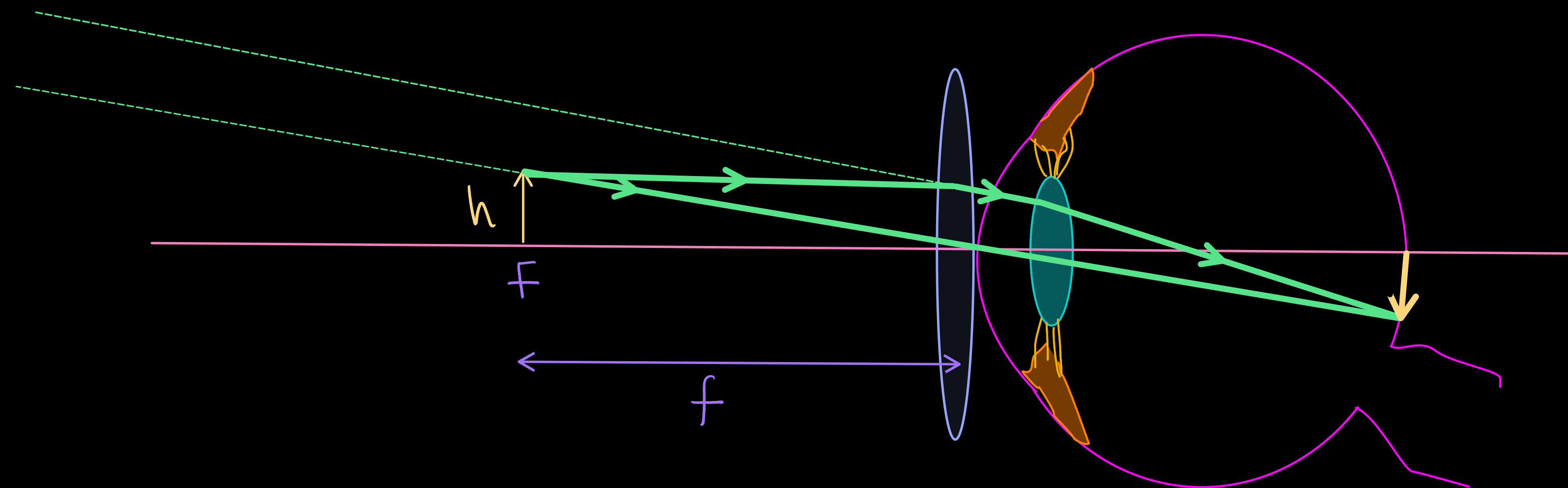
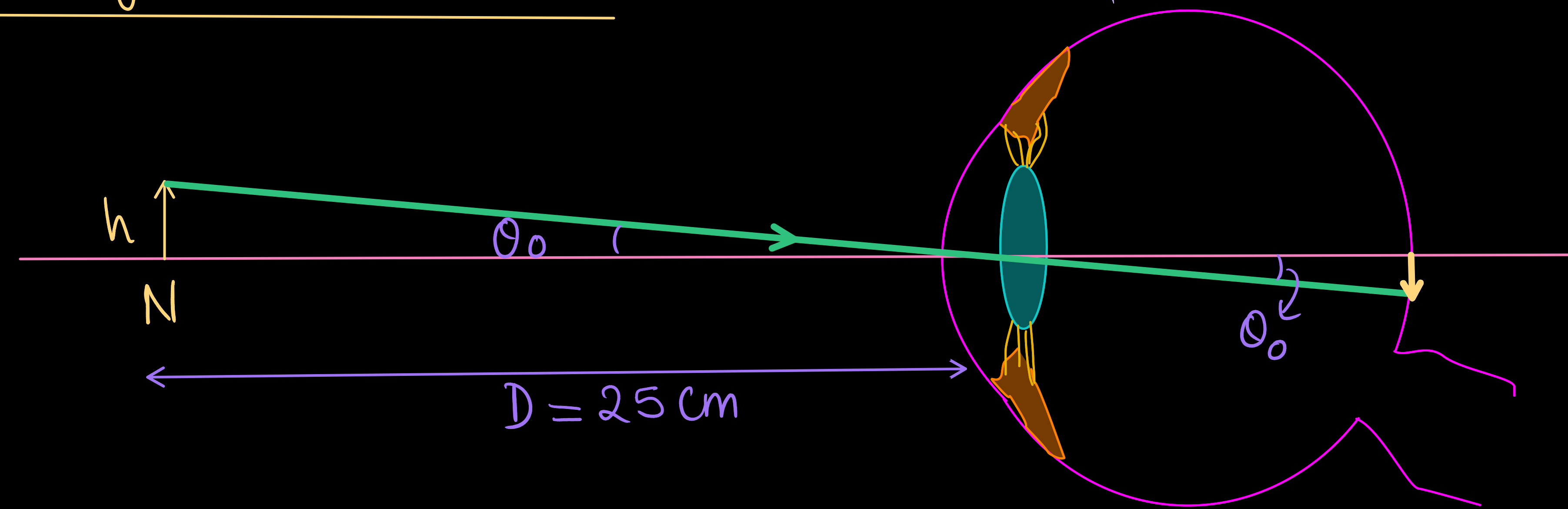


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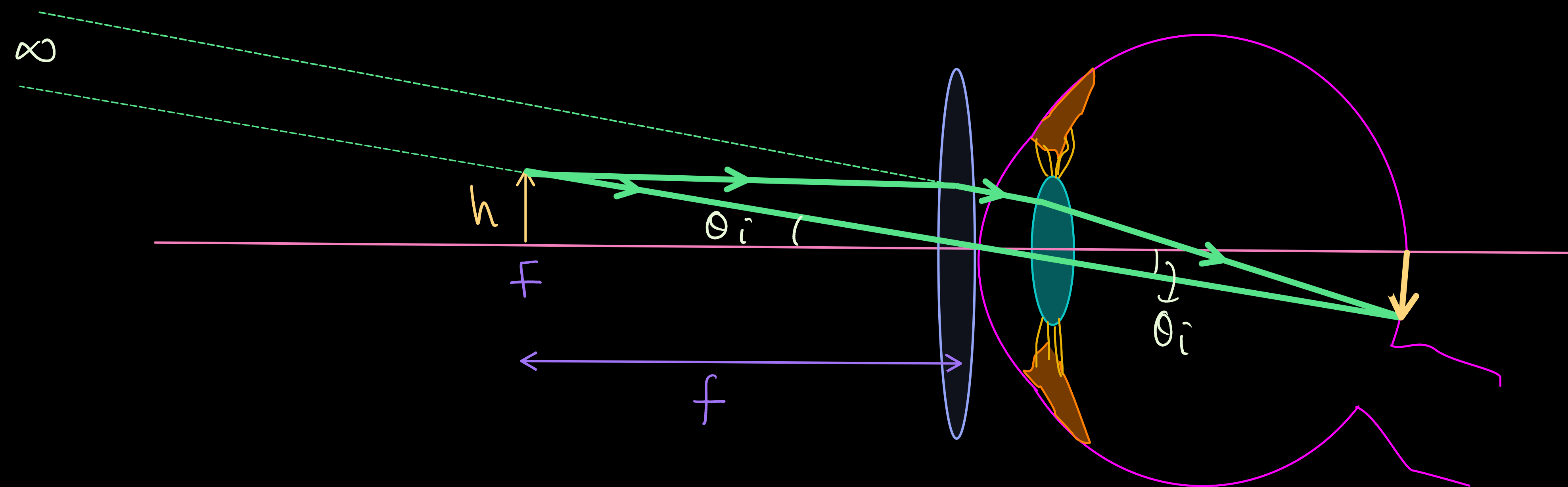
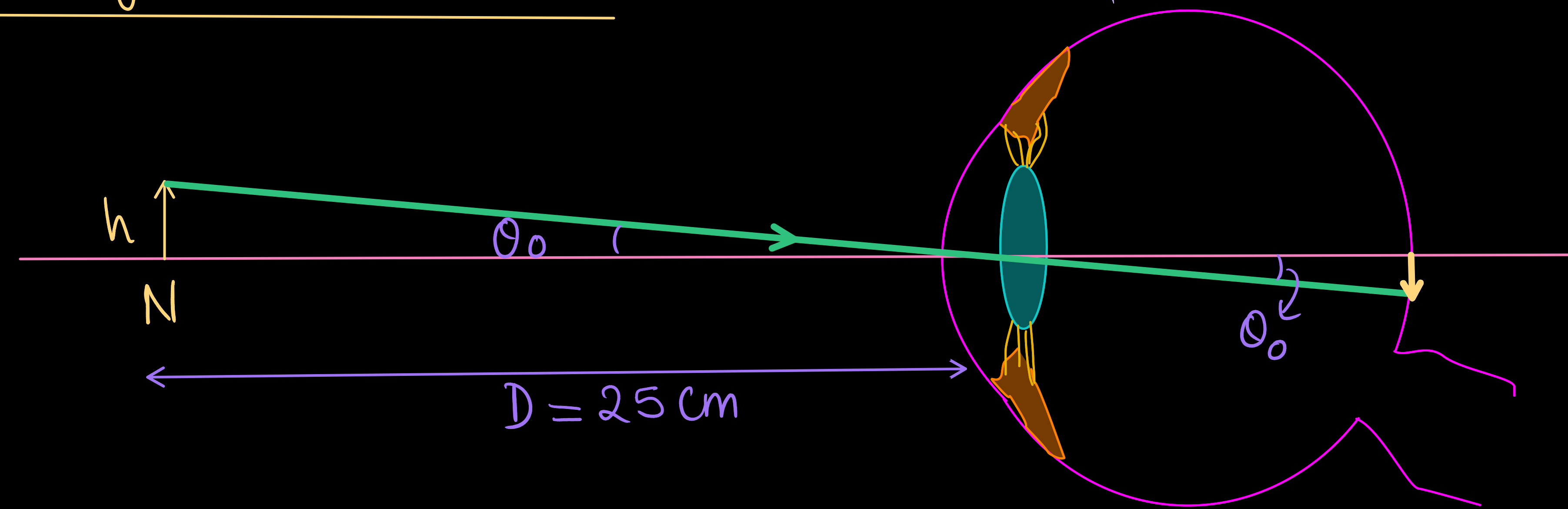


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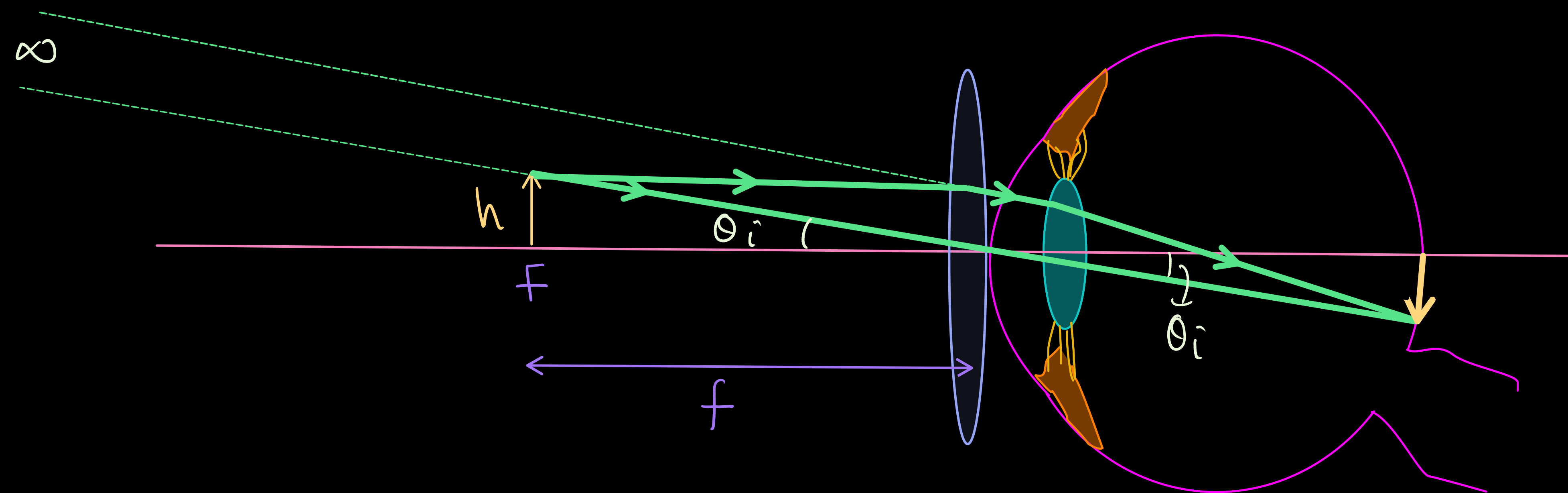
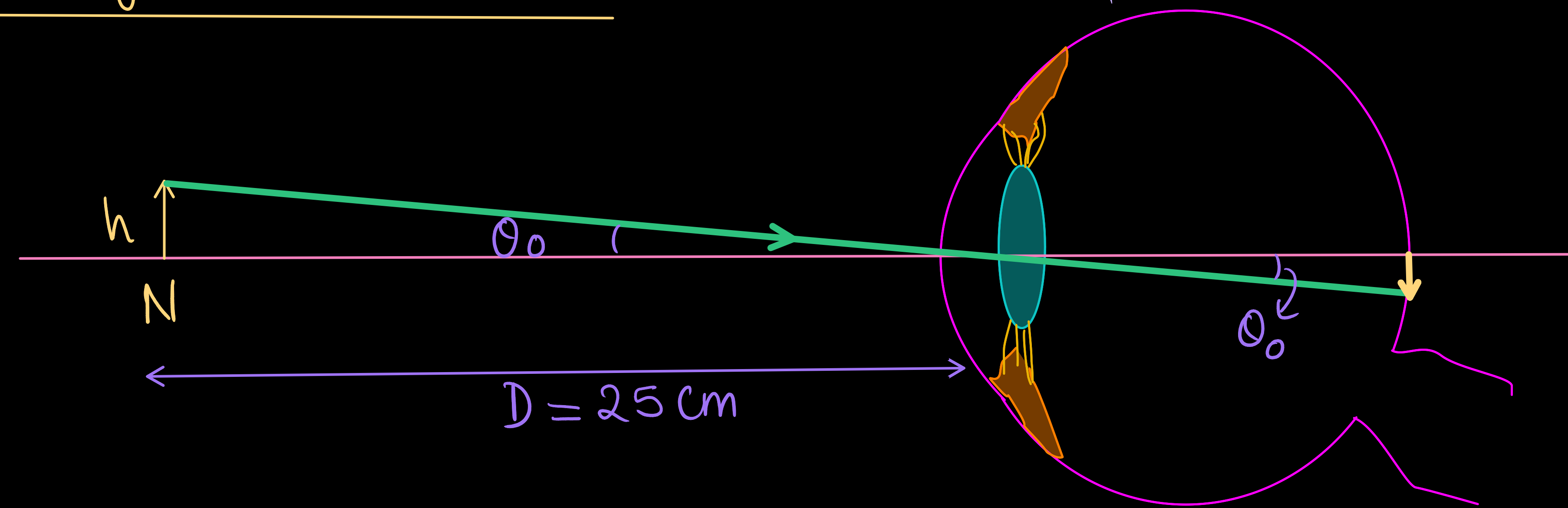
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Angle subtended by object at eye (without lens)

$$\theta_o = \frac{h}{D} \quad \text{--- (1)}$$



Simple Microscope

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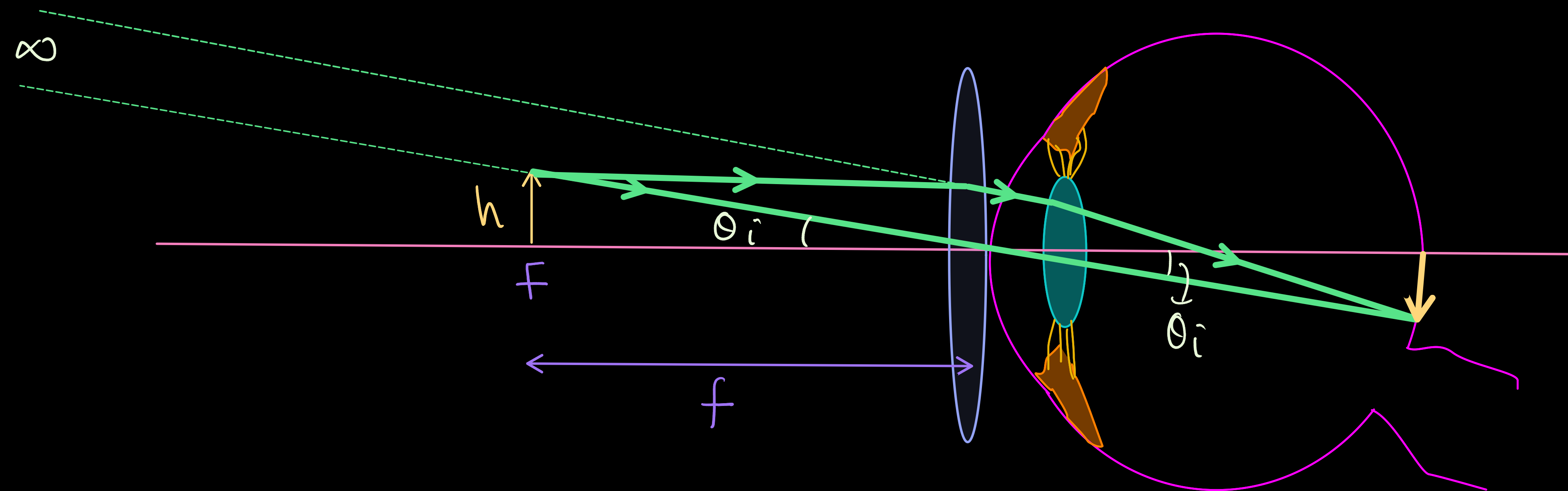
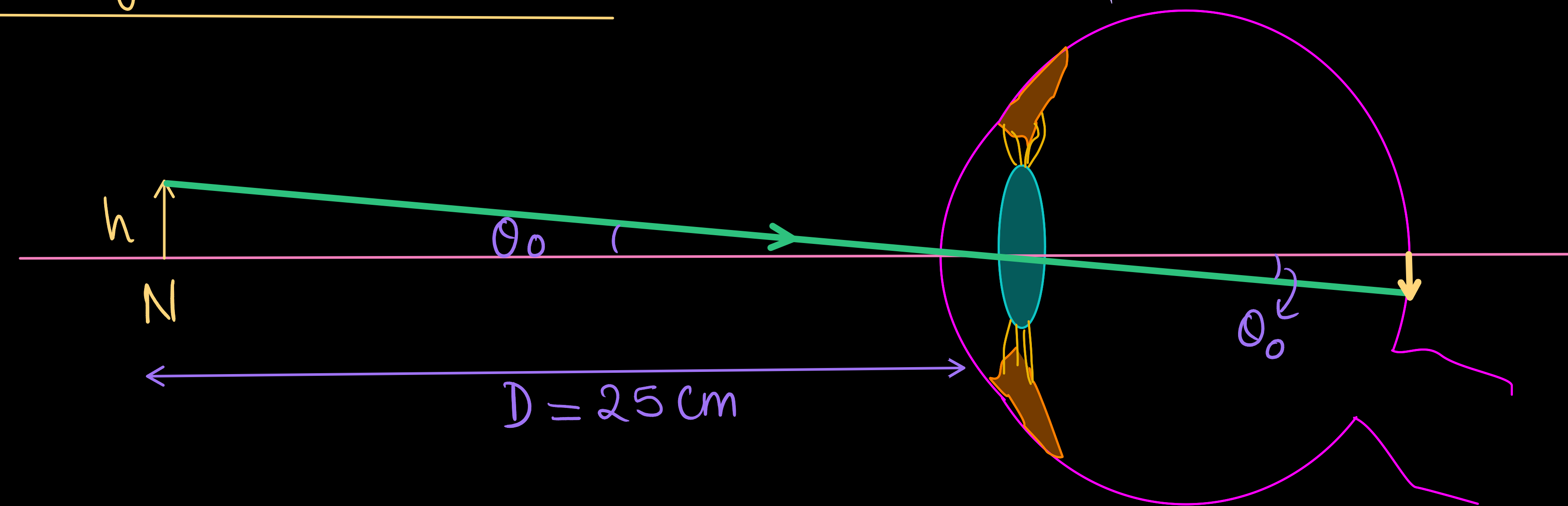
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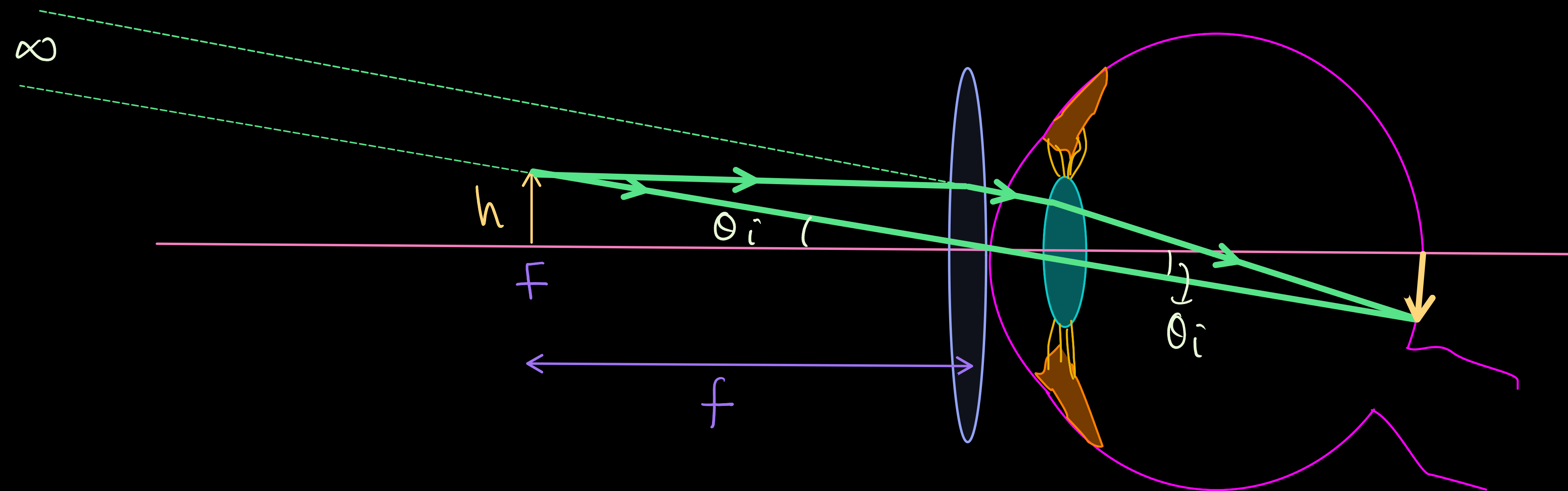
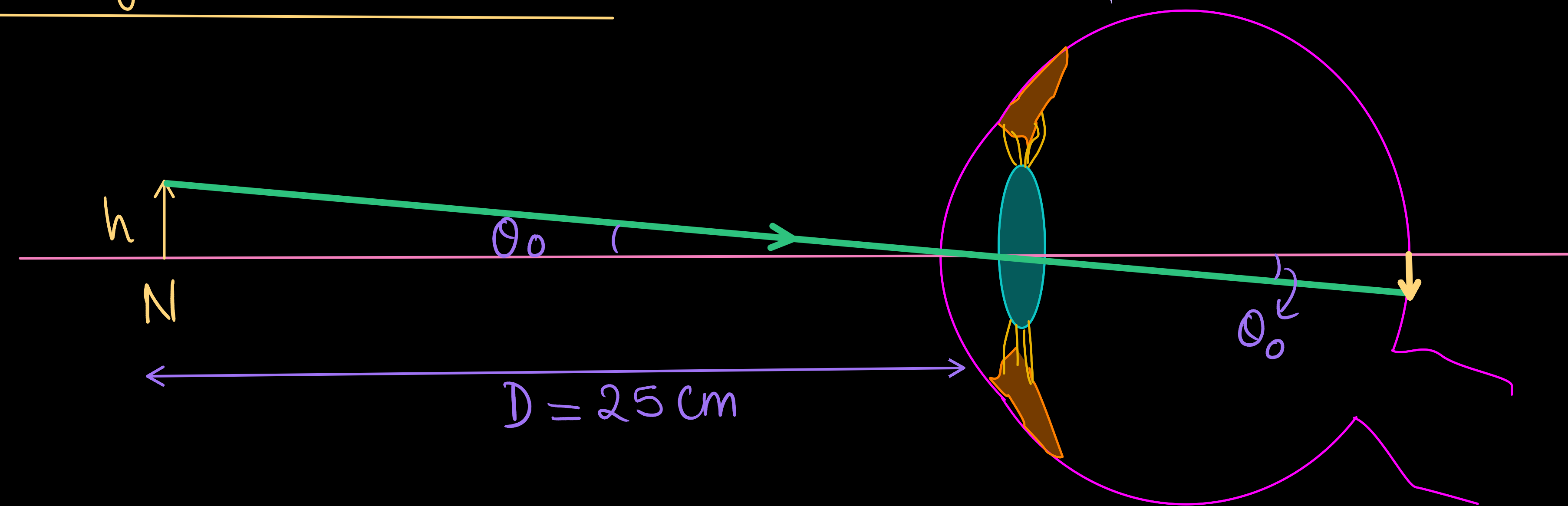
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$\therefore$ Angular magnification produced by simple microscope in normal adjustment -



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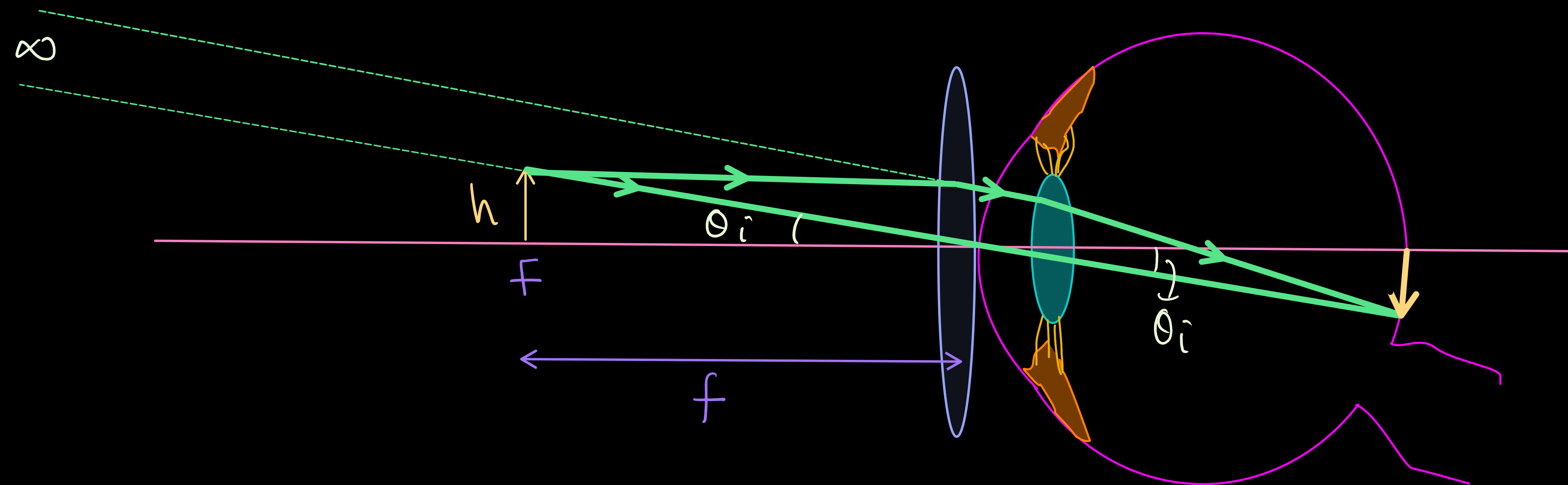
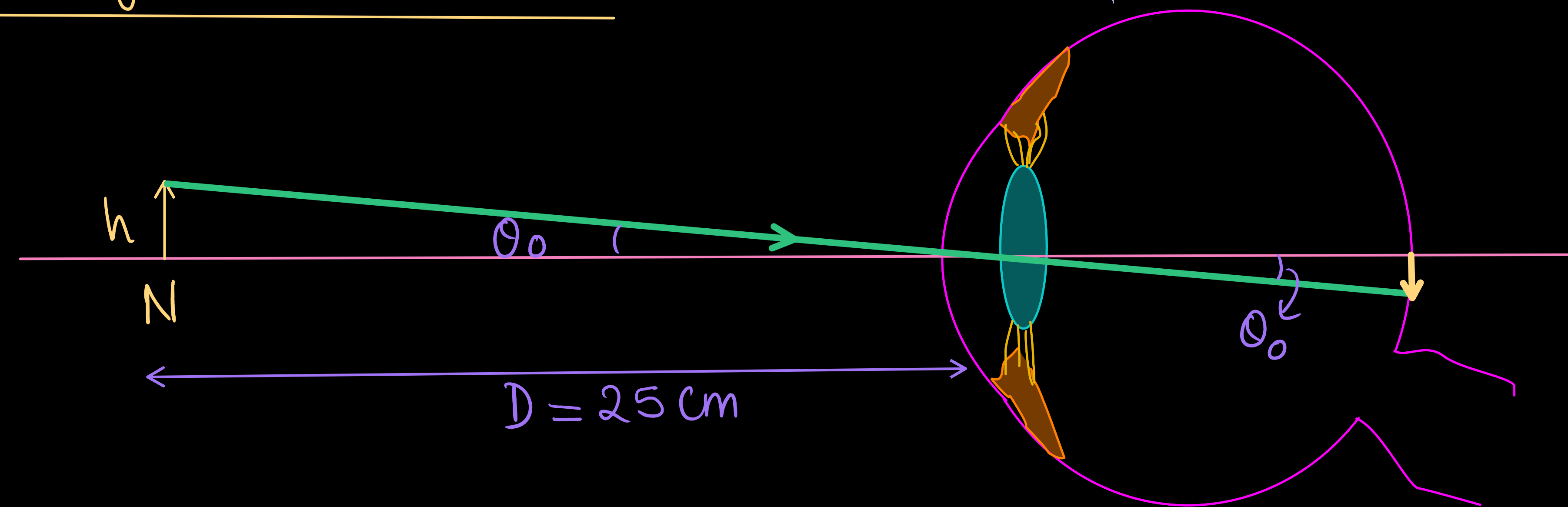
$$\theta_o = \frac{h}{D} \quad - (1)$$

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$$m = \frac{\theta_i}{\theta_o}$$



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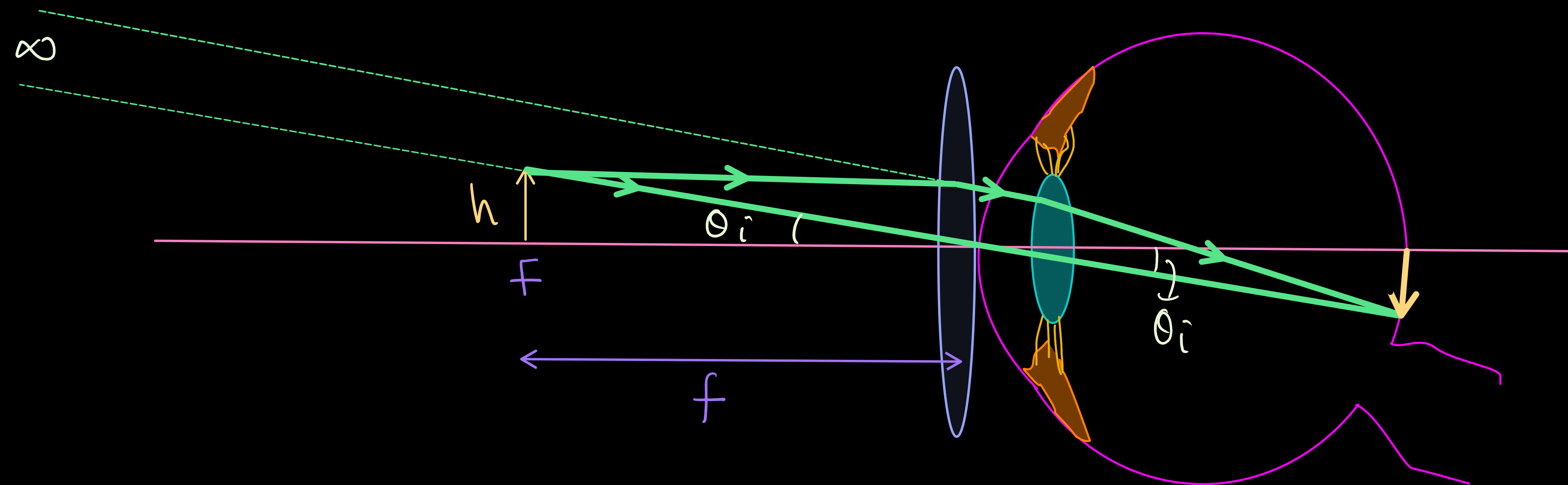
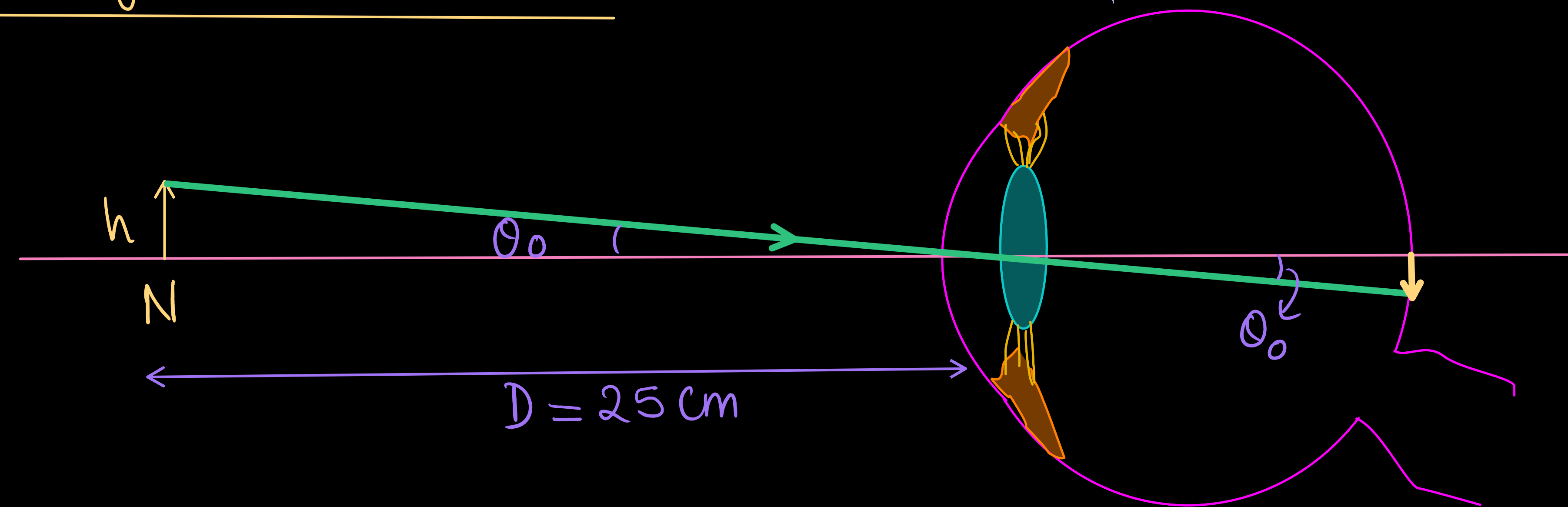
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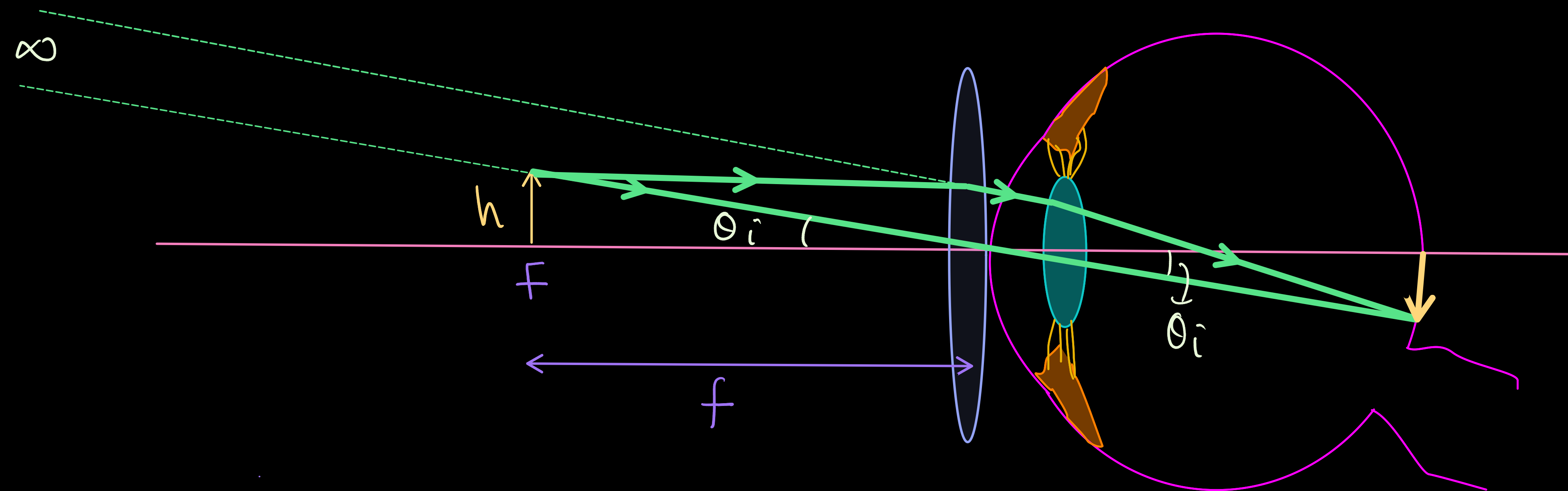
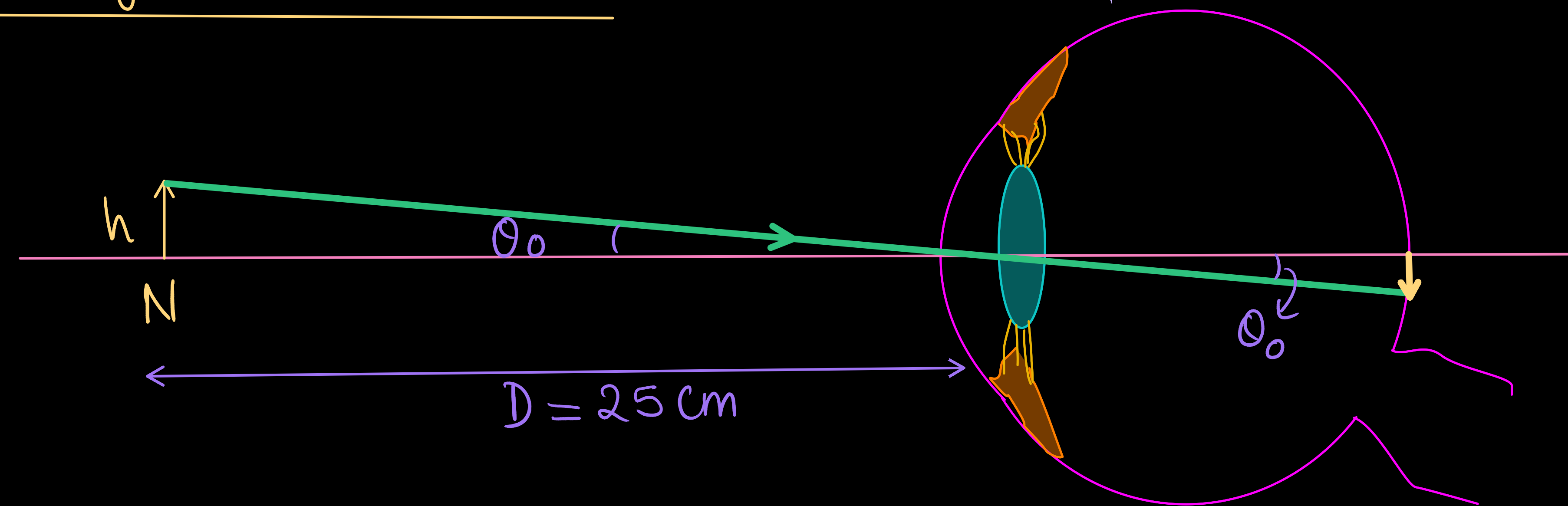
$$\theta_i = \frac{h}{f} \quad - (2)$$

$\therefore$ Angular magnification produced by simple microscope in normal adjustment -

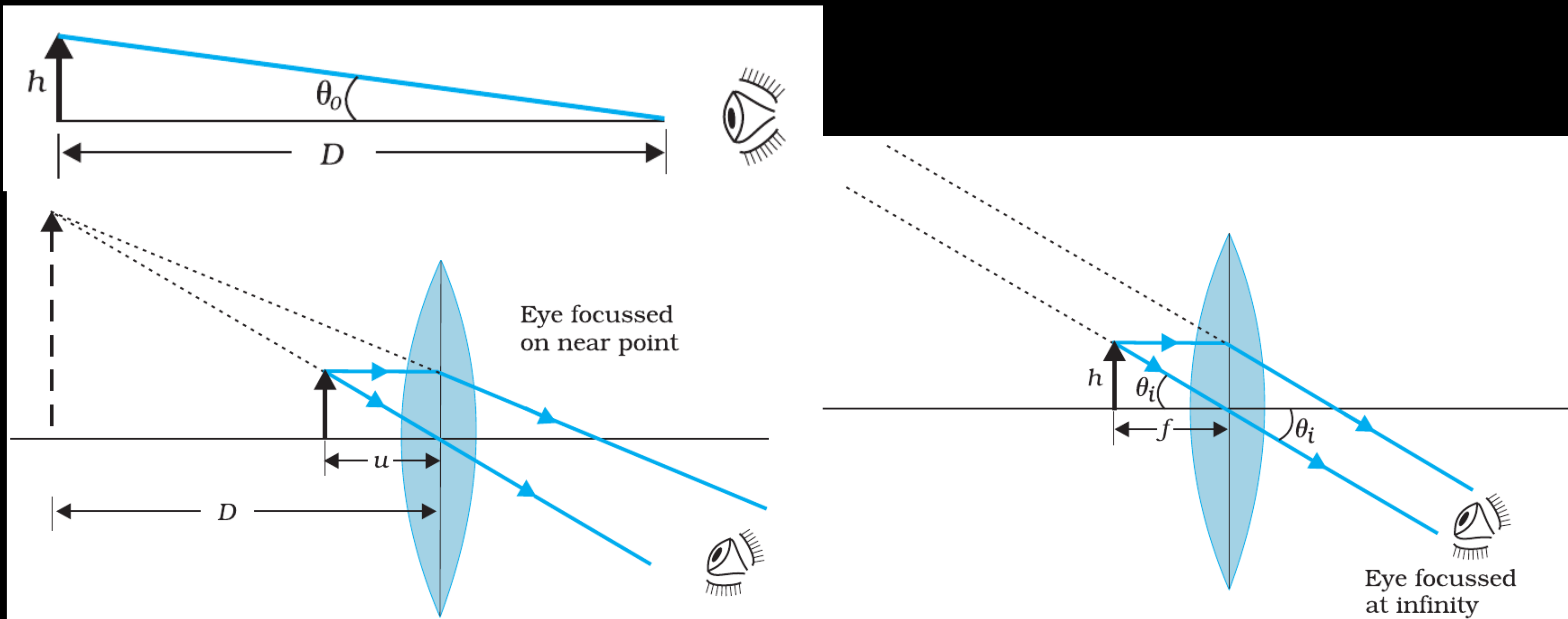
$$m = \frac{\theta_i}{\theta_o} = \frac{\frac{h}{f}}{\frac{h}{D}}$$

$$m = \frac{D}{f}$$

→ 1 less than when image is formed at near point.
But, viewing is more comfortable.



Simple Microscope



Compound Microscope

Compound Microscope

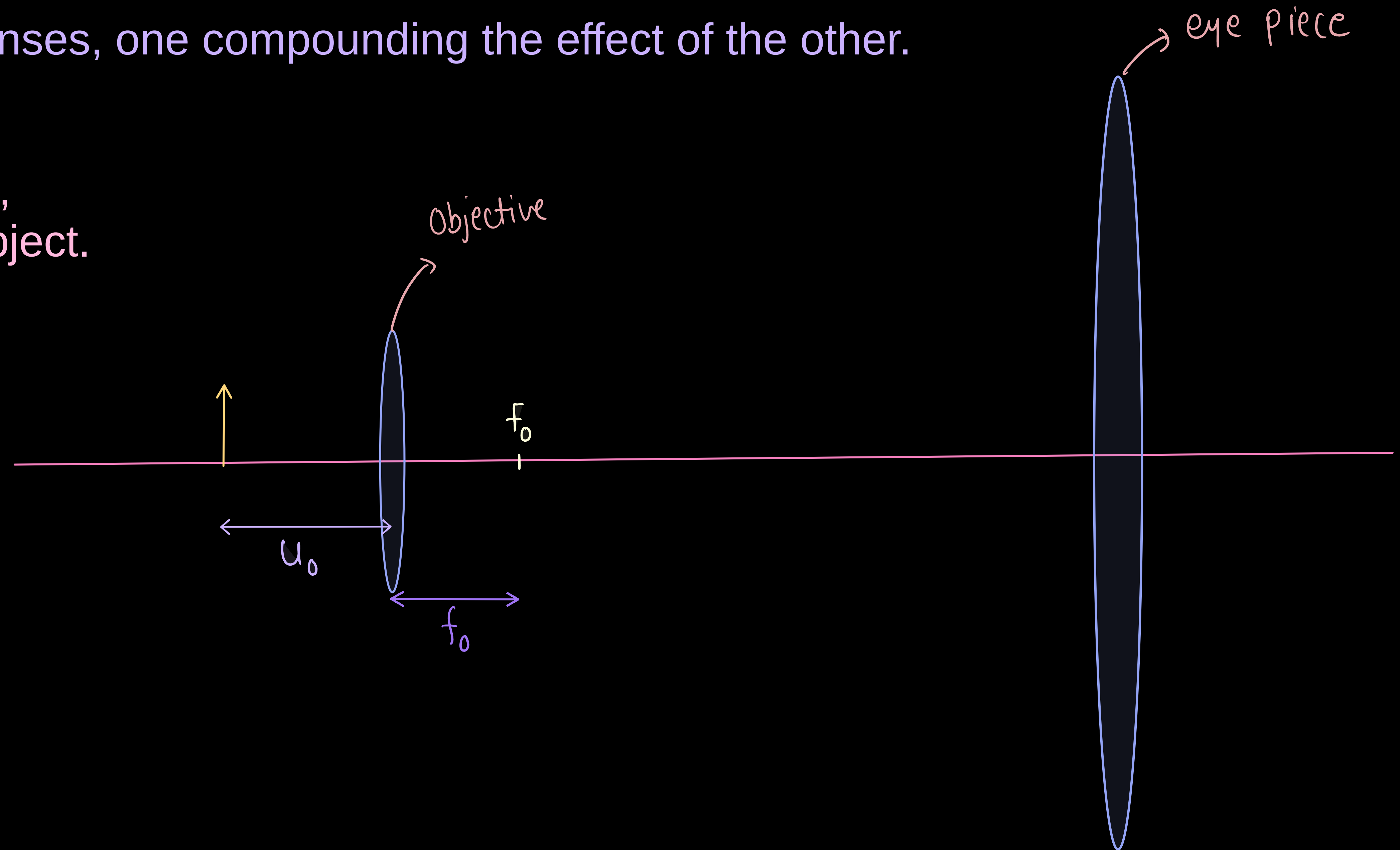
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Compound Microscope

- * A simple microscope has a limited maximum magnification (≤ 9) for realistic focal lengths.
- * For much larger magnifications, one uses two lenses, one compounding the effect of the other. This is known as a compound microscope.

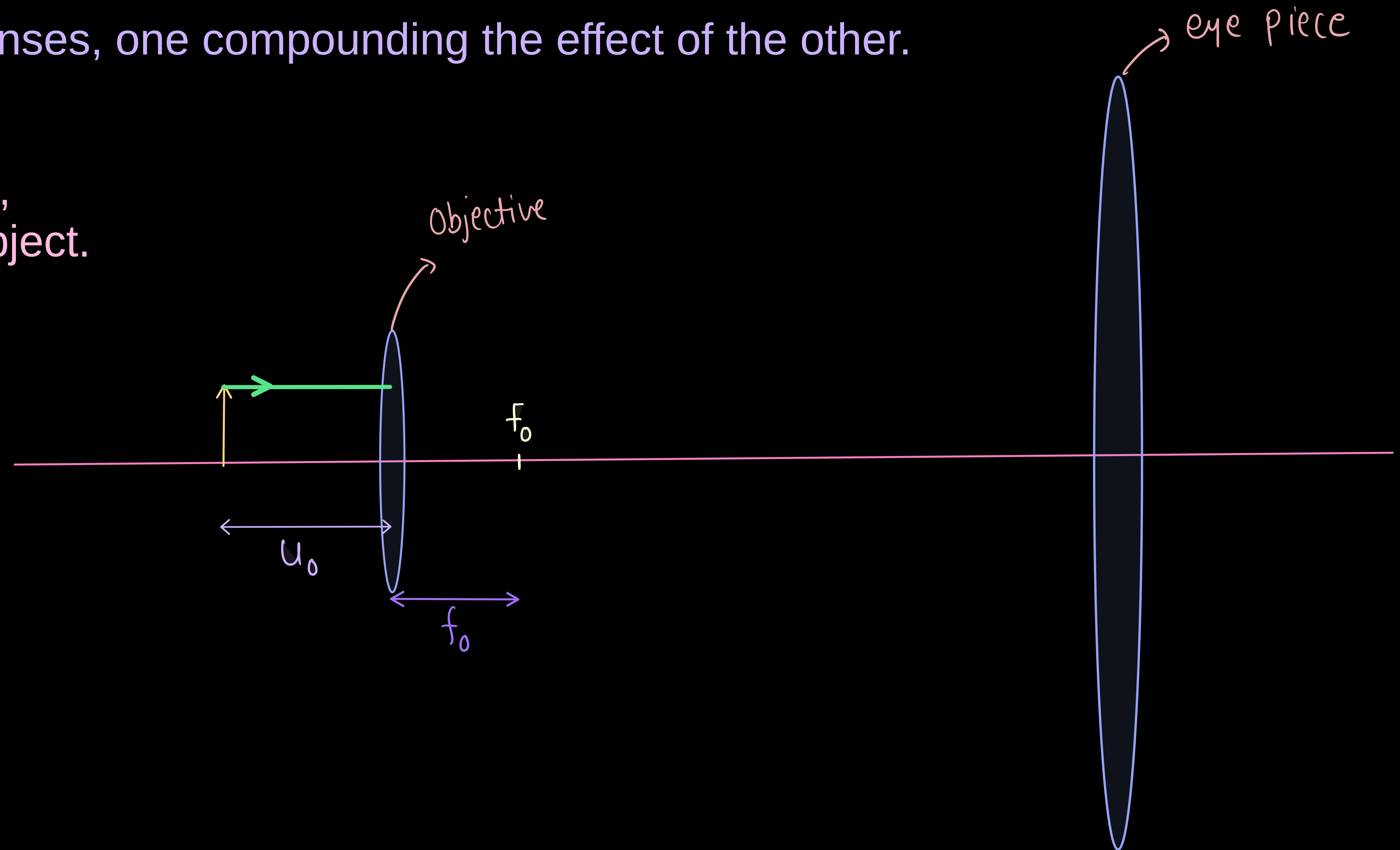
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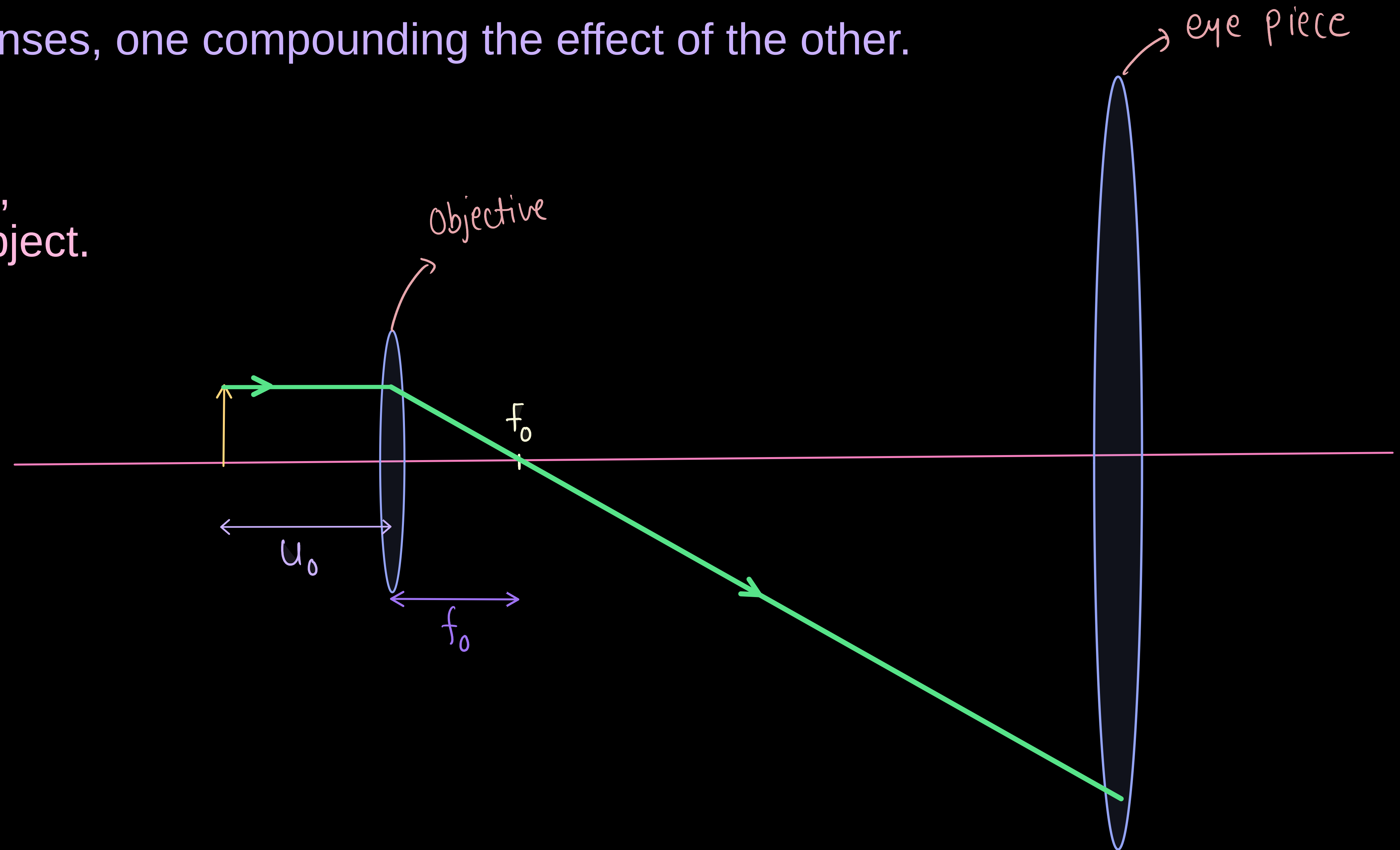
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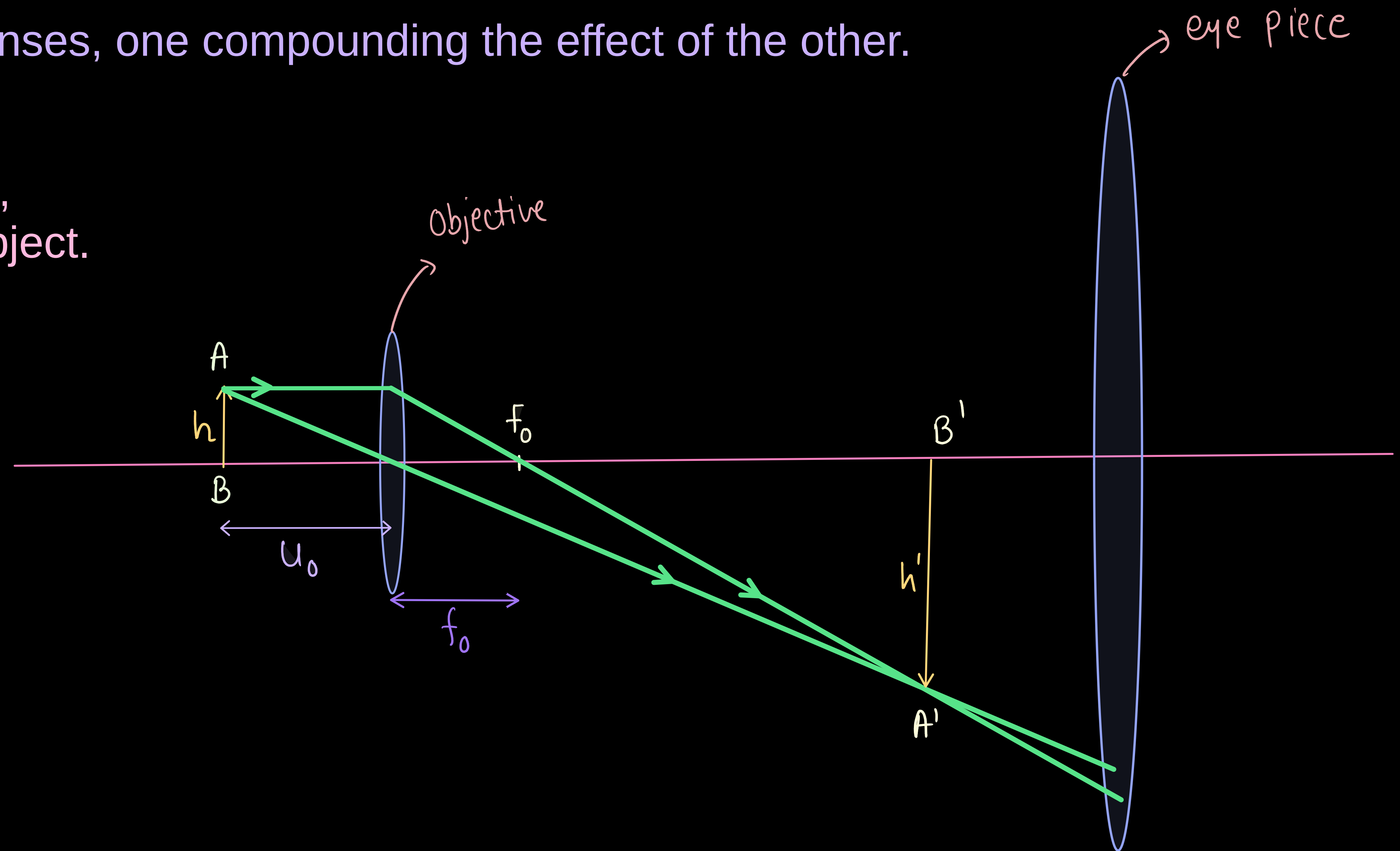
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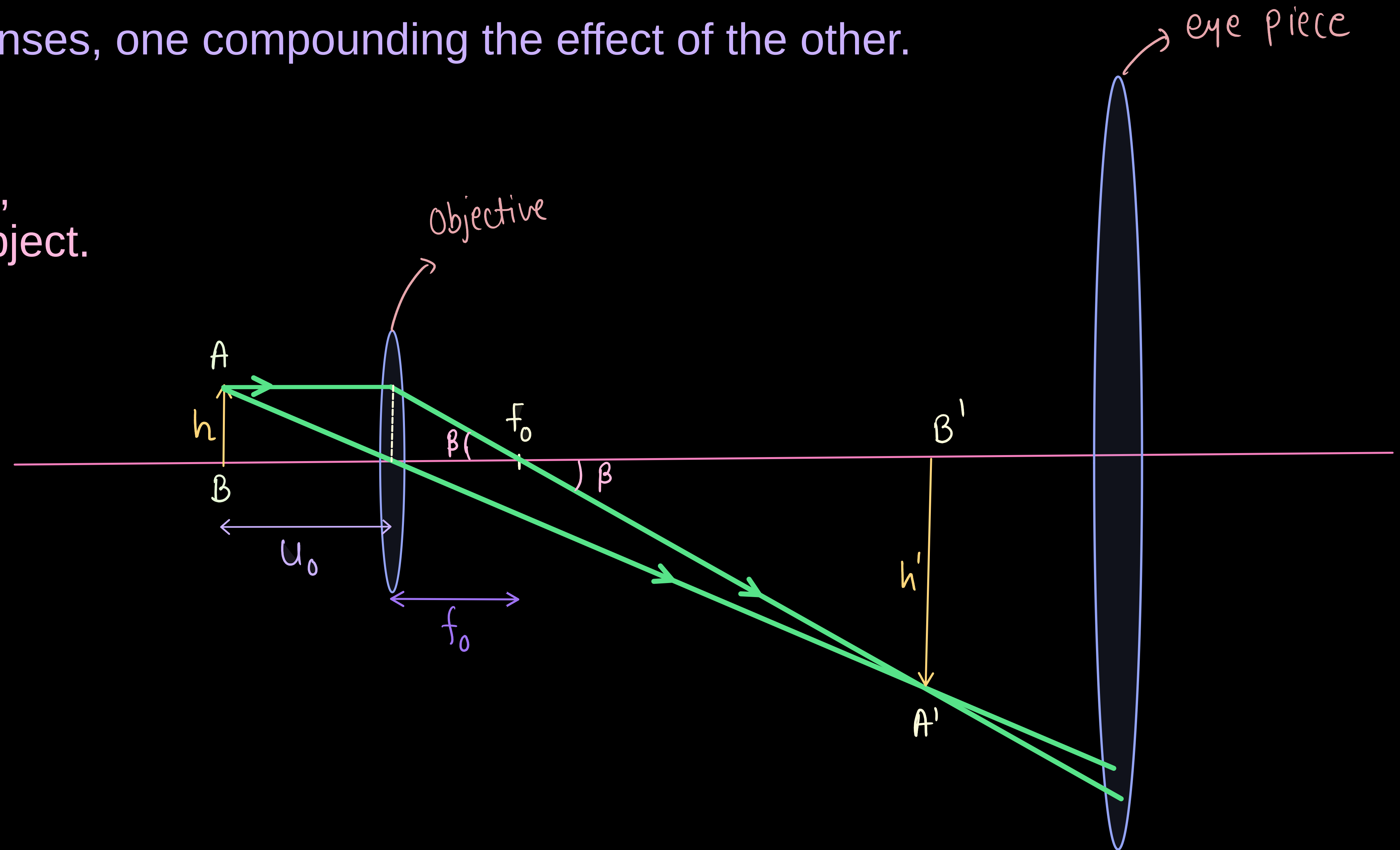
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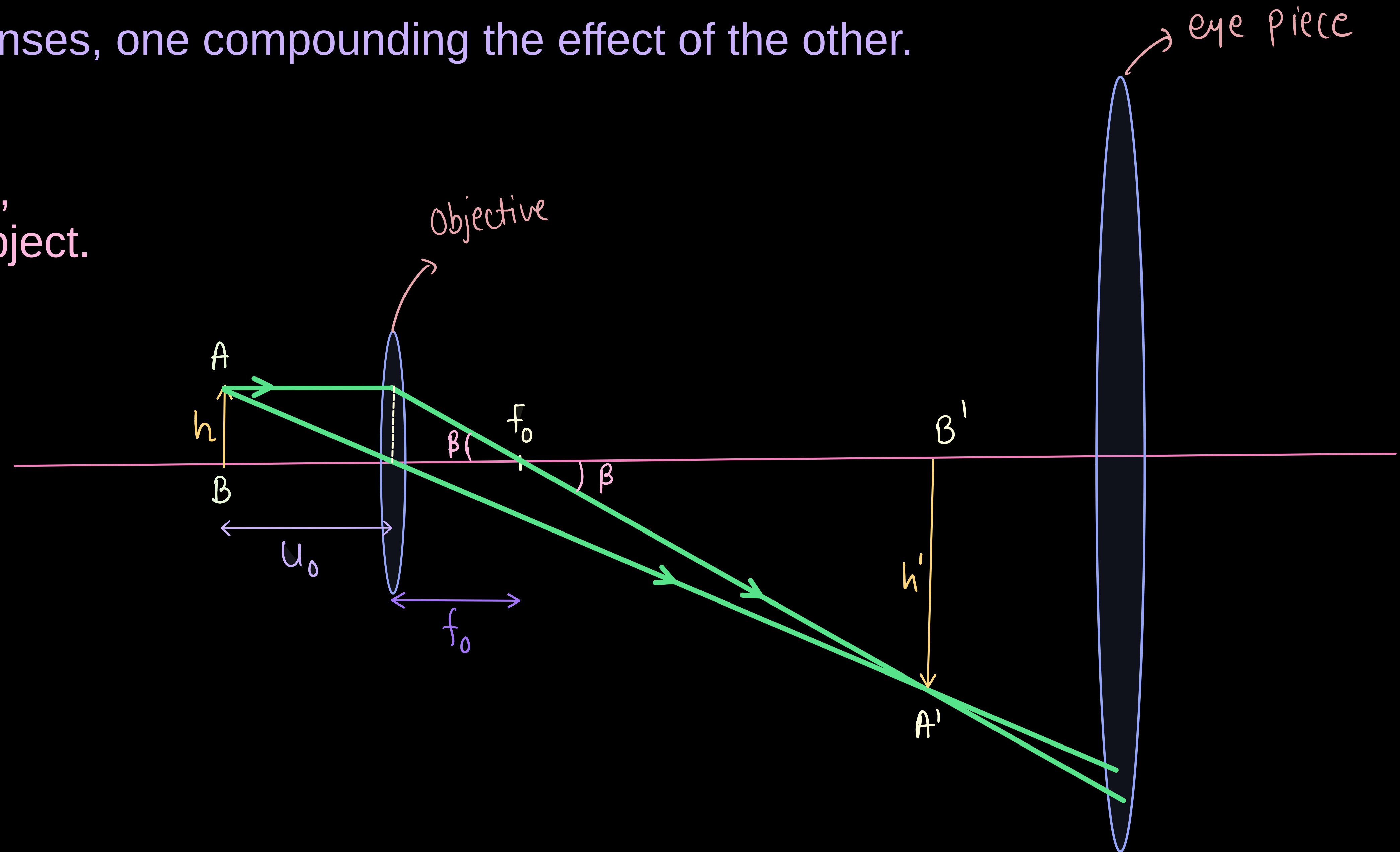
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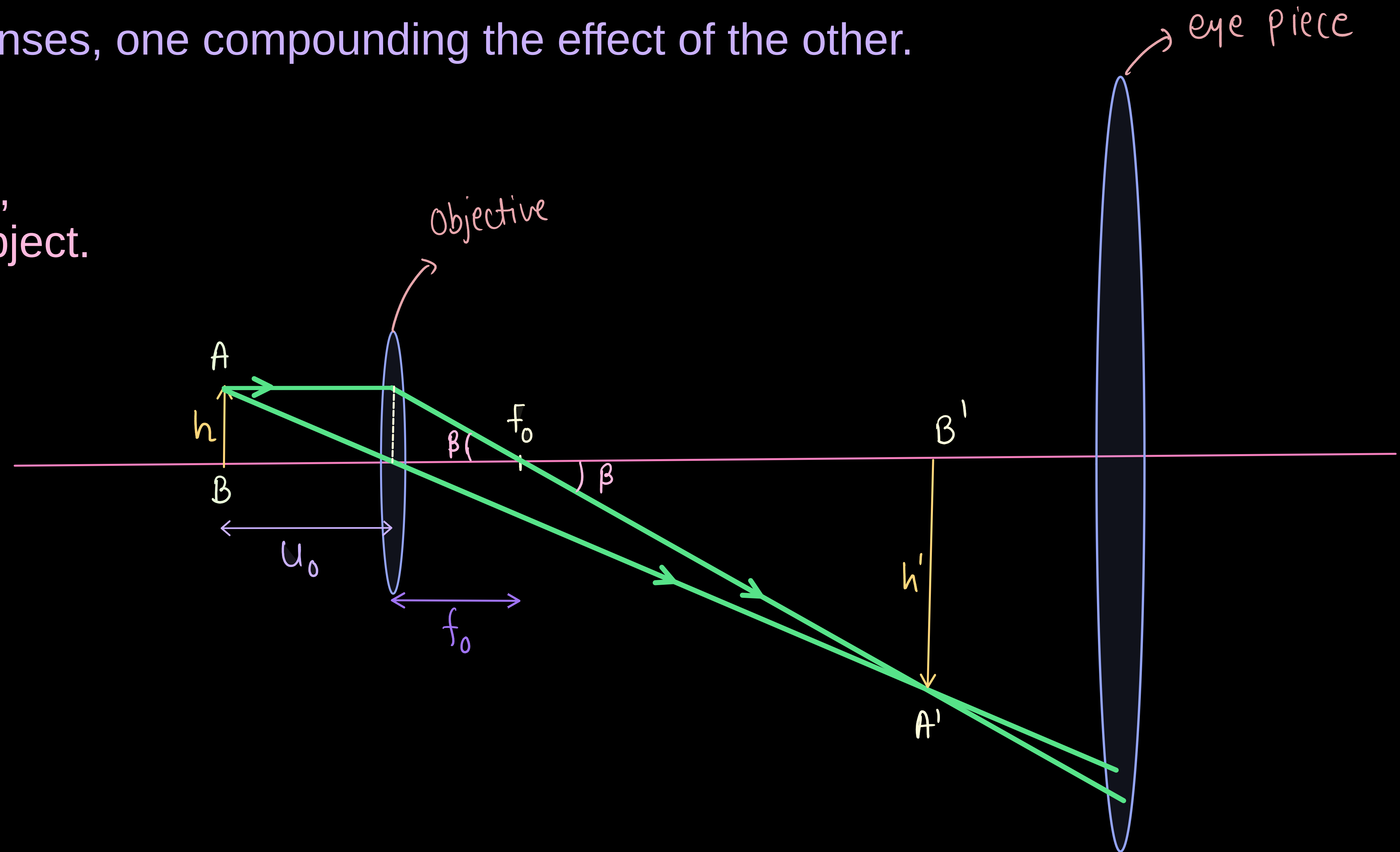


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$$m_o = -\frac{h'}{h}$$



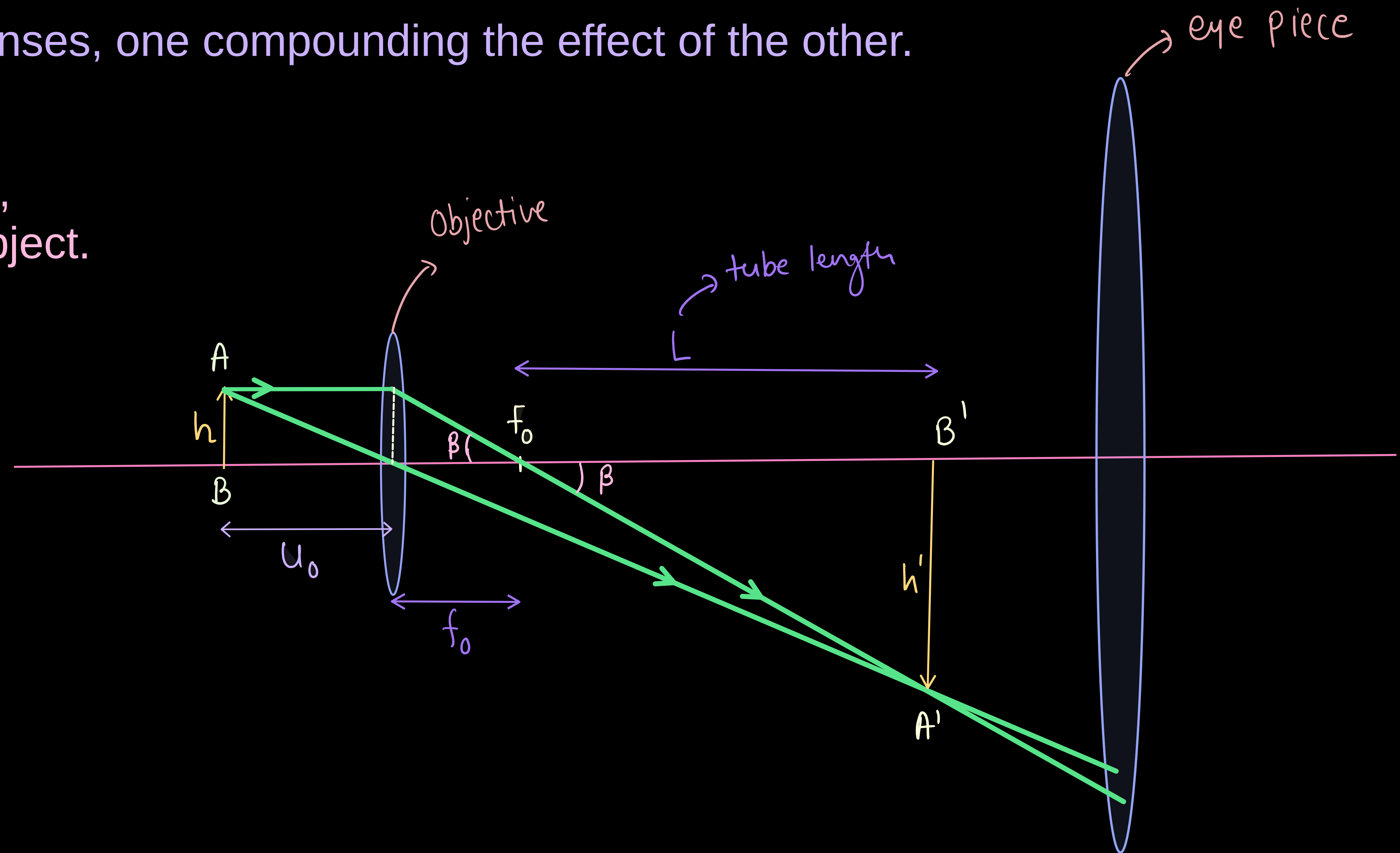
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$$\tan \beta = \frac{h}{f_o} = \frac{h'}{L}$$



Compound Microscope

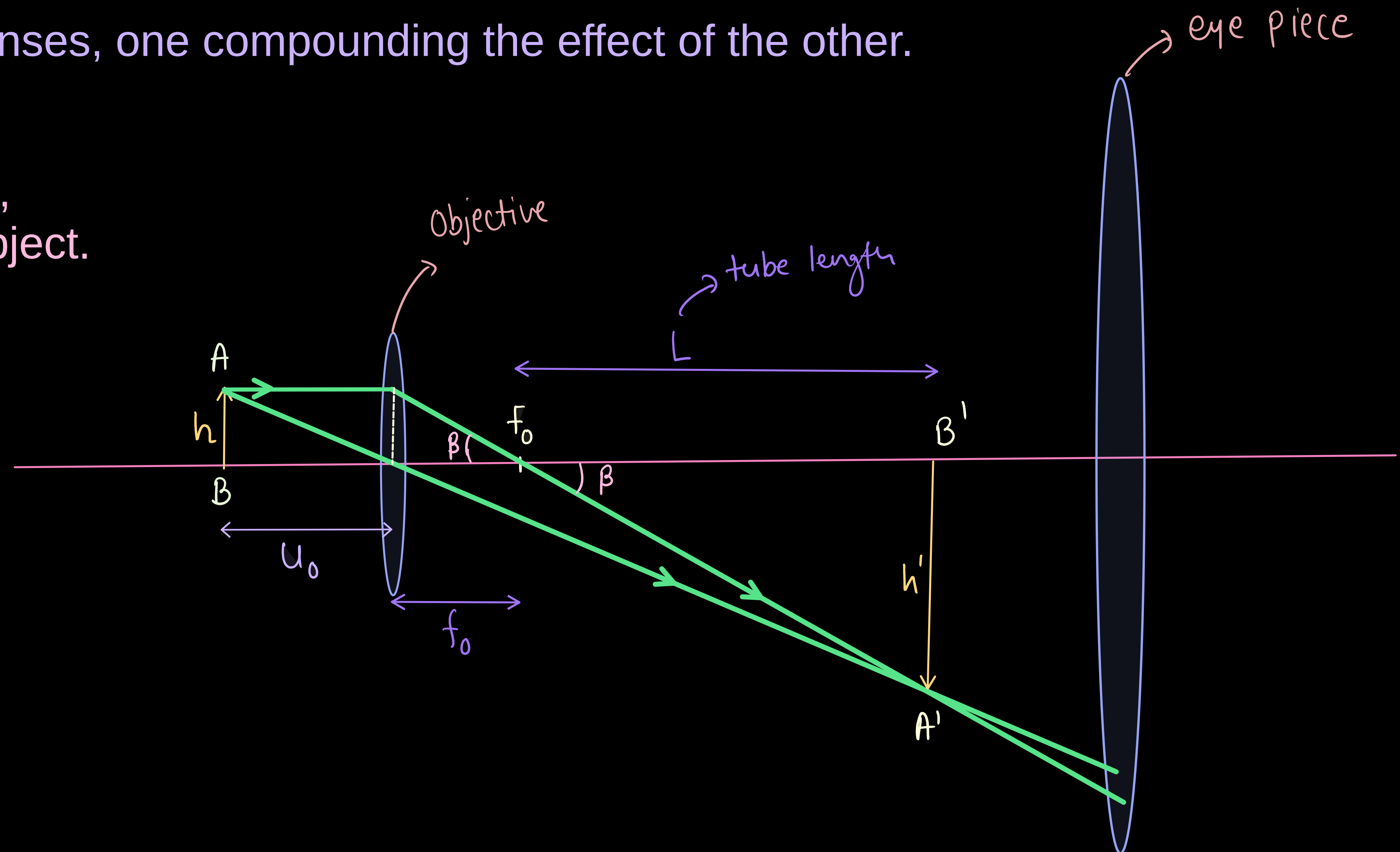
- * A simple microscope has a limited maximum magnification (≤ 9) for realistic focal lengths.
- * For much larger magnifications, one uses two lenses, one compounding the effect of the other. This is known as a compound microscope.
- * The lens nearest the object, called the objective, forms a real, inverted, magnified image of the object.

* Linear magnification (m_o) due to objective lens-

$$m_o = -\frac{h'}{h}$$

$$\tan \beta = \frac{h}{f_o} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_o}$$



Compound Microscope

- * A simple microscope has a limited maximum magnification (≤ 9) for realistic focal lengths.
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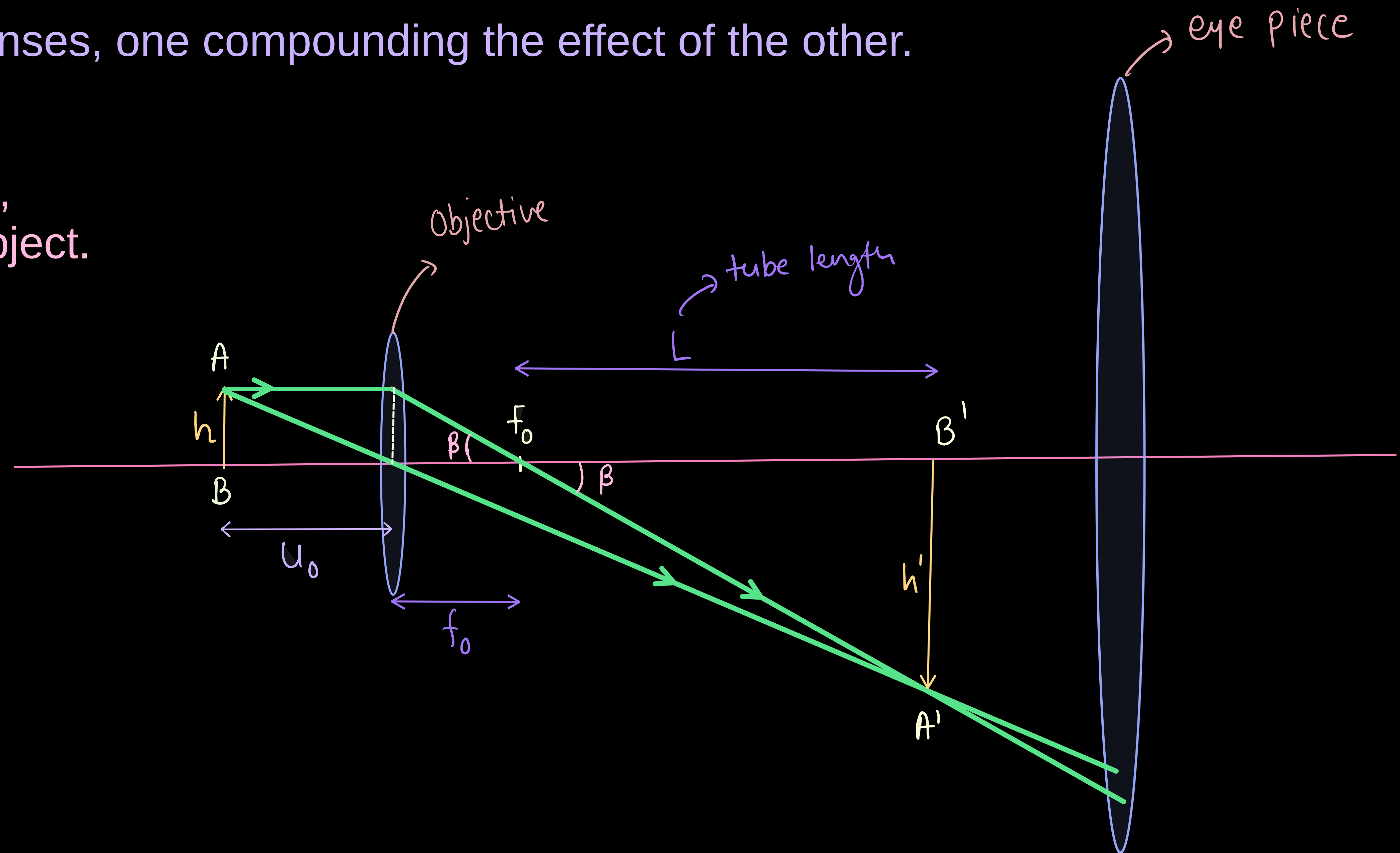
* Linear magnification (m_o) due to objective lens-

$$m_o = -\frac{h'}{h}$$

$$\tan \beta = \frac{h}{f_o} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_o}$$

$$m_o = -\frac{L}{f_o} \quad \text{--- (1)}$$



Compound Microscope

* Linear magnification (m_o) due to objective lens-

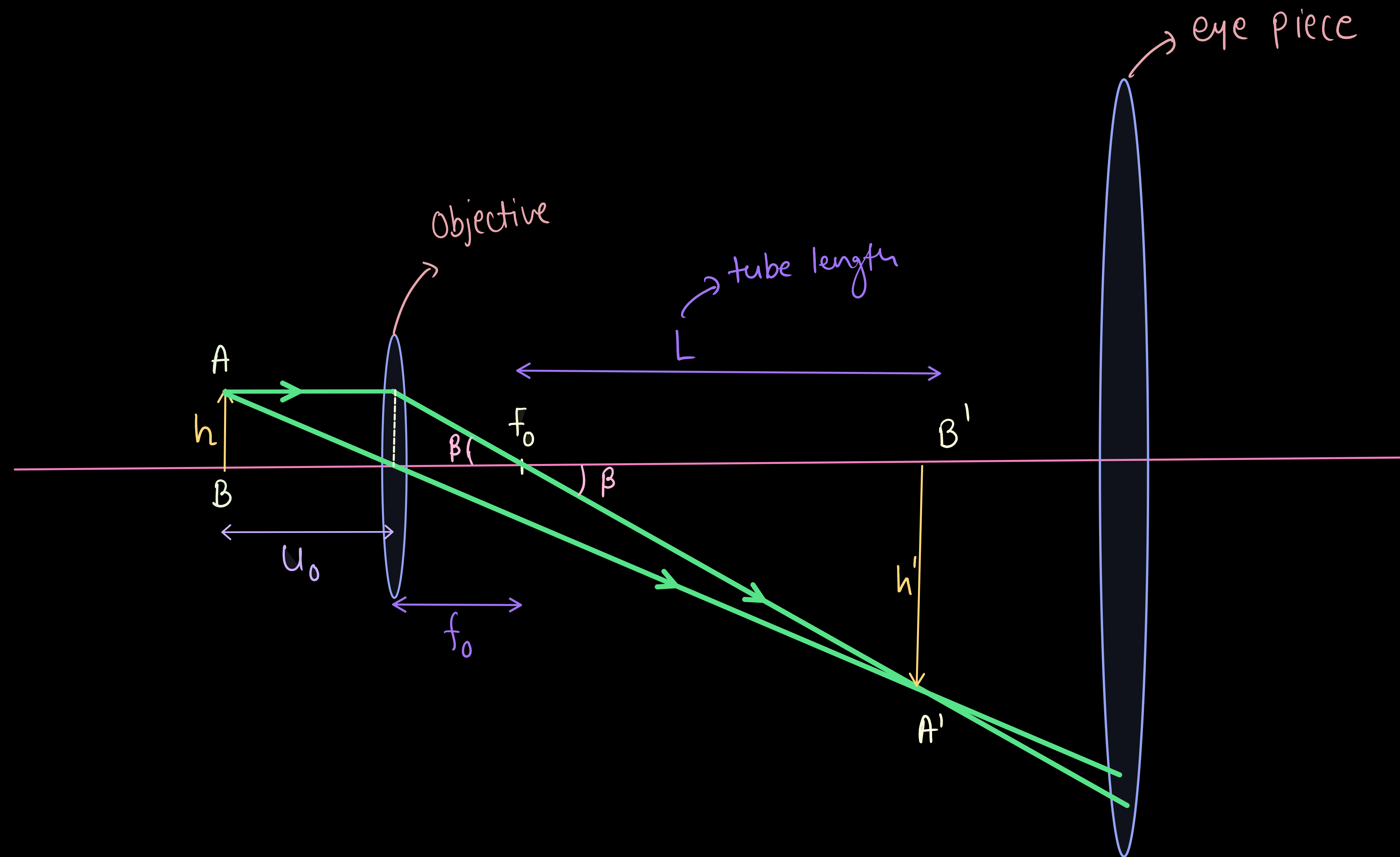
$$m_o = -\frac{h'}{h}$$

$$\tan \beta = \frac{h}{f_o} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_o}$$

$$m_o = -\frac{L}{f_o} \quad \text{--- (1)}$$

* Case I - When image is formed at near point (LDDV)



Compound Microscope

* Linear magnification (m_o) due to objective lens-

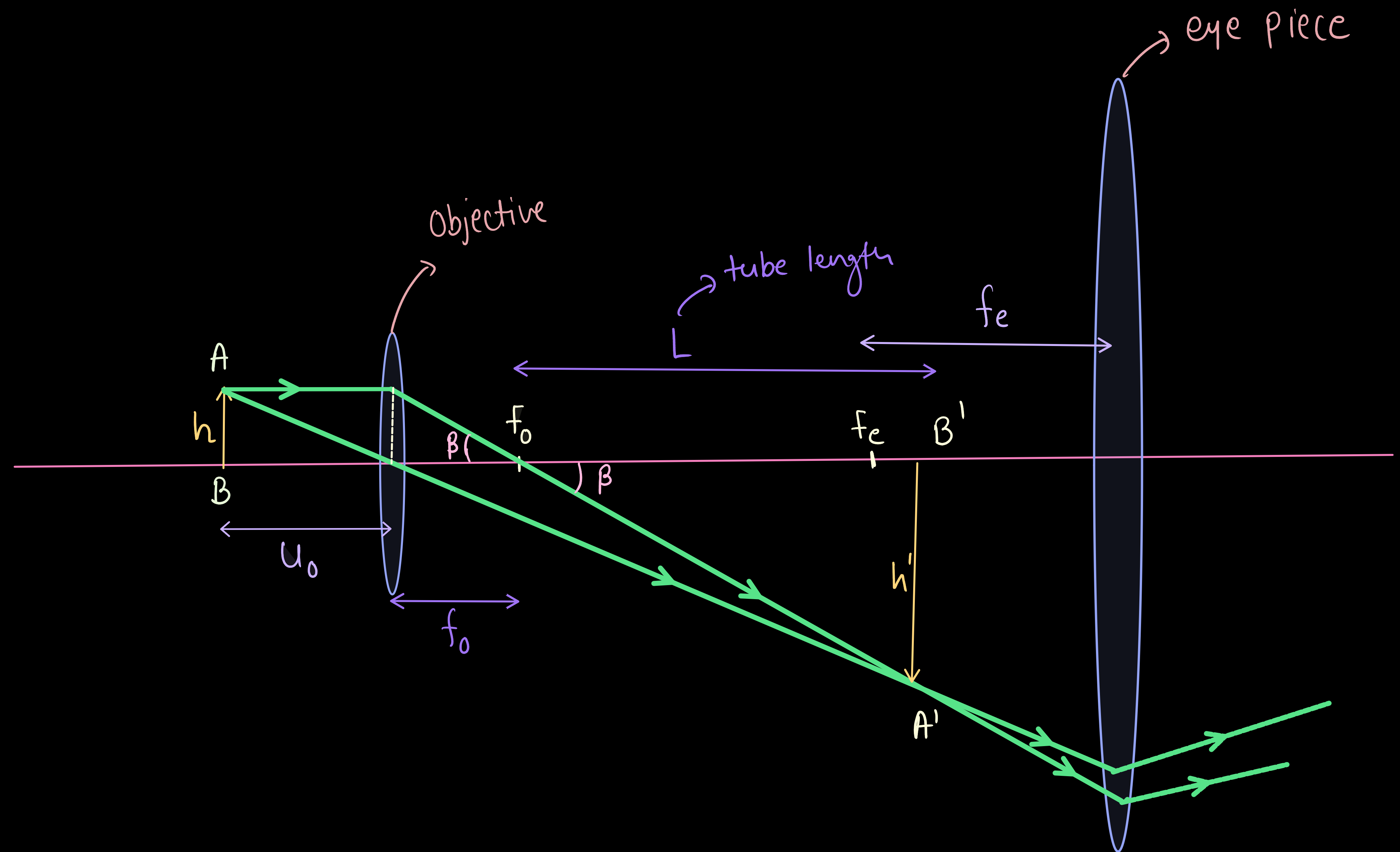
$$m_o = -\frac{h'}{h}$$

$$\tan \beta = \frac{h}{f_o} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_o}$$

$$m_o = -\frac{L}{f_o} \quad \text{--- (1)}$$

* Case I - When image is formed at near point (LDDV)



Compound Microscope

* Linear magnification (m_o) due to objective lens-

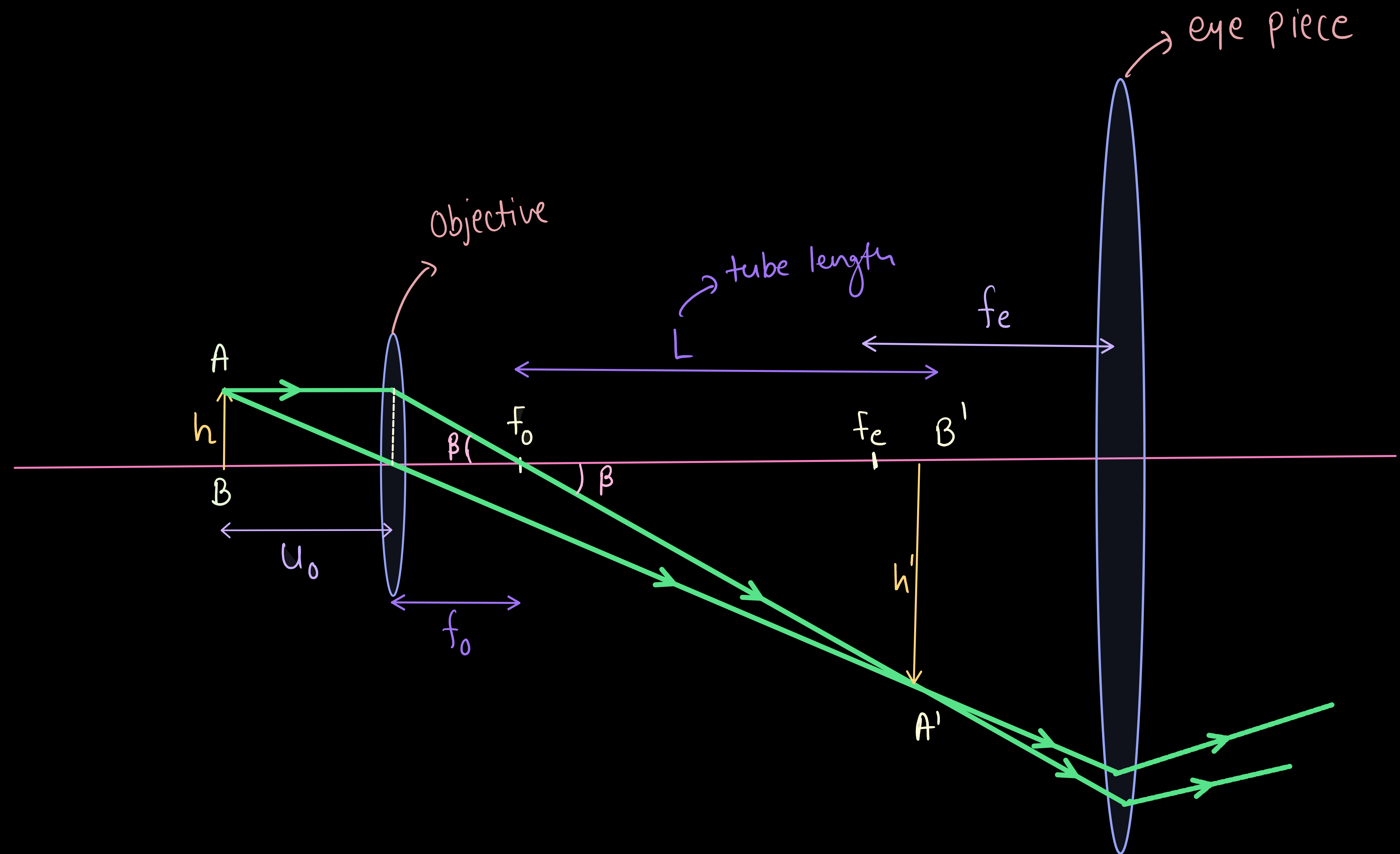
$$m_o = -\frac{h'}{h}$$

$$\tan \beta = \frac{h}{f_o} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_o}$$

$$m_o = -\frac{L}{f_o} \quad \text{--- (1)}$$

* Case I- When image is formed at near point (LDDV)



Compound Microscope

* Linear magnification (m_o) due to objective lens-

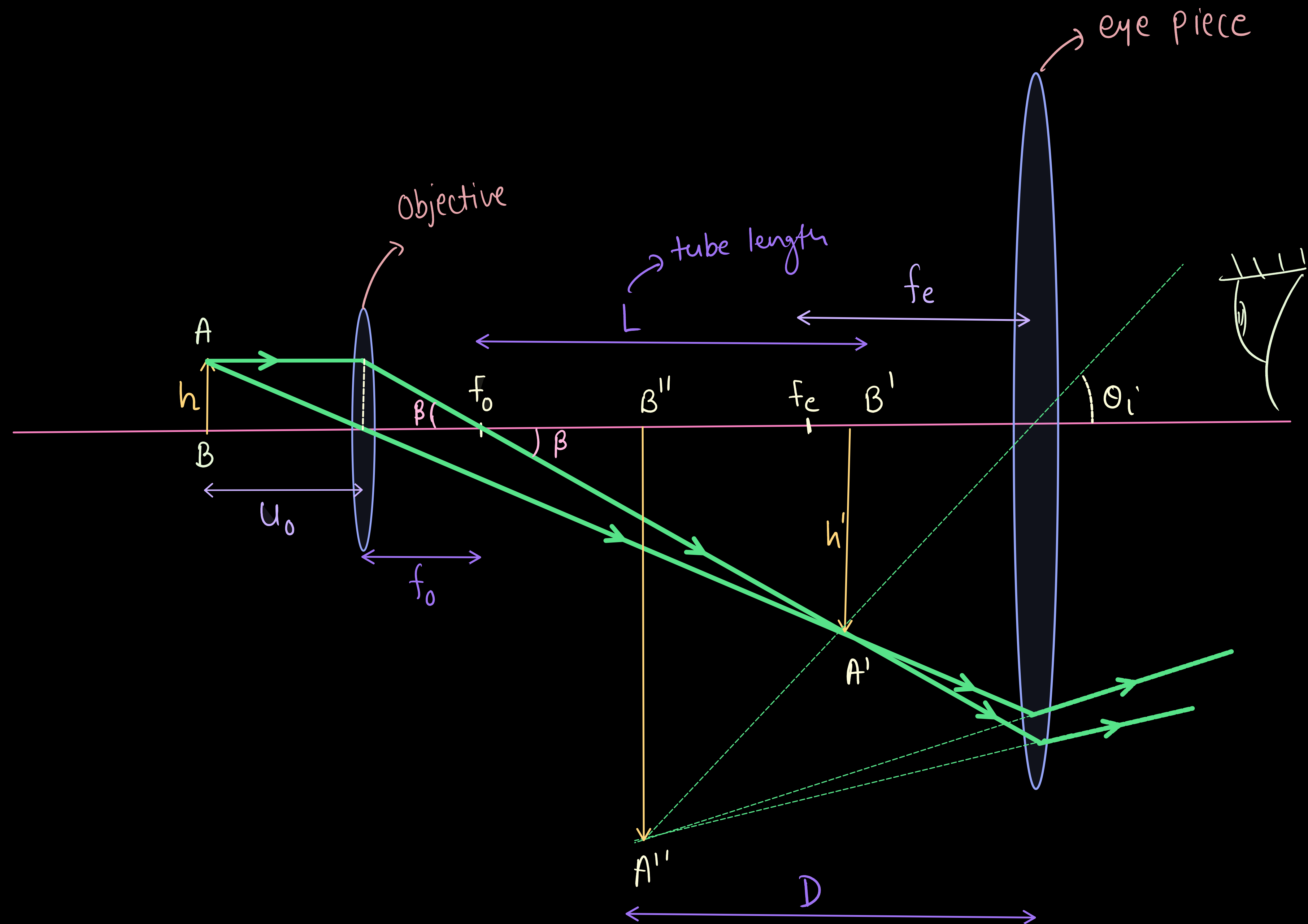
$$m_o = -\frac{h'}{h}$$

$$\tan \beta = \frac{h}{f_o} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_o}$$

$$m_o = -\frac{L}{f_o} \quad \text{--- (1)}$$

* Case I- When image is formed at near point (LDDV)



Compound Microscope

* Linear magnification (m_o) due to objective lens-

$$m_o = -\frac{h'}{h}$$

$$\tan \beta = \frac{h}{f_o} = \frac{h'}{L}$$

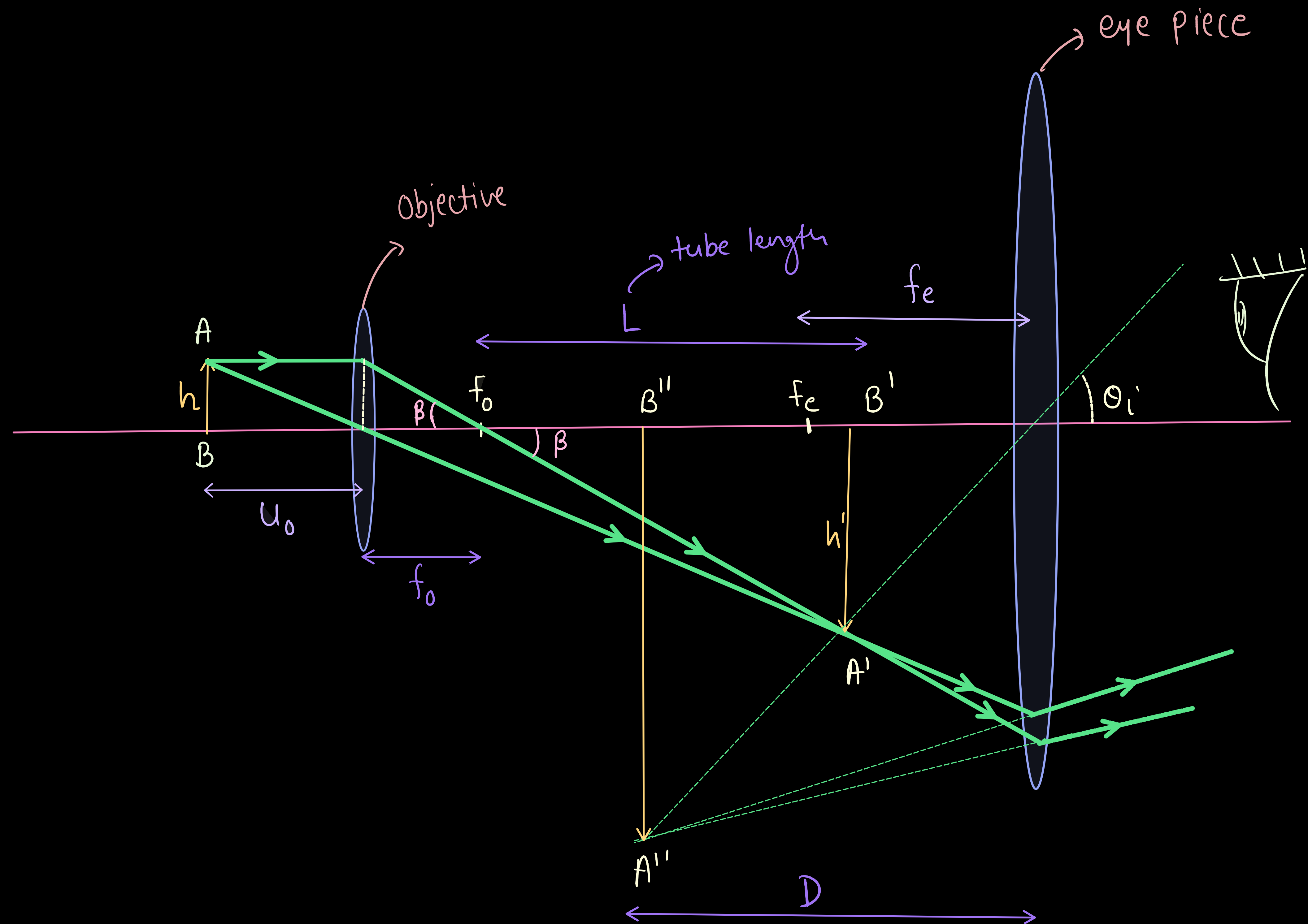
$$\frac{h'}{h} = \frac{L}{f_o}$$

$$m_o = -\frac{L}{f_o} \quad \text{--- (1)}$$

* Case I- When image is formed at near point (LDDV)

magnification by eye-piece -

$$m_e = 1 + \frac{D}{f_e} \quad \text{--- (2)}$$



Compound Microscope

* Linear magnification (m_o) due to objective lens-

$$m_o = -\frac{h'}{h}$$

$$\tan \beta = \frac{h}{f_o} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_o}$$

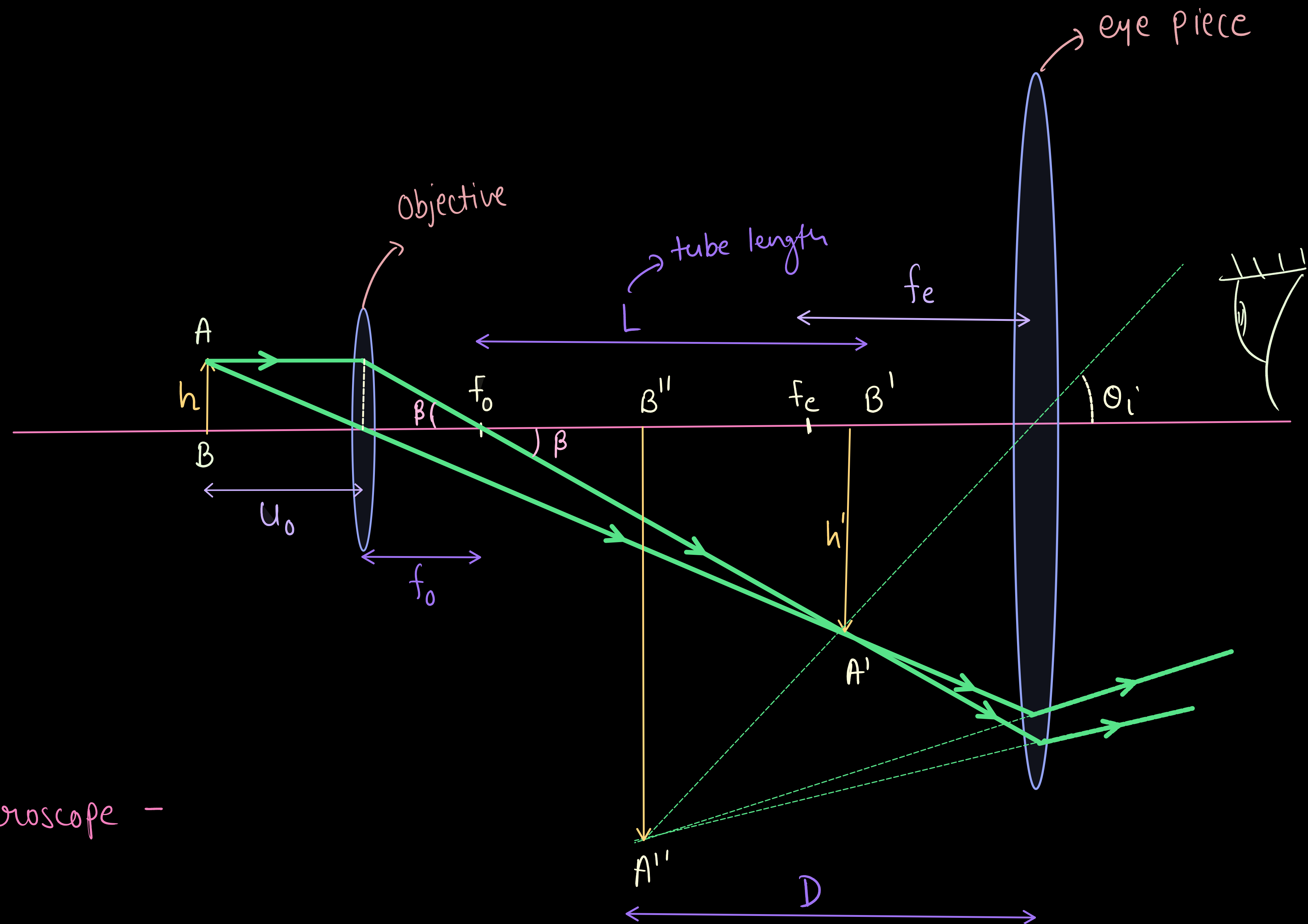
$$m_o = -\frac{L}{f_o} \quad \text{--- (1)}$$

* Case I- When image is formed at near point (LDDV)

magnification by eye-piece -

$$m_e = 1 + \frac{D}{f_e} \quad \text{--- (2)}$$

⇒ Total Magnification produced by compound microscope -



Compound Microscope

* Linear magnification (m_o) due to objective lens-

$$m_o = -\frac{h'}{h}$$

$$\tan \beta = \frac{h}{f_o} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_o}$$

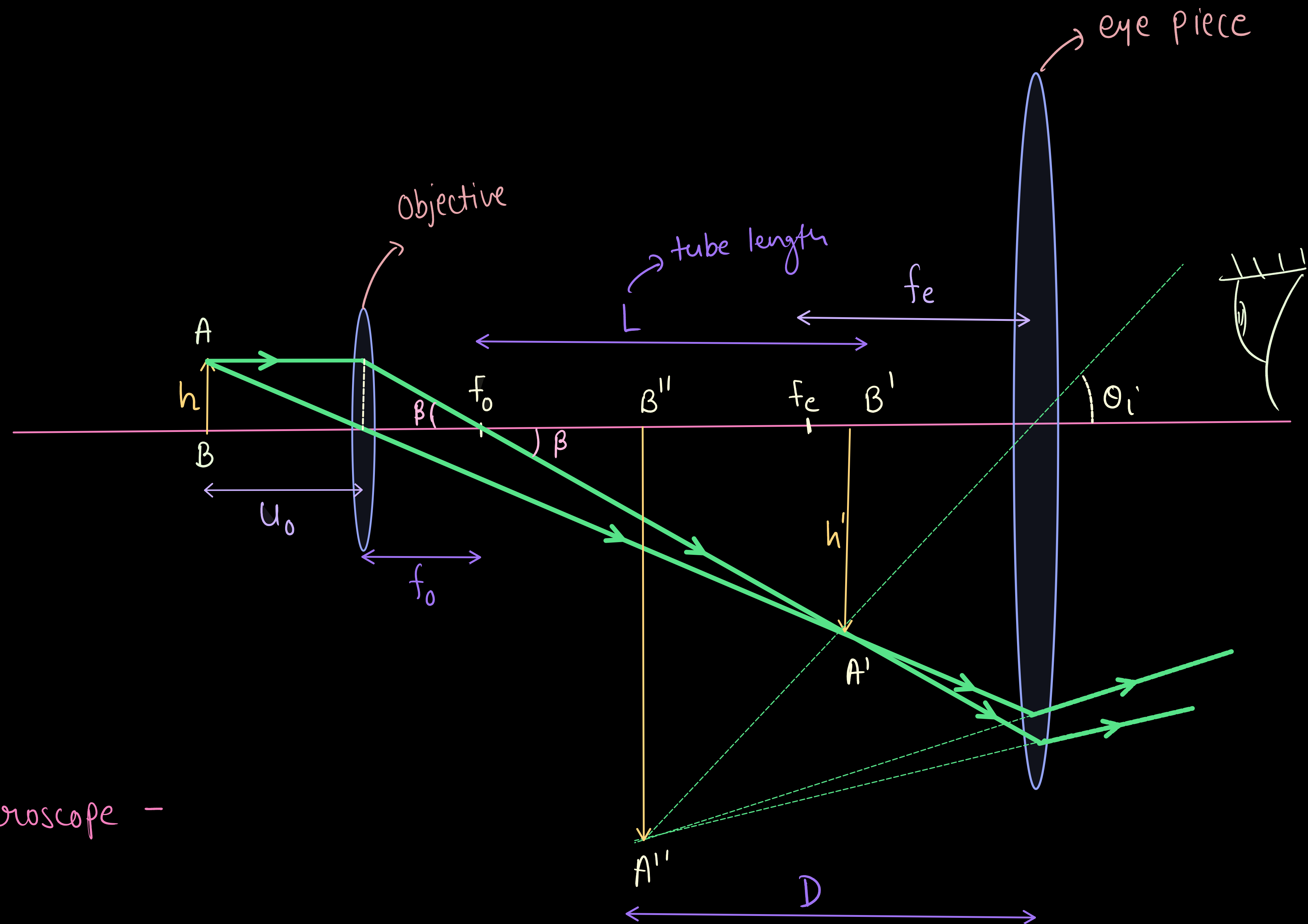
$$m_o = -\frac{L}{f_o} \quad \text{--- (1)}$$

* Case I- When image is formed at near point (LDDV)

magnification by eye-piece -

$$m_e = 1 + \frac{D}{f_e} \quad \text{--- (2)}$$

⇒ Total Magnification produced by compound microscope -
 $m = m_o \times m_e$



Compound Microscope

* Linear magnification (m_o) due to objective lens-

$$m_o = -\frac{h'}{h}$$

$$\tan \beta = \frac{h}{f_o} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_o}$$

$$m_o = -\frac{L}{f_o} \quad \text{--- (1)}$$

* Case I- When image is formed at near point (LDDV)

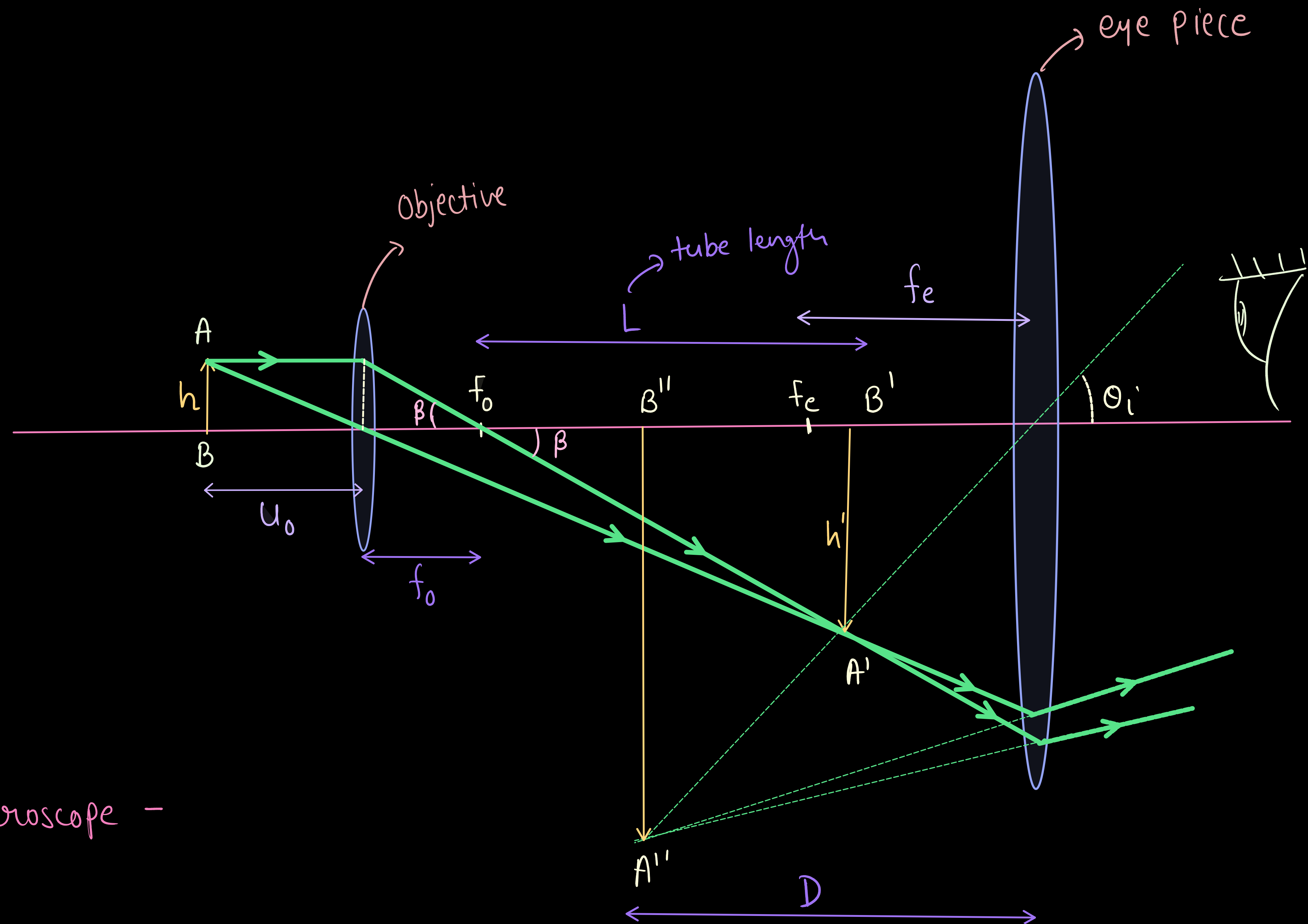
magnification by eye-piece -

$$m_e = 1 + \frac{D}{f_e} \quad \text{--- (2)}$$

⇒ Total Magnification produced by compound microscope -

$$m = m_o \times m_e$$

$$m = -\frac{L}{f_o} \left(1 + \frac{D}{f_e} \right) \quad \text{--- (3)}$$



Compound Microscope

* Linear magnification (m_o) due to objective lens-

$$m_o = -\frac{h'}{h}$$

$$\tan \beta = \frac{h}{f_o} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_o}$$

$$m_o = -\frac{L}{f_o} \quad \text{--- (1)}$$

* Case I- When image is formed at near point (LDDV)

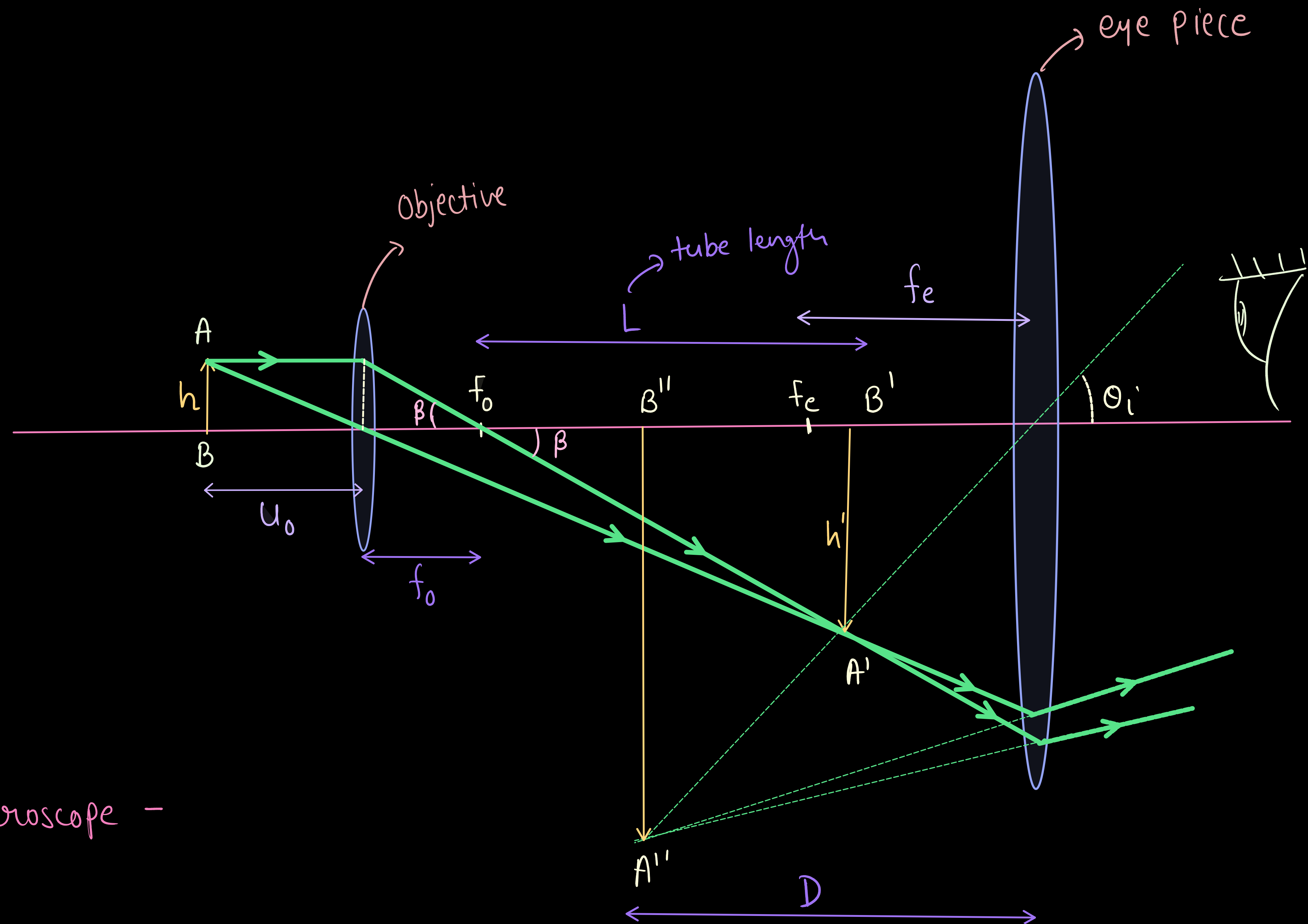
magnification by eye-piece -

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⇒ Total Magnification produced by compound microscope -

$$m = m_o \times m_e$$

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Compound Microscope

* Linear magnification (m_o) due to objective lens-

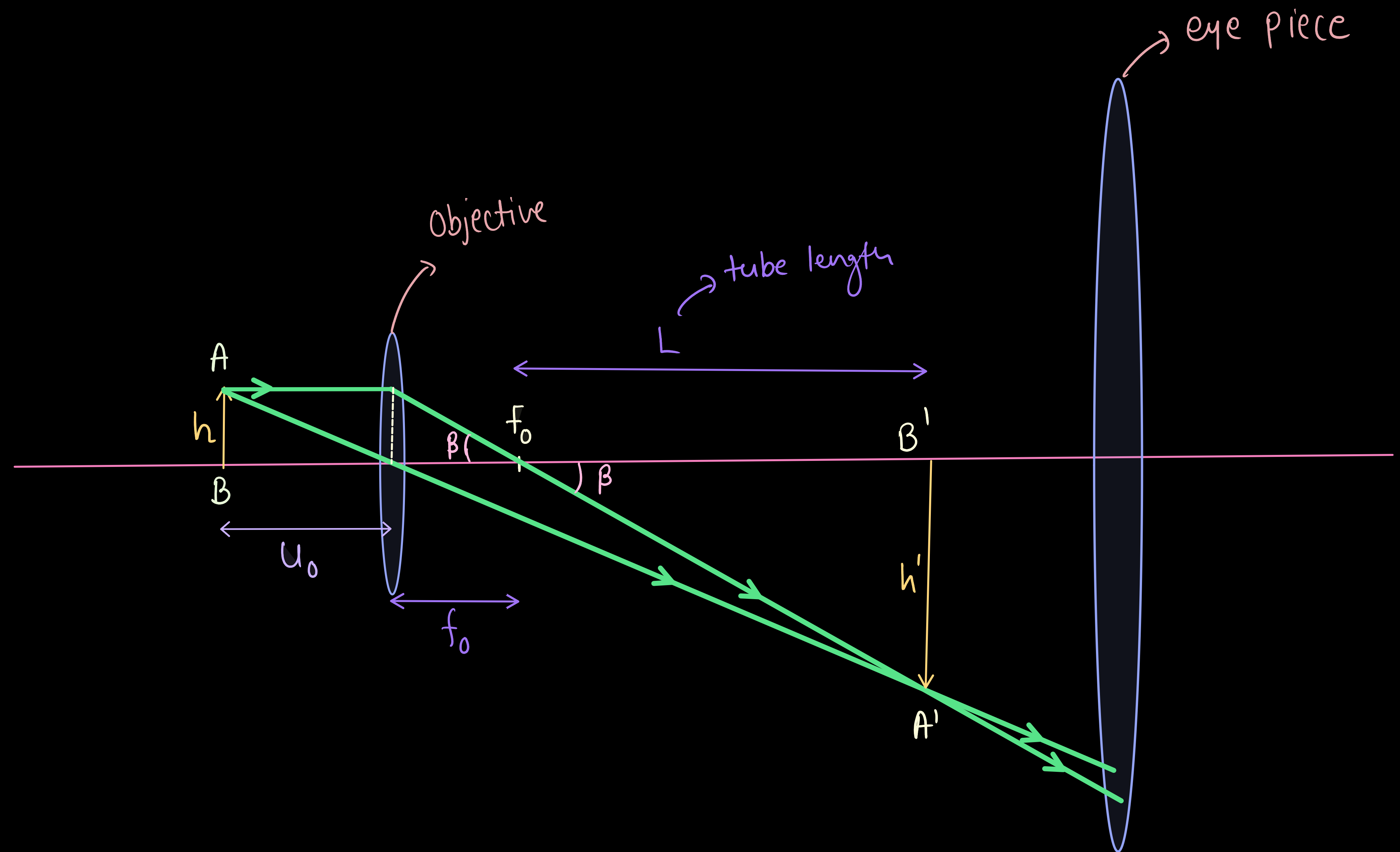
$$m_o = -\frac{h'}{h}$$

$$+amp = \frac{h}{f_o} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_o}$$

$$m_o = -\frac{L}{f_o} \quad \text{--- (1)}$$

Case II- when image is formed at infinity-



Compound Microscope

* Linear magnification (m_o) due to objective lens-

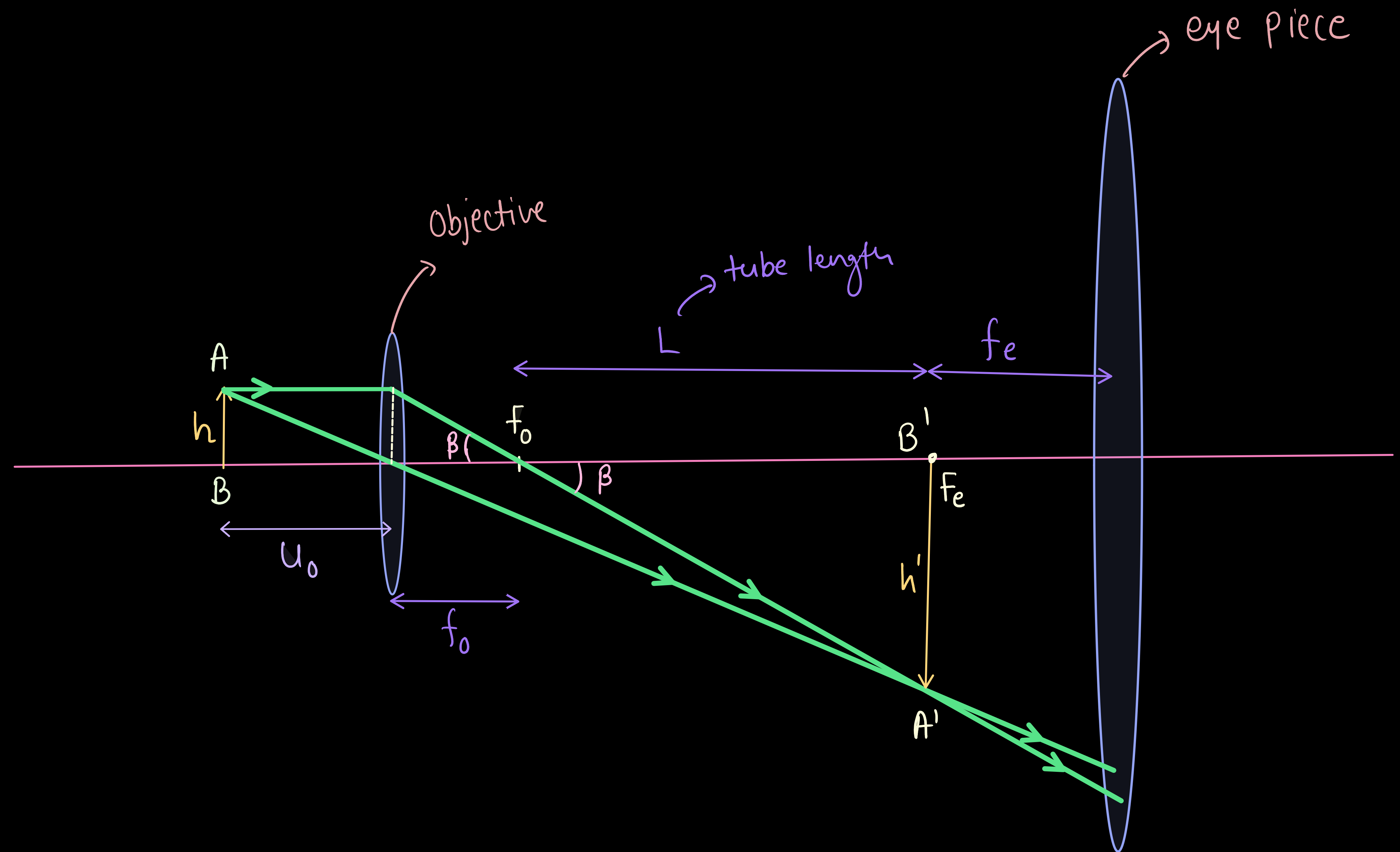
$$m_o = -\frac{h'}{h}$$

$$f_{\text{amp}} = \frac{h}{f_o} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_o}$$

$$m_o = -\frac{L}{f_o} \quad \text{--- (1)}$$

Case II- when image is formed at infinity-



Compound Microscope

* Linear magnification (m_o) due to objective lens-

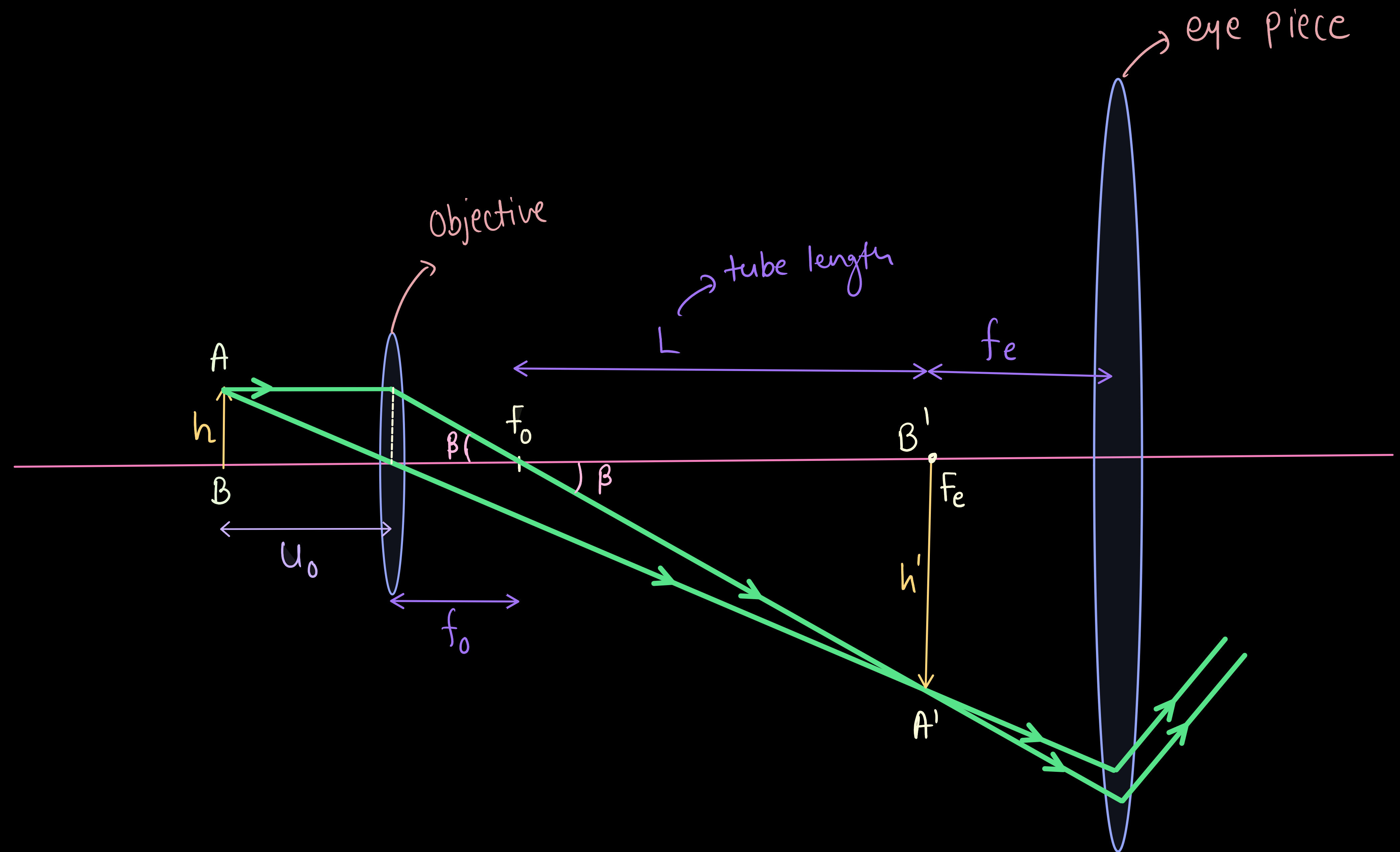
$$m_o = -\frac{h'}{h}$$

$$f_{\text{amp}} = \frac{h}{f_o} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_o}$$

$$m_o = -\frac{L}{f_o} \quad \text{--- (1)}$$

Case II- when image is formed at infinity-



Compound Microscope

* Linear magnification (m_o) due to objective lens-

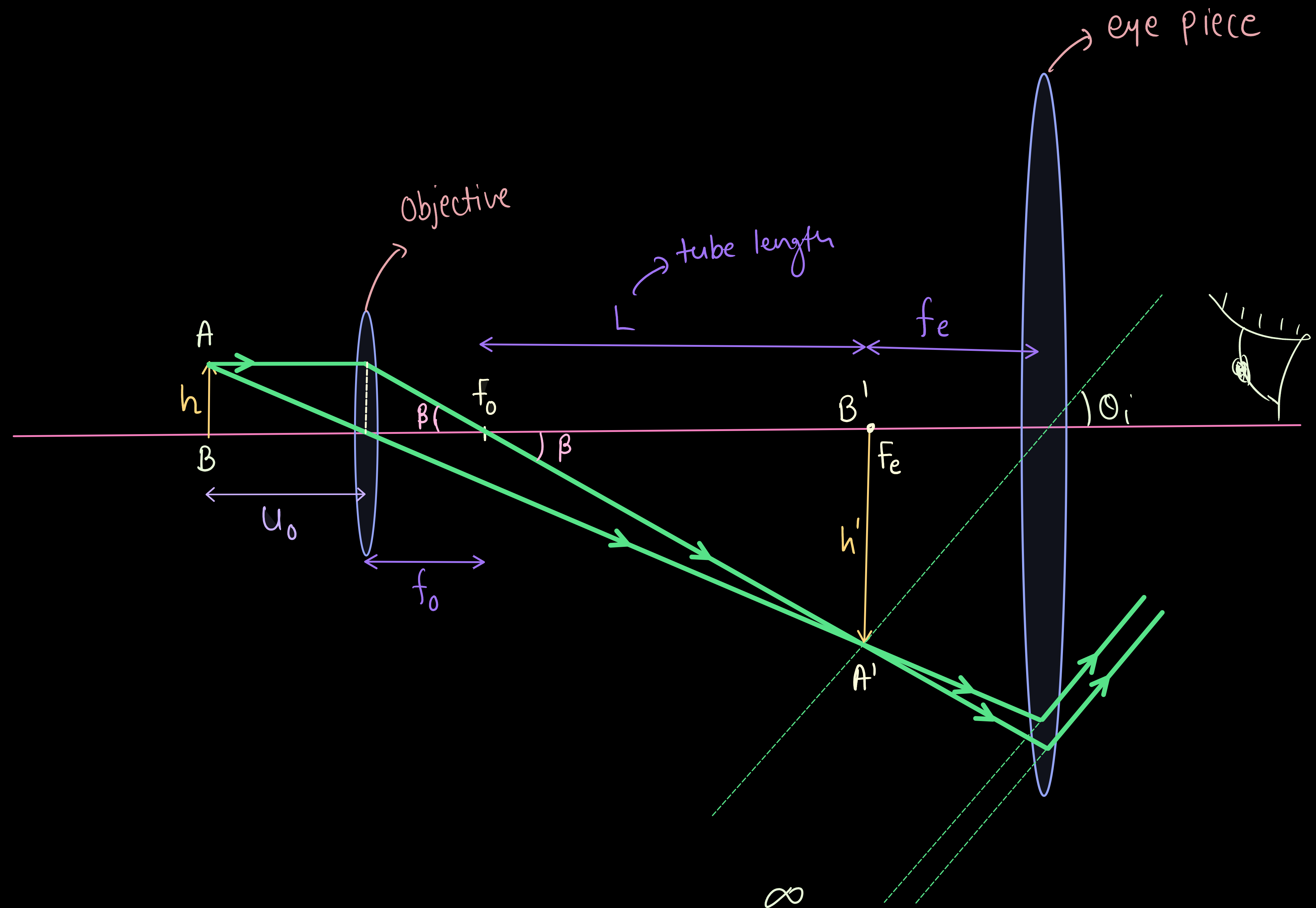
$$m_o = -\frac{h'}{h}$$

$$f_{\text{amp}} = \frac{h}{f_o} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_o}$$

$$m_o = -\frac{L}{f_o} \quad \text{--- (1)}$$

Case II- when image is formed at infinity-



Compound Microscope

* Linear magnification (m_o) due to objective lens-

$$m_o = -\frac{h'}{h}$$

$$+amp = \frac{h}{f_o} = \frac{h'}{L}$$

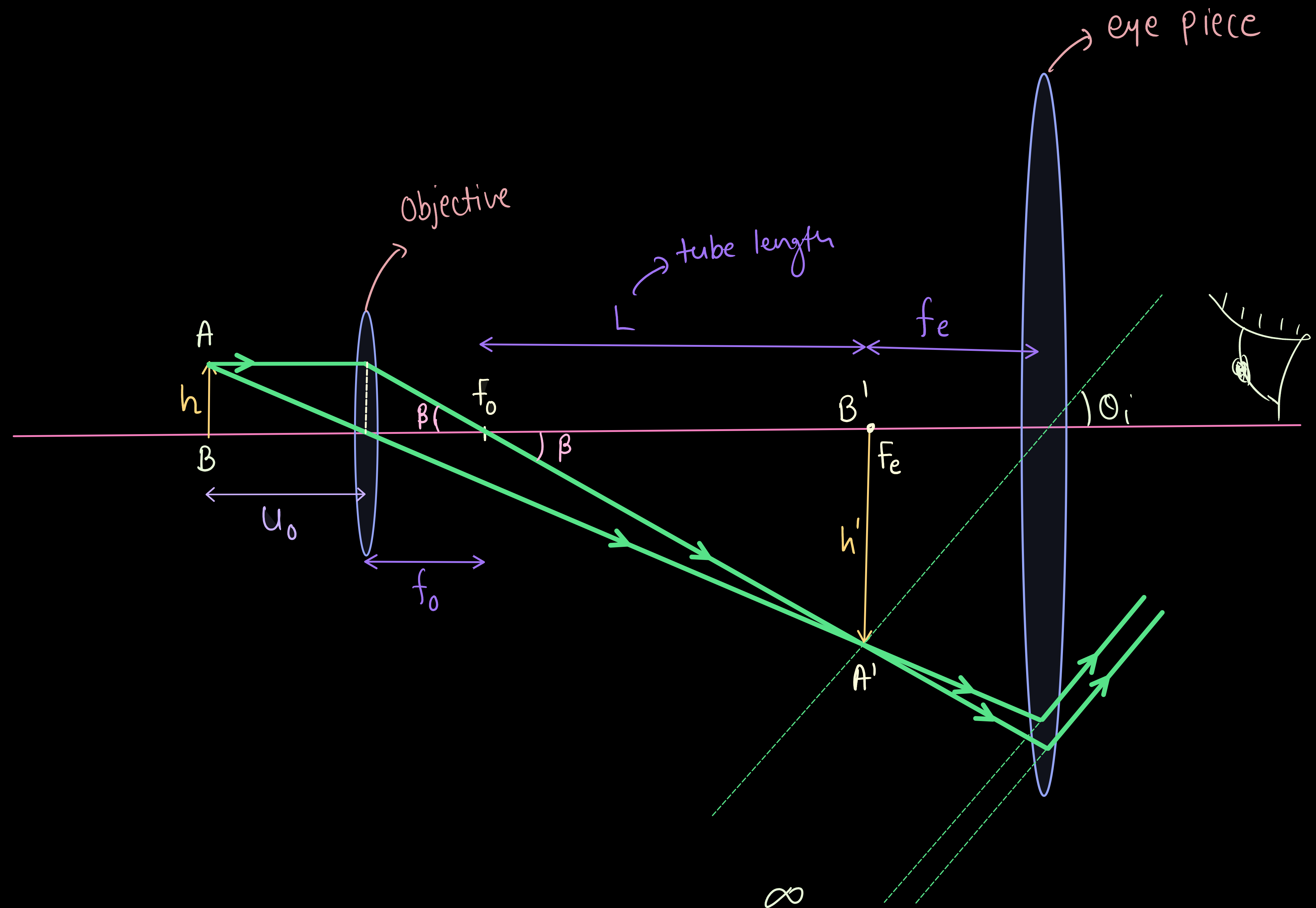
$$\frac{h'}{h} = \frac{L}{f_o}$$

$$m_o = -\frac{L}{f_o} \quad \text{--- (1)}$$

Case II- when image is formed at infinity-

magnification by eyepiece lens

$$m_e = \frac{D}{f_e} \quad \text{--- (2)}$$



Compound Microscope

* Linear magnification (m_o) due to objective lens-

$$m_o = -\frac{h'}{h}$$

$$+amp = \frac{h}{f_o} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_o}$$

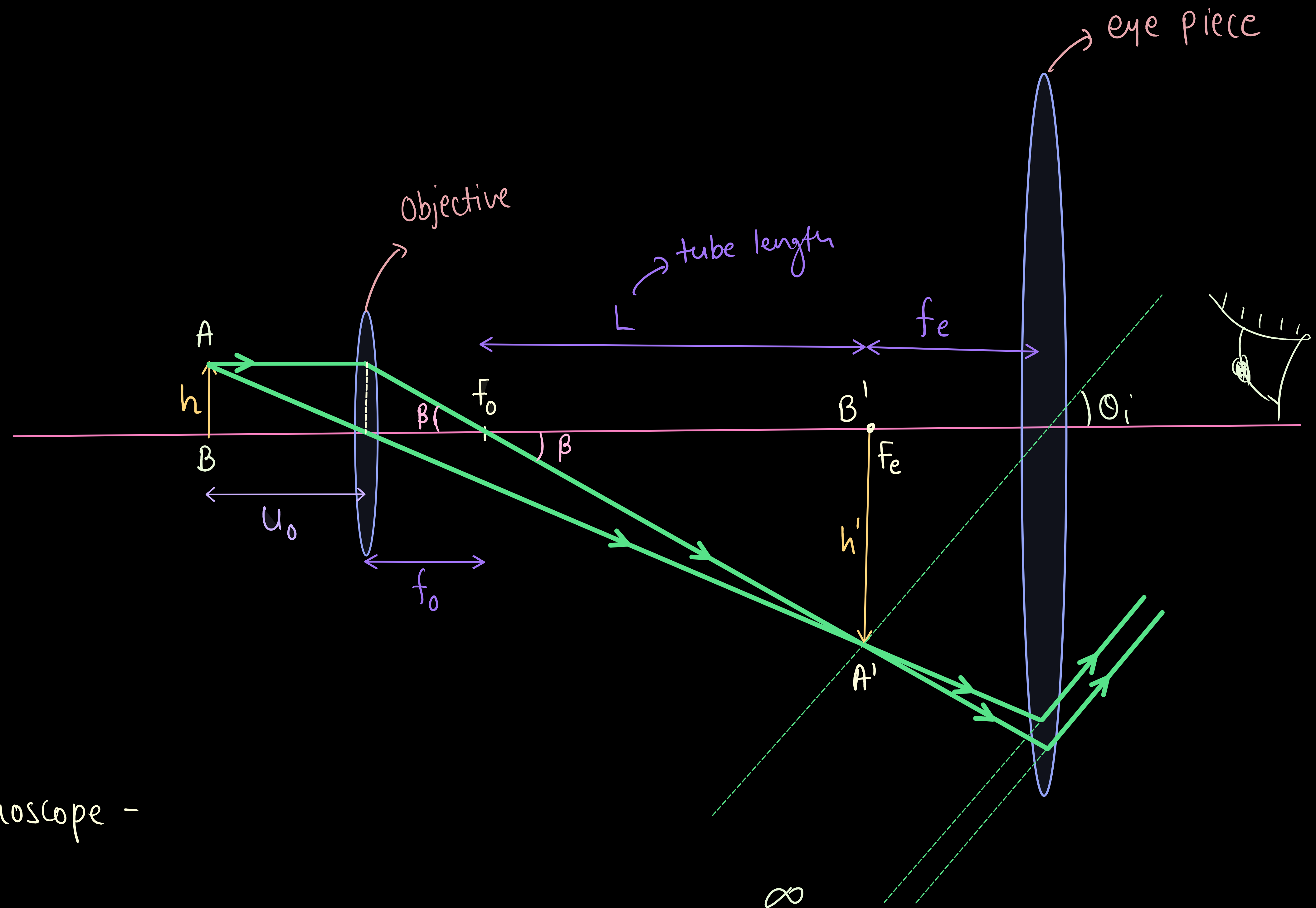
$$m_o = -\frac{L}{f_o} \quad \text{--- (1)}$$

Case II- when image is formed at infinity-

magnification by eyepiece lens

$$m_e = \frac{D}{f_e} \quad \text{--- (2)}$$

Total magnification produced by compound microscope -



Compound Microscope

* Linear magnification (m_o) due to objective lens-

$$m_o = -\frac{h'}{h}$$

$$f_{\text{amp}} = \frac{h}{f_o} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_o}$$

$$m_o = -\frac{L}{f_o} \quad \text{--- (1)}$$

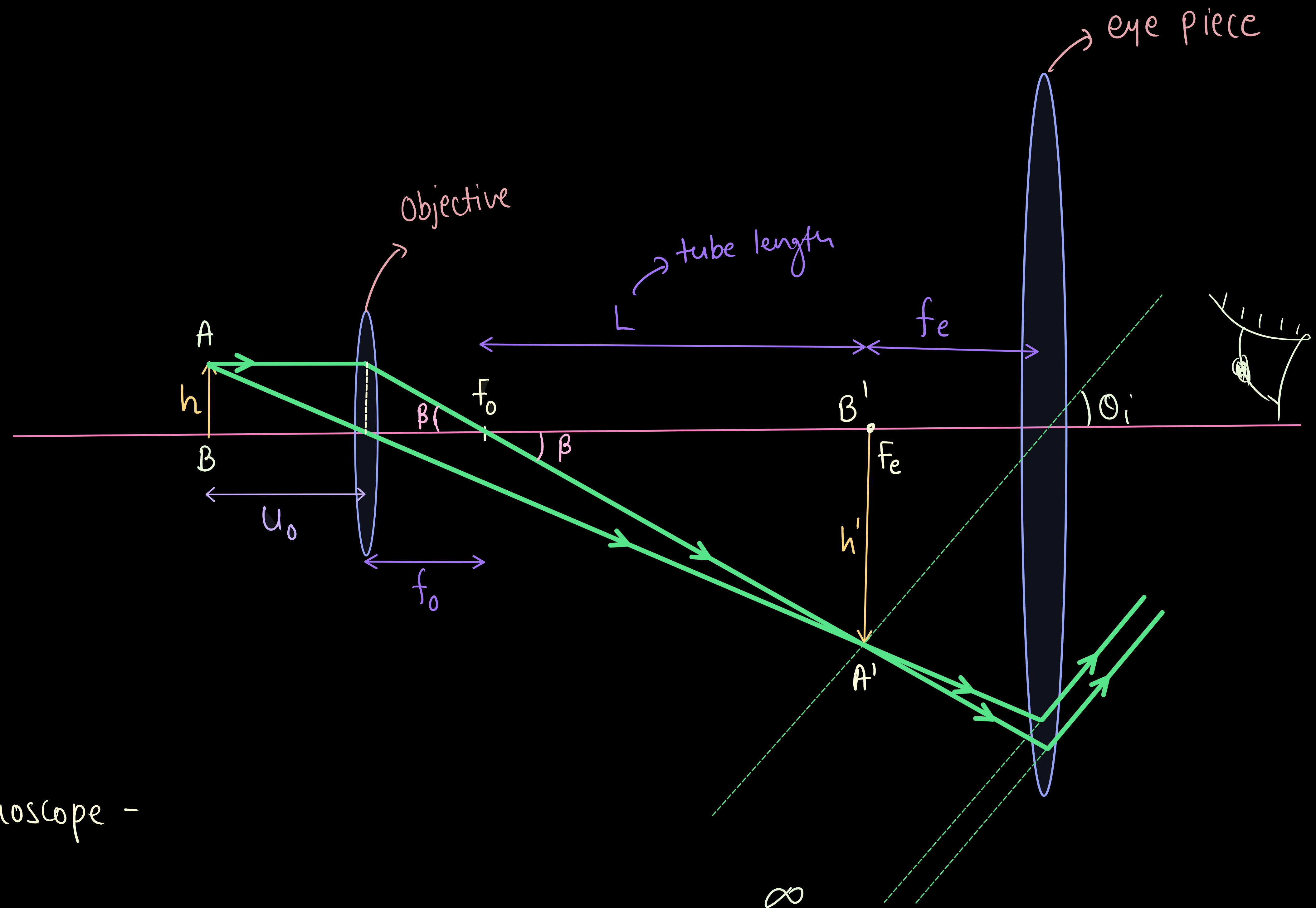
Case II- when image is formed at infinity-

magnification by eyepiece lens

$$m_e = \frac{D}{f_e} \quad \text{--- (2)}$$

Total magnification produced by compound microscope -

$$m = m_o \times m_e$$



Compound Microscope

* Linear magnification (m_o) due to objective lens-

$$m_o = -\frac{h'}{h}$$

$$+amp = \frac{h}{f_o} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_o}$$

$$m_o = -\frac{L}{f_o} \quad \text{--- (1)}$$

Case II- when image is formed at infinity-

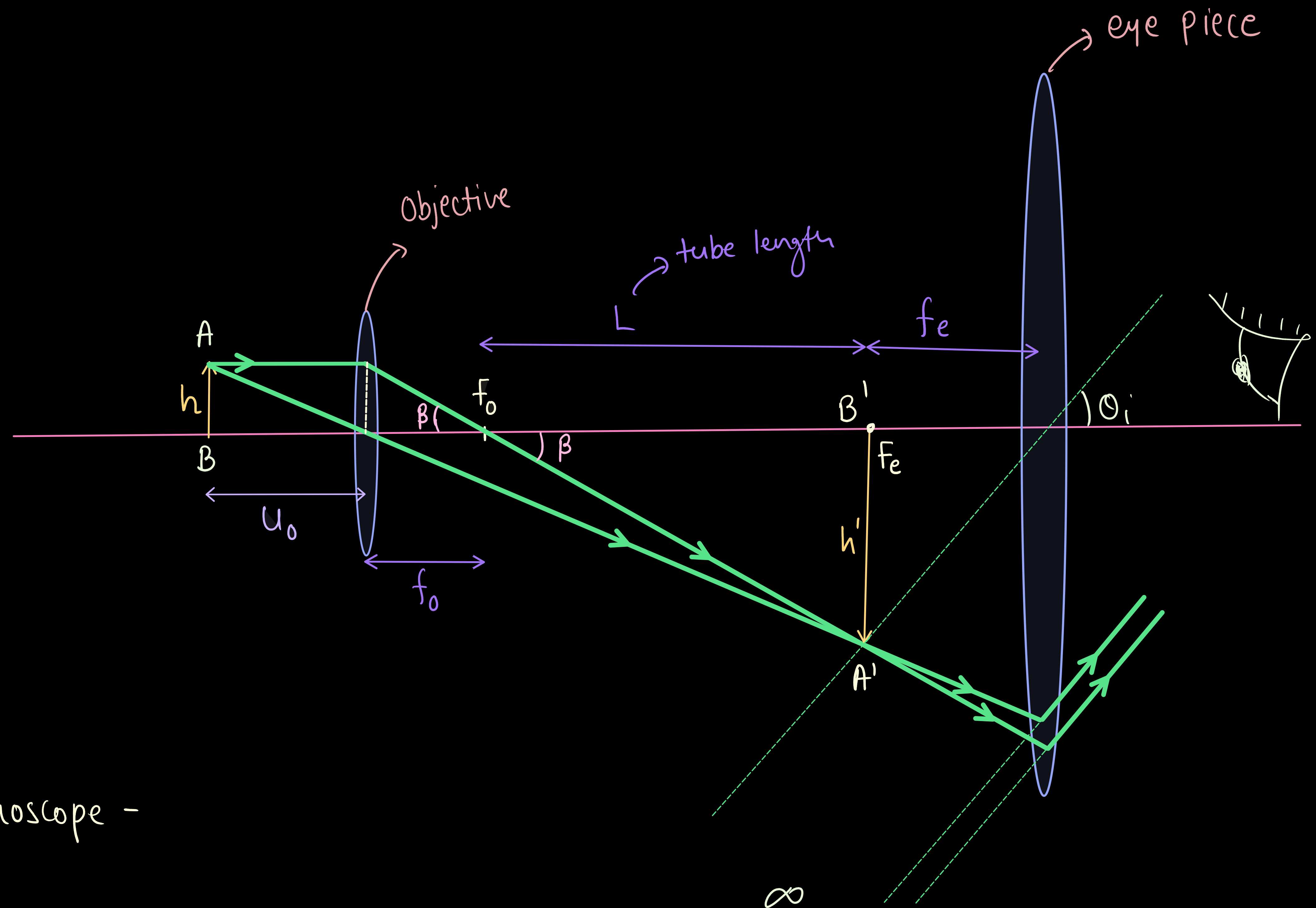
magnification by eyepiece lens

$$m_e = \frac{D}{f_e} \quad \text{--- (2)}$$

Total magnification produced by compound microscope -

$$m = m_o \times m_e$$

$$m = -\frac{L}{f_o} \times \frac{D}{f_e} \quad \text{--- (3)}$$



Compound Microscope

* Linear magnification (m_o) due to objective lens-

$$m_o = -\frac{h'}{h}$$

$$+m_p = \frac{h}{f_o} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_o}$$

$$m_o = -\frac{L}{f_o} \quad \text{--- (1)}$$

Case II- when image is formed at infinity-

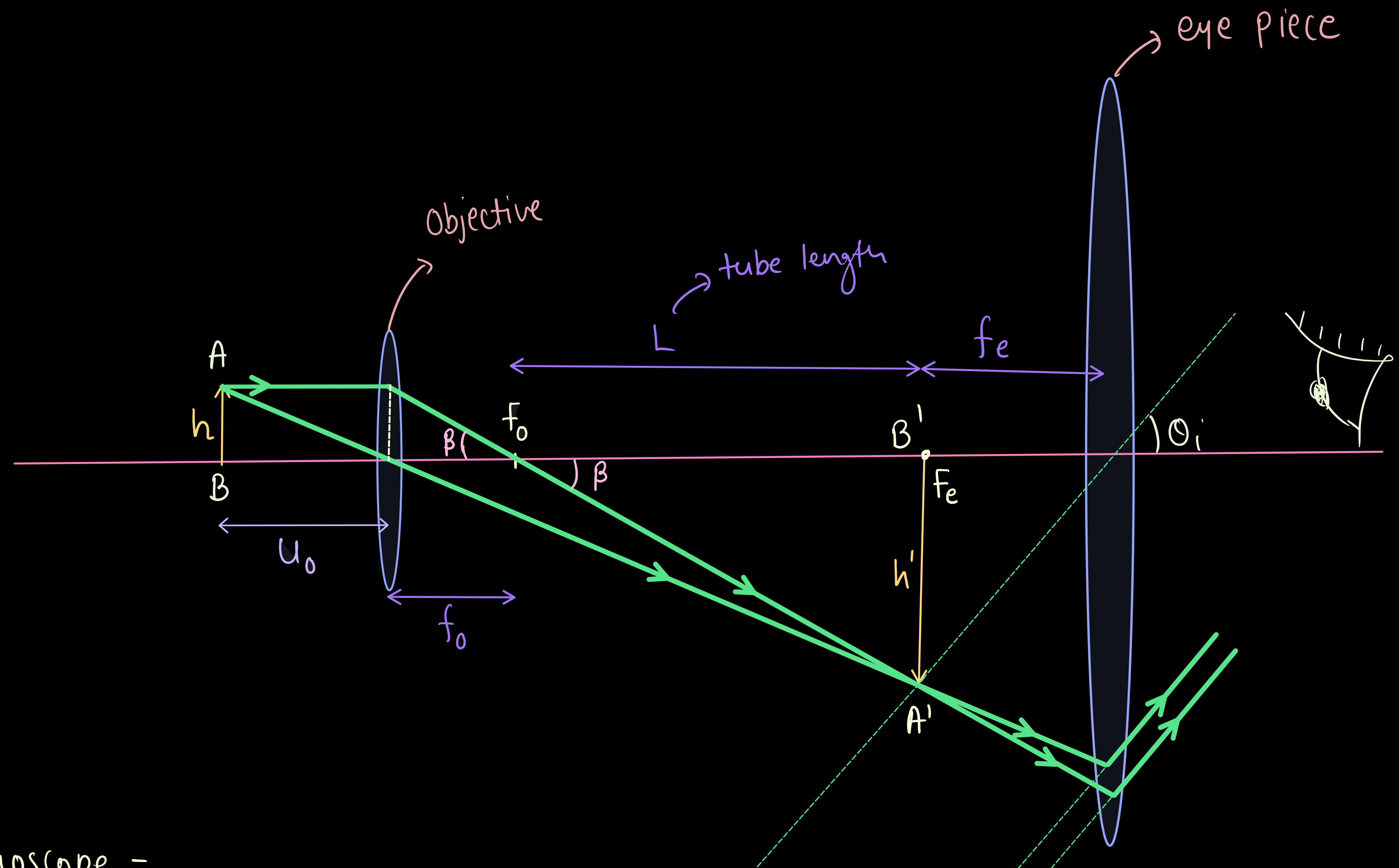
magnification by eyepiece lens

$$m_e = \frac{D}{f_e} \quad \text{--- (2)}$$

Total magnification produced by compound microscope -

$$m = m_o \times m_e$$

$$m = -\frac{L}{f_o} \times \frac{D}{f_e} \quad \text{--- (3)}$$



* Clearly, to achieve a large magnification of a small object (hence the name microscope), the objective and eyepiece should have small focal lengths. In practice, it is difficult to make the focal length much smaller than 1 cm. Also large lenses are required to make L large.

Compound Microscope

Reflecting Telescope
